

**REDUCING CASH CONVERSION CYCLE IN
RENOVATION PROJECTS AT R.E. WEERAKOON
CONSTRUCTIONS (PVT) LTD.**

J. L. Weerakoon

2023

**REDUCING CASH CONVERSION CYCLE IN
RENOVATION PROJECTS AT R.E. WEERAKOON
CONSTRUCTIONS (PVT) LTD.**

By

J. L. Weerakoon

BA (Northumbria University, UK)

Management Field Research Project Report submitted to the Postgraduate
Institute of Management, University of Sri Jayewardenepura, in partial
fulfilment of the requirements of the Master of Business Administration Degree

Colombo – Sri Lanka

2023

This is to certify that the Management Field Research Project report on

**REDUCING CASH CONVERSION CYCLE IN
RENOVATION PROJECTS AT R.E. WEERAKOON
CONSTRUCTIONS (PVT) LTD.**

By

J. L. Weerakoon

(MBA/20/5689)

has been accepted by the Postgraduate Institute of Management of the University of Sri
Jayewardenepura, in partial fulfilment of the requirements of the Master of Business
Administration Degree.

.....

Supervisor

.....

Director

.....

Date

.....

Date

DECLARATION

I certify that this Management Field Research Project report does not incorporate without acknowledgement, any previously submitted for a degree or a diploma in any university, and to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in the text.

.....

J. L. Weerakoon

January 31, 2023.

TABLE OF CONTENT

S

LIST OF ABBREVIATIONS.....	xi
ACKNOWLEDGEMENTS.....	xii
EXECUTIVE SUMMARY.....	xiii
Chapter 1 INTRODUCTION.....	1
1.1 Background.....	1
1.2 Project.....	2
1.3 SignificanceoftheProject.....	2
1.4 Methodology.....	3
1.5 Limitations.....	3
1.6 ChapterFramework.....	4
Chapter 2 ORGANISATIONAL ANALYSIS AND PROBLEM IDENTIFICATION.....	6
2.1 Introduction.....	6
2.2 OrganisationProfile.....	6
2.3 OrganisationAnalysis.....	7
2.4 KeyProblem(s).....	11
2.5 Description of the Problem(s).....	11
2.6 Analysis of the Key Problem.....	19
2.7 Summary.....	22
Chapter 3 LITERATURE REVIEW.....	23
3.1 Introduction.....	23

3.2	Theoretical Background to the Problem.....	23
3.3	StudyFramework.....	42
3.4	DescriptionofTechniques.....	43
3.5	Summary.....	45
Chapter 4 THE PROJECT/METHODOLOGY.....		46
4.1	Introduction.....	46
4.2	FieldProjectObjectives.....	46
4.3	The Project: Project Components.....	47
4.4	ResourceAllocation.....	57
4.5	CosEstimates.....	63
4.6	Summary.....	64
Chapter 5 FINDINGS.....		65
5.1	Introduction.....	65
5.2	Benefit-CostAnalysis.....	65
5.3	Projected Outputs and Outcomes.....	66
5.4	Summary.....	67
Chapter 6 DISCUSSION OF FINDINGS, RECOMMENDATIONS AND CONCLUSION.....		69
6.1	Introduction.....	69
6.2	Discussion of Findings.....	69
6.3	Recommendations.....	72
6.4	Conclusions.....	76

REFERENCES.....	77
APPENDICES.....	90
Annexure I: Summary of the Income Statements.....	90
Annexure II: Organisational Structure.....	91
Annexure III: Cash Conversion Cycle Calculation.....	92
Annexure IV: Summary of the Interviews Conducted with Key Stakeholders.....	94
Annexure V: Ranked Journals Used.....	103
Annexure VI: Project Objective Setting.....	104
Annexure VII: Proposed Inventory Management Process.....	105
Annexure VIII: Proposed Daily Project Report Format.....	106
Annexure IX: Proposed Workshop to Introduce ABC Analysis.....	108
Annexure XI: Proposed Training for Quantity Surveyors.....	128
Annexure XII: Proposed Receivable Policy.....	133
Annexure XIII: Proposed Quality Plan.....	147
Annexure XIV: Proposed Work Breakdown Structure Method.....	158
Annexure XV: Proposed Supplier Evaluation Process.....	162
Annexure XVI: Current Supplier Evaluation Form.....	163
Annexure XVII: Proposed Supplier Evaluation Criterion.....	165
Annexure XVIII: Proposed Supplier Relationship Management Guidelines.....	169
Annexure XIX: Proposed Accounting Software.....	171
Annexure XX: Tangible Benefits.....	181

Annexure XXI: Estimated Costs to Implement Solutions.....	183
---	-----

LIST OF TABLES

Table 2.1: SWOT Analysis.....	8
Table 2.2: Cash Conversion Cycle in Renovation Projects.....	12
Table 2.3: Idle Inventory in Renovation Projects.....	14
Table 2.4: Overdue Receivables in Renovation Projects.....	16
Table 2.5: Early Payment to Suppliers in Renovation Projects.....	18
Table 4.1: Project Component, Objectives and Techniques.....	46
Table 4.2: Resource Allocation.....	57
Table 4.3: Project Components, Proposed Solutions and Budget.....	63
Table 5.1: Estimated Benefits.....	65
Table 5.2: Benefit-Cost Analysis.....	66
Table 5.3: Projected Outputs and Outcomes.....	66

LIST OF FIGURES

Figure 2.1: Functions of REWC.....	7
Figure 2.2: Location of the Problem.....	12
Figure 2.3: Cash Conversion Cycle of Renovation Projects.....	13
Figure 2.4: Idle Inventory in Renovation Projects.....	15
Figure 2.5: Overdue Receivables in Renovation Projects.....	17
Figure 2.6: Early Payment to Suppliers in Renovation Projects.....	19
Figure 2.7 Root Cause Analysis.....	20
Figure 3.1: Study Framework.....	42
Figure 4.1: Idle Inventory at the Warehouse.....	48
Figure 4.2: Supplier Delivering Inventory to the Construction Site.....	49
Figure 4.3: Current Inventory Management Process of REWC.....	51
Figure 4.4: Current Invoicing Process for Client Payments.....	53
Figure 4.5 Current Supplier Evaluation Process.....	56
Figure 4.6: Project Team Structure.....	62

LIST OF ABBREVIATIONS

Abbreviation	Full form
BOQ	Bills of Quantities
CCC	Cash Conversion Cycle
CIDA	Construction Industry Development Authority
FIFO	First In First Out
GDP	Gross Domestic Product
ICP	Inventory Conversion Period
ICTAD	Institute of Construction Training and Development
IPC	Interim Payment Certificate
KPIs	Key Performance Indicators
LKR	Sri Lankan Rupee
POP	Payable Outstanding Period
QS	Quantity Surveyor
RCP	Receivable Conversion Period
REWC	R.E. Weerakoon Constructions
SLSI	Sri Lanka Standard Institution
SMART	Specific, Measurable, Achievable, Realistic and Time bounded
SME	Small and Medium-sized Enterprise
SWOT	Strengths, Weaknesses, Opportunities and Threats
TO	Technical Officer
WACC	Weighted Average Cost of Capital
WBS	Work Breakdown Structure
WCM	Working Capital Management

ACKNOWLEDGEMENTS

It was a fascinating experience where I had the great opportunity to integrate everything, I had learned during my two years of MBA coursework to use and experience the pleasure of seeing a realistic, practical outcome. I express gratitude to Mr. Jithendra Gunathilake for the guidance and encouragement provided during this period as my supervisor. His assistance made it possible to complete the project successfully.

Secondly, I would like to express gratitude to Dr. Senaka Kelum Gamage for the invaluable guidance provided as the director of postgraduate institute of management. I also would like to sincerely thank the PIM faculty, including Dr. A.K.L. Jayewardene, Dr. Samantha Rathnayake, senior assistant registrar, and administrative staff of PIM for every support provided during the MBA.

Thirdly, I would like to thank the managing director and all the staff members of R.E. Weerakoon Constructions from top to bottom for the support given by providing the opportunity to gain the knowledge and skill required to perform the project. The project would have been a failure without those discussions, interviews, guidance, and other forms of information provided amidst their busy schedules.

Finally, I am grateful and indebted to my family and friends for being a genuine encouragement in complex challenges in this journey even with an encouraging word. It would not have been possible to complete this project without your sacrifices and support.

EXECUTIVE SUMMARY

R.E. Weerakoon Constructions started its operations in 1998 as a construction company. It has only involved building construction for the past eight years, identifying it as its core competence graded with C2 level. REWC is undertaking new building projects, extension building projects, additions, modifications, and renovation projects under government, semi-government, and private sectors as clients. Currently, 57% of the revenue from construction activities of REWC is from renovation projects. Renovation projects have a much shorter project life cycle, yet one of the key problems faced by REWC is having a high CCC.

Conducting SWOT revealed the company's current position and performance gap and explained the high CCC in renovation projects compared with the industry average. Three associated problems identified were idle inventory, overdue receivables, and early payment for payables. Root cause analysis was performed with the qualitative inputs gained from interviews. Seven root causes identified were no inventory planning system, no inventory monitoring mechanism, no receivable policy, lack of training, inadequate project controls, improper supplier evaluation, and absence of accounting software.

Literature was reviewed using several ranked journal articles, conference proceedings, and book sections under cash conversion cycle, inventory management, receivables management, and payables management. Thereafter study framework was developed. Solutions were identified from past studies conducted on similar-natured problems in different industries. In addition, techniques to be used in the Project, such as process flowcharts, ABC analysis, training need analysis, policy documents, work breakdown structure, and KPIs, were also discussed in the literature.

Project objectives were set based on the project components, and the current situation of each component was discussed. Solutions were developed for each project component. Developing a new inventory management process, conducting a workshop to introduce ABC analysis, introducing daily project report format, introducing a store policy, conducting training for quantity surveyors, developing a receivable policy, designing a quality plan, preparing WBS, introducing improved supplier evaluation process, developing well-formulated supplier

evaluation criteria sheet, prepare set of supplier relationship management guidelines and introducing QuickBooks software to manage payables were the solutions under three project components.

Further required resources for each proposed solution and relevant cost estimates were presented for developed solutions. Cost savings were identified to derive the total tangible benefits for one year. After that, estimated costs incurred for each solution under project components were used to calculate the project's benefit-cost ratio. The benefit-cost ratio is estimated at 12.1:1, which indicates the Project's viability. Next, project outputs and outcomes were illustrated against project objectives in percentages and numbers. Additionally, qualitative outputs that can be itemized were presented, linking to the solutions.

Finally, a connection between the literature review and the conclusion of the Project report was made by discussing the identified components of the Project. The components were evaluated concerning the status of the literature reviewed in Chapter 3. Further, the author's comprehensive understanding of the concepts and their application in the project context was demonstrated while utilising theoretical references and practical examples. Following the discussion of the components, recommendations were provided focusing on short-term and medium-term.

Chapter 1

INTRODUCTION

1.1 Background

Growth of the construction industry was steady in all regions around the world in 2021 despite the Covid 19 pandemic's effects and the change in global investment priorities. While most nations regarded construction as necessary, it decreased from 6.1% to 3.6% in 2022. The construction sector is resilient, but to continue growing in the future, it must solve the problems created by supply shortages and price hikes and take a global shift towards sustainable infrastructure. In 2021, the construction market was estimated to be worth US\$ 7.28 trillion, and it is anticipated to be worth US\$ 14.41 trillion before 2030 (Parada & Rutgers, 2021).

The annual report of Sri Lanka (2021) stated that the construction industry decreased from 6.2% to 6.1% in 2021. Construction companies had to deal with a lack of labour and building materials. Interest rates on all services, including bonds and guarantees, overdrafts, term loans, and leasing offered to businesses, will significantly rise due to the Central Bank's recently announced most significant policy rate increase. However, the moratorium on loans granted by the government and central bank to construction companies due to the COVID-19 pandemic and the recession was a great support in difficult times (Razeek, 2022).

The SMEs sector in Sri Lanka accounts for 90% of businesses, employs over 45% of the labour force, primarily in the construction and industry sectors, and generates over 52% of the country's GDP (Advocata, 2019). R.E. Weerakoon Constructions (referred to as REWC hereafter) is an SME providing job opportunities to many full-time and part-time employees. Although SMEs play a significant role in the economy, they most frequently experience a slow death due to macroeconomic factors. REWC is one such SME that has experienced a consistent fall in performance in the past few years due to continuous challenges.

1.2 Project

This project intends to examine the high Cash Conversion Cycle (referred to as CCC hereafter) in renovation projects at REWC and provide solutions to the identified project components to improve the current situation by reducing the CCC. Currently, renovation projects at REWC have a CCC of 140 days, indicating a higher number of days than the industry average. The project intends to reduce the CCC from 140 days to 87 days which will give tangible and intangible benefits to REWC. Past studies have revealed that high CCC brings unfavourable financial impact in terms of liquidity and profitability of companies in numerous ways. Therefore, this project will conduct a comprehensive analysis of the problem directly.

In depth review of the literature will provide an understanding of similar situations faced by companies and proposed managerial solutions for the problem based on past studies conducted by researchers. Root causes identified through root cause analysis will provide solutions to project components. Resource allocation and benefit-cost analysis will be carried out to identify the project's viability. Finally, short-term, and medium-term recommendations will aid in effectively achieving the presented solutions.

1.3 Significance of the Project

This project pursues to analyse high CCC in renovation projects at REWC. The author will conduct an extensive study to propose solutions for identified root causes utilizing data inputs. This study illustrates the performance gap of high CCC against company benchmarks and industry standards. Having a high CCC results in cash tied up unnecessarily. When cash is tied up in idle inventory, overdue receivables and early payment to payables result in the company facing a cash shortage to manage its day-to-day operations. Payables are trade creditors, utility payments, salary payments, loan instalments and lease.

Some penalties come together when the payments are not settled on time. Further, this tarnishes the company's reputation, increasing labour turnover, causing disruptions in day-to-day work, unavailability of inventory, delays in project completion, and finding new suppliers due to tarnished relationships with existing suppliers. Therefore, REWC has taken the help of external funding. These external funding are bank overdraft facilities, bank loans, and loans from non-

financial institutes. However, external funding comes with a higher interest, which is a finance cost in the income statement and has reduced the profitability of REWC (Annexure I).

1.4 Methodology

This project will use a problem-solving management consultancy report approach utilizing qualitative and quantitative research methods. Problem within the REWC is first identified, and it is then it will be illustrated with company data. Financial reports, invoices, and reports from construction sites will be used to present the project's data. SWOT analysis will assist in analysing the organisation. After that, a root cause analysis will be conducted with the support of the inputs gained from interviews and brainstorming sessions conducted with REWC staff. Thereafter root causes will be identified up to the minimum fourth level to ensure the real causes of the problem.

Thereafter the literature will be comprehensively reviewed using ABDC-ranked journals based on past studies conducted with the identified root causes. This will support the development of the study framework for the project. Next, solutions for each project component will be proposed regarding new processes redesigned against the current processes. The resource allocation and cost estimates concerning identified project components will be presented to achieve project outputs and outcomes. Subsequently, a benefit-cost ratio will be calculated to identify the project's viability, followed by recommendations to effectively achieve the proposed outputs and outcome.

1.5 Limitations

Conducting this management field research project has several limitations identified. Most importantly, the REWC has different types of projects, but this project will focus only on renovation projects conducted by the REWC. This step was taken to give more clarity about the problem, as reasons for high CCC in different projects can have different reasons due to the nature of the projects. Therefore, the different types of projects need to be analysed separately to recognize the real cause of the problem. Further, there are restrictions in getting the industry

benchmark as there are limited published reports issued by the construction companies which are available to the general public.

Furthermore, there are limitations in getting data related to construction sites, while some data are available only for a specific period. Therefore, the project will use only five years of data to explain the problem's persistence in the company. Data obtained through interviews can sometimes be subjective to the respondents' opinions. Some respondents will be reluctant to give inputs regarding the errors made by the operational staff. Finally, there will be time restrictions as this project needs to be concluded on 31st January 2023. However, the author is making her maximum effort to minimize limitations within the direct control boundary.

1.6 Chapter Framework

Chapter 1 illustrated the context of the project, describing the macro, industry, and organisational level scenarios. It highlighted the project's intention to reduce the high CCC in renovation projects at REWC. Thereafter it described the importance to the company in reducing unnecessarily tied up cash. Further, the methodology adopted in conducting the project was briefly explained. Anticipated constraints the author has to face when conducting the project were discussed subsequently. Finally, the chapter concluded with an outline of the respective chapter contents in conducting the Project.

Chapter 2 will introduce the chapter and then summarize the organisation. Thereafter, the organisation's internal and external environment analysis will be explained using SWOT analysis to understand the broader context adding a flavour to the problem. Next, the entry for the problem is set by explaining the main problem and other problems the company faces, followed by a numerical and graphical illustration of the performance gaps. Afterwards, root cause analysis will be conducted using the quantitative and qualitative inputs obtained. The chapter will end with a summary.

Chapter 3 will commence with the introduction and include extensive literature using the previous research conducted on similar-natured problems and the authors' critical review of previous research findings of the scholars. The literature will be reviewed, linking the flow to the main problem, project components, and root causes identified through the cause-and-effect

analysis. Thereafter a study framework will be developed using the warranted knowledge gained. Next, management techniques will be identified and described, and used in proposed solutions for the root cases. Finally, the chapter will end with a summary.

Chapter 4 will initiate with the introduction, and thereafter it will present the overall main objective for the Project and sub-objectives for the project components. These objectives will be stated together with proposed techniques linking to the project components. Solutions must be developed simultaneously with the loopholes identified in the company's current processes. Thereafter, the chapter will focus on the resource requirement to carry out the Project. Next, the chapter will brief out cost estimates and relevant references in the appendix of the proposed solutions that will be developed. Ultimately chapter will end with a summary.

Chapter 5 of the Project will begin with an introduction. Next, it will illustrate the benefit-cost analysis to recognize the net estimated tangible and intangible gains that can be realised through implementing the proposed solutions. This will indicate the Project's viability against the estimated implementation cost. Costs incurred in the Project will be supported with quotations in the annexures to justify the amounts presented. This chapter will also present quantitative and qualitative outputs and outcomes that can be achieved for each component after implementing the Project. Lastly, it will provide an outline of the chapter.

Finally, Chapter 6 will start with a brief introduction and discuss the findings conducting the Project related to each component. These agreements or disagreements of findings will be compared against the discoveries from the literature review performed in Chapter 3. Subsequently, each project component will propose relevant mid and short-term recommendations for the problem. Lastly, the chapter will summarise a conclusion to recap the overall project's line-up, intended outputs and outcomes to be achieved, and the author's views in the Project's context.

Chapter 2

ORGANISATIONAL ANALYSIS AND PROBLEM IDENTIFICATION

1.7 Introduction

The chapter intends to introduce REWC and conduct an internal and external environment analysis using SWOT concerning organisations' core competencies, adding flavour to the main problem. Next, the critical problem will be identified, supported, and illustrated with relevant data. A particular problem will be further analysed with three associated problems. Subsequently, a root cause analysis will be conducted, and root causes will be recognised based on inputs gained from primary data and interviews and brainstorms conducted. Finally, the chapter will end with details narrated.

1.8 Organisational Profile

REWC was founded by Mr. R.E. Weerakoon and started its operations in 1998 as a construction company. It is registered as a private limited company and obtained registration from Construction Industry Development Authority (CIDA). Further, it obtained ISO 9001 certification in 2009 from Sri Lanka Standard Institution (SLSI). REWC has several revenue streams besides construction activities such as hotel and restaurant, rent income from buildings, hire of construction equipment and machinery, and sale of construction material. However, more than 75% of the revenue is from construction activities. Figure 2.1 illustrates functions of REWC.

Even though there is a list of functions under REWC, it has only involved building construction for the past eight years, identifying building construction as its core competence graded with C2 level. REWC is undertaking new building projects, extension building projects, and renovation projects under government, semi-government, and private sectors as clients. The mission of REWC is to provide high customer satisfaction while satisfying employees and other stakeholders. It has provided job opportunities for over 25 employees and more than 30 labours

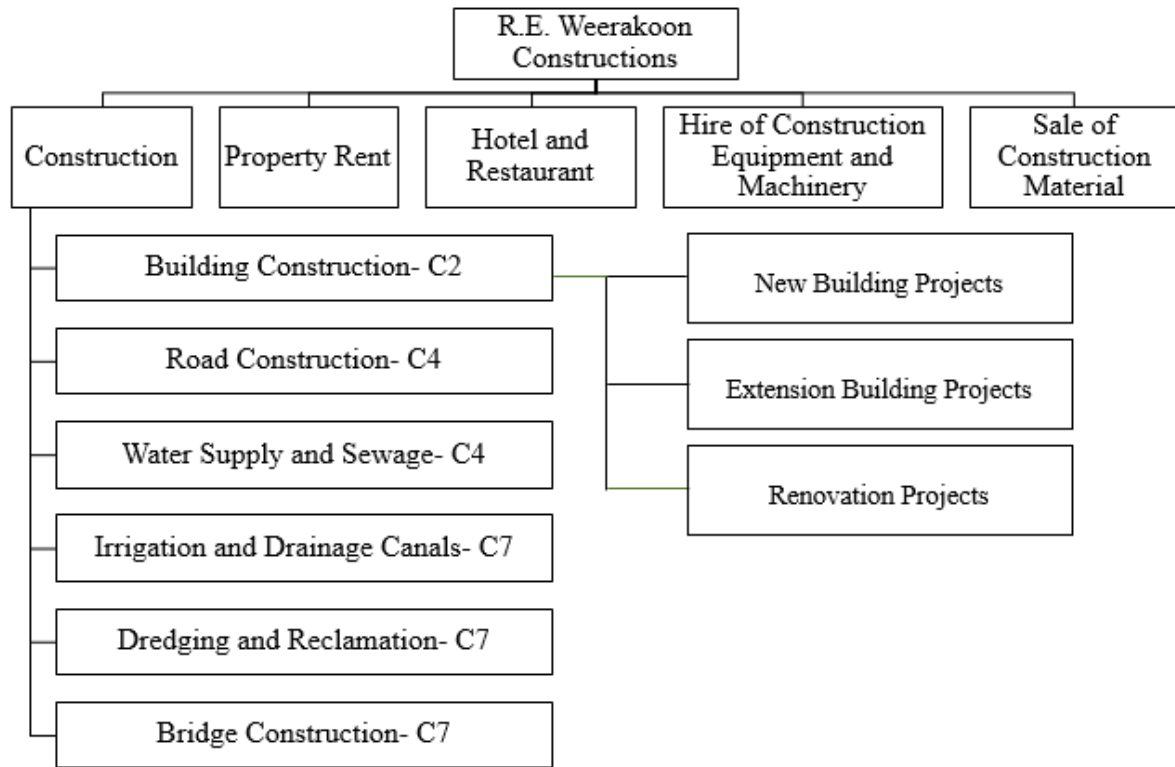


Figure 2.1: Functions of REWC

Source: Primary data.

The managing director holds the highest position, followed by the department managers. REWC has five departments: Finance, Accounts, Procurement, HR, and Administration and Operations. Each department has managers, senior executives, executives, and trainees. The operations department has project managers, site engineers, Quantity Surveyors (referred to as QS hereafter), and Technical Officers (referred to as TO hereafter) for each project. Additionally, projects with a broad scope are assigned to trainee QS and TO. Altogether these team leaders direct the construction workers and the storekeepers (refer to Annexure II for organisational structure).

1.9 Organisational Analysis

This section represents the organisational analysis to understand the impact of internal and external environments. Table 2.1 represents the SWOT analysis carried out for REWC.

Table 2.1: SWOT Analysis

Strengths	Weaknesses
i. Experience over 20 years in the construction industry, including infrastructure development projects. ii. Clients from the government, semi-government, and private sector. iii. C2 level grading by CIDA for building construction projects. iv. It has been certified to ISO 9001 since the year 2009. v. Individual capacity to undertake significant projects. vi. Diversified revenue streams from the hotel, sale of building construction materials, and rent from properties. vii. Developed long-term relationships with financial institutions.	i. Out of several functions, it has undertaken only building construction projects for the past eight years. ii. High idle inventory. iii. Increase in overdue receivables. iv. Short-dated suppliers are resulting in early payment to payables. v. Decreased profitability over the years and recorded a loss for the financial year 2021/22. vi. Increase in finance cost. vii. Highly geared company due to external funding.
Opportunities	Threats
i. A significant amount of contribution from the construction industry to Sri Lankan GDP. ii. It can be diversified into construction-related other businesses, such as selling construction materials. iii. Government providing moratoriums in challenging times.	i. Delays in client payments lead to cash flow problems. ii. The rising cost of labour. iii. The cost of raw materials increases due to import restrictions and the rupee devaluation. iv. Changes are happening to government policies within a short

iv. Opportunity to recruit foreign employees.	period.
v. Government's effort to get support from the international.	v. Increase in competition from foreign construction firms and local major players like Access engineering.
	vi. Migration of skilled professionals into other countries.
	vii. Power cuts increase the idle labour and machinery time resulting in project delays.

Source: Author.

1.9.1 Strengths

REWC was established in 1998 and has experience building construction and other functions over 20 years in government, semi-government, and private sector projects. It is graded as a C2-level construction company by CIDA and can undertake large-scale projects. Further, it has obtained ISO 9001 certification since the year 2009. REWCs' revenue is not limited to construction activities as well as it sells construction materials and hires construction equipment and machinery to use its assets to the maximum. It has been a customer of several banks and other financial institutions, building long-term relationships.

1.9.2 Weaknesses

REWC has undertaken different functions such as road construction, water supply, and sewage. However, it has not taken on projects other than building construction for the past eight years. REWC has a lot of idle inventory, increasing overdue receivables, and suppliers with short payment dates. As a result, tied-up cash unnecessarily has pressured REWC to seek external funding sources to fulfil the working capital requirement. REWC is highly geared due to the external funds. This has led to the company recording a loss in renovation projects in the financial year 2021/22.

1.9.3 Opportunities

One of the main contributors to the GDP in Sri Lanka is the construction industry, which provides job opportunities in the market. As construction is linked with many other businesses, it can quickly diversify into other businesses. Providing a moratorium for contractors as support during a time when the industry was affected by COVID-19 was an encouragement to continue the business. Costs of construction activities are hiking, lowering profitability. It is a relief to use cheap foreign labour in construction-related activities in such a situation. Further government is mid-way to a discussion with International Monetary Fund to get support in building back the economy.

1.9.4 Threats

As the government is the client for most large-scale new building and extension projects, there are delays in getting payments for the projects completed. Further, the cost of labour and the price of the material is rising because of the scarcity of raw materials in the market. Additionally, government changes its policies in a short period resulting in uncertainties. At the same time, the company has high competition from foreign construction companies and major local players in the market with advanced technology and long-term relationships with suppliers. Besides that, the skilled workforce migrates to foreign countries, creating a colossal brain drain in Sri Lanka.

1.9.5 Conclusions from the SWOT Analysis

There are highlighting areas from the SWOT analysis to pay attention to in the internal and external environment. With experience in the construction industry, with various certifications and clients, REWC can undertake projects of different natures. Further diversifying into different customer segments can reduce the industry's current risk. However, despite several strengths and opportunities, REWC faces cash flow problems because it has a high CCC. REWC needs to be efficient in its operations to overcome cash being unnecessarily tied up in inventories and receivables, creating shortages in cash to settle the payables.

1.10Key Problem(s)

REWC is undertaking different projects, such as construction, extension buildings, and renovation projects. Out of these three, renovation projects have a much lesser project life cycle. Yet, one of the critical problems faced by REWC is having a high CCC. Currently, renovation projects at REWC have a CCC of 140 days, indicating that it has 140 days short with cash. This is 38% higher than the industry average. REWC also has other problems, such as high idle inventory, increasing overdue receivables, and early payment to suppliers resulting in cash flow problems as cash is tied up unnecessarily in idle inventory and uncollected receivables.

Also, this creates problems in finding other working capital requirements such as paying salaries, paying utility bills, and continuing operations. As a short-term solution, REWC uses a bank overdraft facility with one of the highest interest rates in lending. REWC tends to go for bank loans when a bank overdraft facility cannot fulfil the requirement but also to the extent of selling its non-current assets. When the payment is delayed to suppliers, the relationship between the REWC and suppliers tends to be tarnished, bringing a bad reputation. Suppliers can withhold sending raw materials, which delays the construction activities.

1.11Description of the Problem(s)

The most significant problem within REWC is having a high CCC. This indicates the period a company is short with cash from when inventories are purchased to when payables are settled. Contribution to the significant part of the revenue of REWC consists of construction activities. The contribution of revenue from construction activities for the past five years is illustrated in Figure 2.2 below.

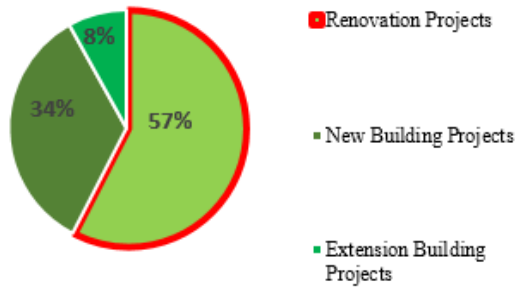


Figure 2.2: Location of the Problem

Source: Company data.

The highest contribution to revenue from construction activities is from renovation projects (57%) and then new building projects (34%) and extension projects (8%), respectively. Therefore, the Author will focus only on CCC of renovation projects.

High CCC contributes directly to the profitability of the company as well as directed to other problems. CCC of Access engineering is considered the benchmark for the current performance of REWC. There can be differences in experience, government support, skilled labour, and new technology, but external factors such as the Easter attack, Covid 19 pandemic, and economic problems in the country directly hit down the actual performance of the construction industry as a whole. Therefore, taking CCC of Access Engineering as the benchmark can be considered a realistic parameter. Table 2.2 and Figure 2.3 illustrate the CCC of REWC in renovation projects.

Table 2.2: Cash Conversion Cycle in Renovation Projects

Year	2017/18	2018/19	2019/20	2020/21	2021/22
REWC	115	127	150	160	147
Company Average	140	140	140	140	140
Industry Average	87	87	87	87	87
Variance in Days	28	40	63	73	60
Variance in %	24%	31%	42%	46%	41%

Source: Company data; Access Engineering PLC (2022, 2021, 2020, 2019, 2018).

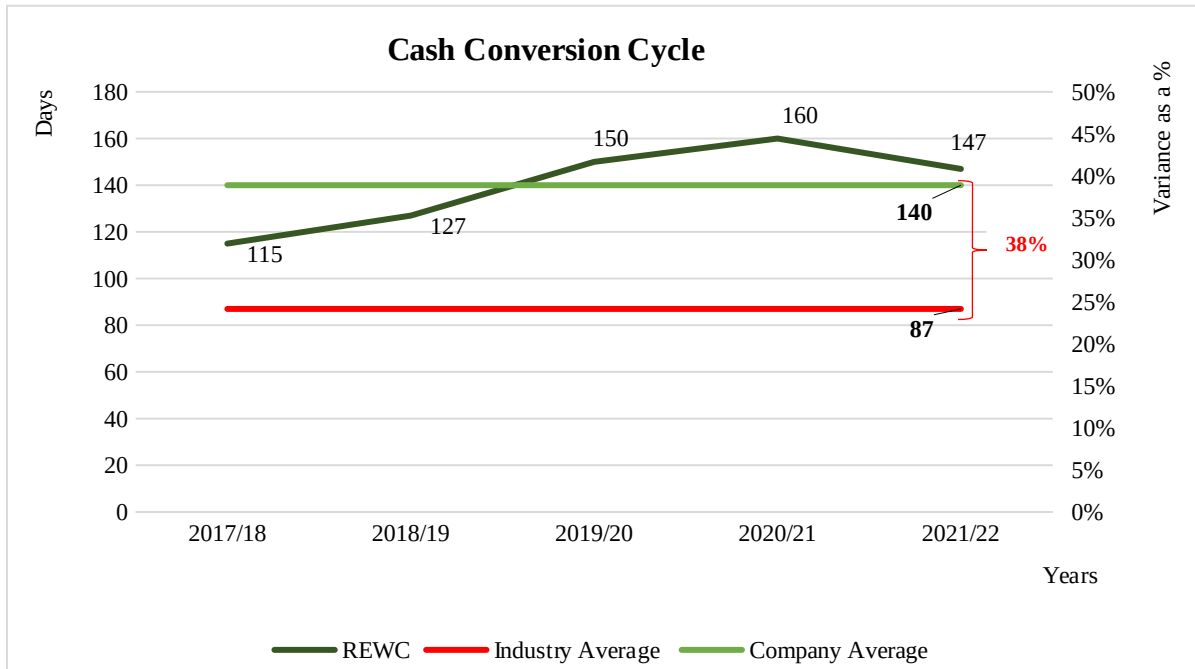


Figure 2.3: Cash Conversion Cycle of Renovation Projects

Source: Company data; Access Engineering PLC (2022, 2021, 2020, 2019, 2018).

There is a high deviation when the company average for CCC is compared against the industry average (which is the average of Access Engineering considered here) for the past five years. The company average for renovation projects shows a CCC of 140 days (Annexure III), but the industry average is only 87 days. Out of five years, the lowest CCC days is 115 days, and the highest is calculated as 160 days. All five years show a deviation higher from the industry average. The industry average for CCC is 54%, but the company average is 88%. This is a gap of 38%. As the problem of high CCC has been persistent throughout the past five years, the following problems were identified associated with the critical problem.

1.11.1 Idle Inventory

The first associated problem identified with high CCC is idle inventory. According to the site engineer, “there is a lot of remaining inventory in the construction site stores at the end of the period. Inventory sits idle for longer periods”. The total closing stock is not considered an idle inventory in this situation. Generally, REWC is targeting to submit an invoice once in 30 days.

That company benchmark for the Inventory Conversion Period (referred as ICP hereafter) is 30 days. Similarly, inventory should be converted into a sellable condition in 30 days cycle. Therefore, inventory that remains in the warehouse purchased within the last 30 days is deducted from the closing inventory to arrive at the idle inventory figure. Idle inventory in renovation projects is illustrated in Table 2.3 and Figure 2.4 below.

Table 2.3: Idle Inventory in Renovation Projects

Year	2017/18	2018/19	2019/20	2020/21	2021/22
Closing inventory	1,551,783	1,802,373	2,096,139	2,015,937	2,378,588
Remaining inventory purchased within the last 30 days	986,248	897,709	883,452	689,403	998,510
Idle inventory	565,535	904,664	1,212,687	1,326,534	1,380,078
Idle inventory as a % of closing inventory	36%	50%	58%	66%	58%
Company average	54%	54%	54%	54%	54%
Company benchmark	5%	5%	5%	5%	5%
Variance	86%	90%	91%	92%	91%

Source: Company data.

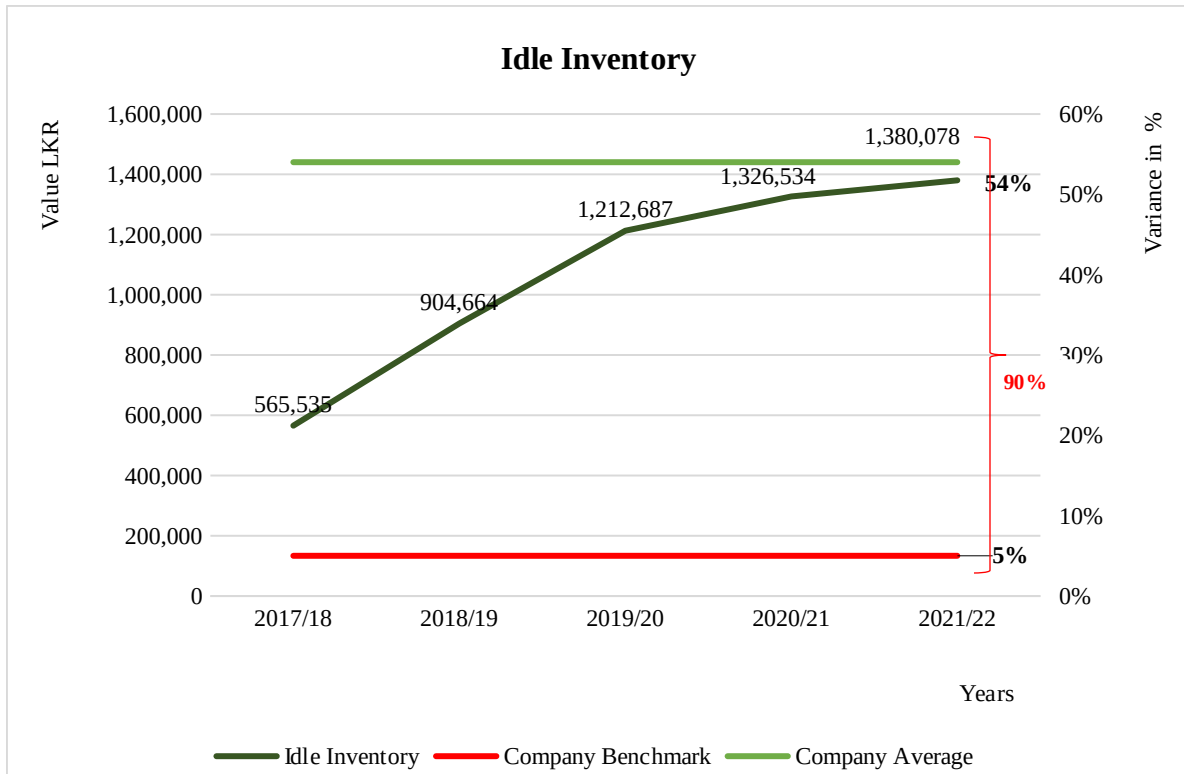


Figure 2.4: Idle Inventory in Renovation Projects

Source: Company data.

In this scenario, the company benchmark is used rather than the industry average as internal factors affect the inventory conversion period and idle inventory, which are different from company to company, even within the same industry. For example, inventory usage for renovation projects differs from building construction projects. REWC average shows 54% in the past five years, which is at the lowest of 36% in the financial year 2017/18 and highest of 66% in the financial year 2020/21. As per the management, the company benchmark is set at 5%. This is a gap of 90% from the company benchmark

1.11.2 Overdue Receivables

The second associated problem identified with high CCC is overdue receivables. As per the finance manager, “clients are taking longer than the provided credit period. Because of this, we currently have overdue receivables older than six months.” Working capital problems arise when the receivables are not collected within the credit period”. In this situation, the total

closing receivables are not considered overdue. According to the ICTAD publication, payment by the client should be paid within 35 days from the date invoice is submitted. Therefore, the company benchmark for the Receivables Collection Period (referred as RCP hereafter) is 35 days. Accordingly, receivables within the credit period were deducted from the closing receivables to arrive at the overdue receivables figure. Overdue receivables in renovation projects is illustrated in Table 2.4 and Figure 2.5 below.

Table 2.4: Overdue Receivables in Renovation Projects

Year	2017/18	2018/19	2019/20	2020/21	2021/22
Closing receivables	11,342,281	12,025,748	12,904,254	13,036,408	13,168,239
Receivables within the credit period	7,939,597	7,936,994	7,871,595	7,039,660	7,505,896
Overdue receivables	3,402,684	4,088,754	5,032,659	5,996,748	5,662,343
Overdue receivables as a % of closing receivables	30%	34%	39%	46%	43%
Company average	38%	38%	38%	38%	38%
Company benchmark	20%	20%	20%	20%	20%
Variance	33%	41%	49%	57%	53%

Source: Company data.

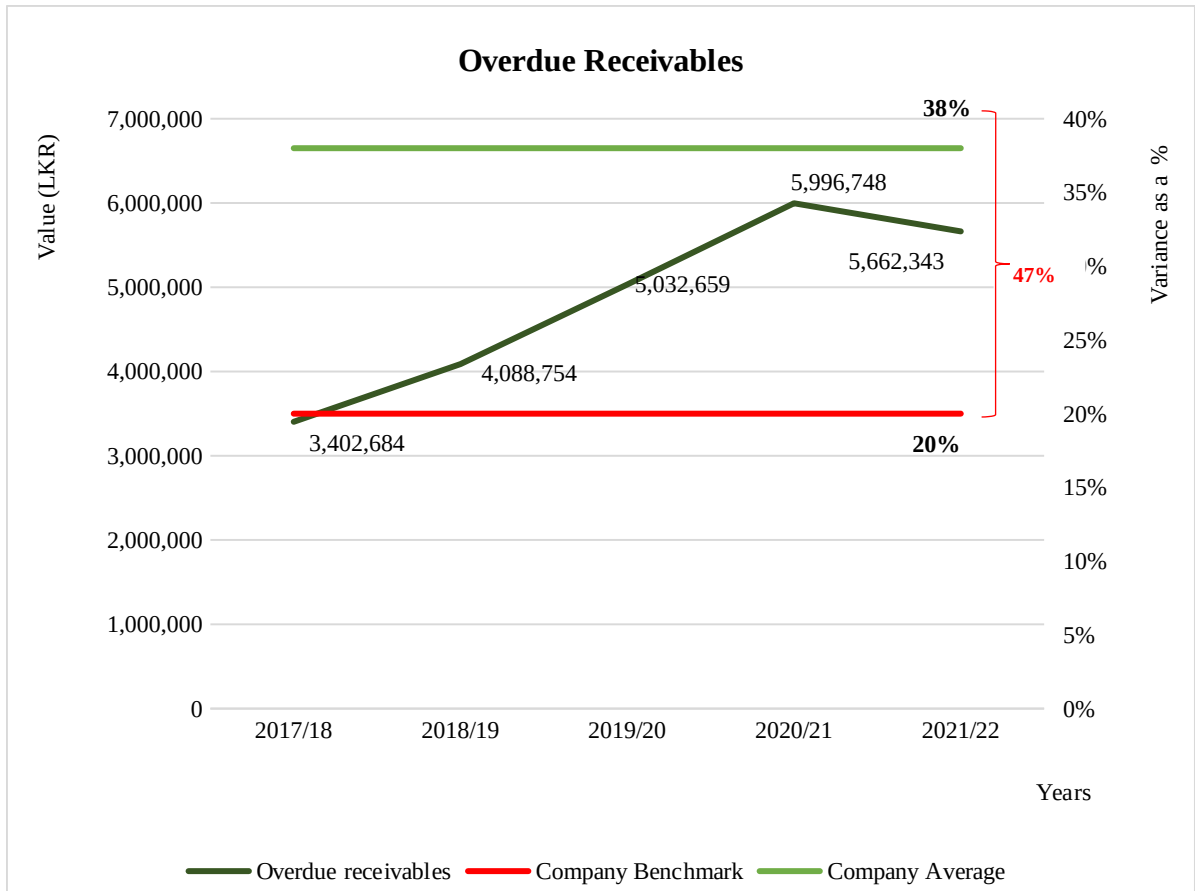


Figure 2.5: Overdue Receivables in Renovation Projects

Source: Company data.

In this scenario, the specified receivable conversion period provided by ICTAD is used as the company benchmark. REWC's average for receivables shows as 38% in the past five years, and the company benchmark is set at 20% as late payment can be charged with interest claims for the number of days the payment got delayed. All five years show a deviation higher than 20% showing the lowest overdue receivable percentage of 30% in the financial year 2017/18 and the highest of 46% in the financial year 2020/21. This is a gap of 47% from the company benchmark. Overdue receivables have been increasing compared to the increase in receivables within the period.

1.11.3 Early Payment to Suppliers

The third associated problem identified with high CCC is early payment to suppliers. According to the procurement executive, “payables mostly consist of suppliers of the inventory to the construction sites, and most of these suppliers provide short credit period.” This increases the CCC, and the company will not have enough time till receivable pays cash. According to the list of suppliers of REWC, the average days to settle the dues is 45 days. Therefore, the company Payable Outstanding Period (referred as POP hereafter) is considered as 45 days here. Accordingly, payables giving more than 45 days credit period were deducted from the closing payables to arrive at the early payment to suppliers figure. Early payment to suppliers in renovation projects is illustrated in Table 2.5 and Figure 2.6 below.

Table 2.5: Early Payment to Suppliers in Renovation Projects

Year	2017/18	2018/19	2019/20	2020/21	2021/22
Payments to payables	3,091,606	4,614,26 4	4,070,922	3,861,654	5,928,223
Payables dated more than 45 days	1,916,796	2,999,27 2	2,401,844	2,085,293	3,379,087
Early payment to suppliers	1,174,810	1,614,99 2	1,669,078	1,776,361	2,549,136
Early payment to suppliers as a % of closing payables	38%	35%	41%	46%	43%
Company average	41%	41%	41%	41%	41%
Company benchmark	10%	10%	10%	10%	10%
Variance	74%	71%	76%	78%	77%

Source: Company data.

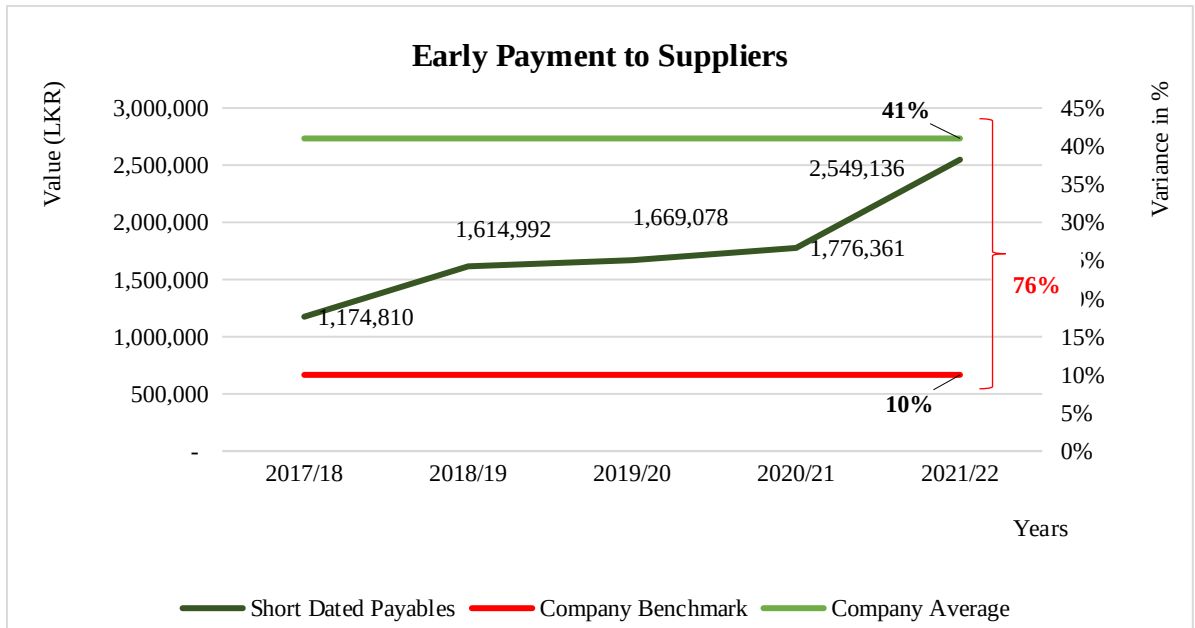


Figure 2.6: Early Payment to Suppliers in Renovation Projects

Source: Company data.

In this scenario, the specified payables settlement period is considered as 45 days, which was found by taking the average credit period provided by the current list of suppliers of REWC. REWC's average for early payments to suppliers shows 41% in the past five years, and the company benchmark is set at the lowest of 10% for the purpose of obtaining discounts. All five years show a deviation higher than 10%; the lowest percentage is marked at 35% in the financial year 2018/19, and the highest is 46% in the financial year 2020/21. This is a gap of 76% from the company benchmark. Early payment to suppliers has been increasing compared to the increase in payables.

1.12 Analysis of the Key Problem

To delve deeper into the issue, the REWC team was consulted to determine the potential factors contributing to high CCC in renovation projects at REWC. A root cause analysis diagram was created using the information gathered from the interviews (Annexure IV) is illustrated in Figure 2.7 below.

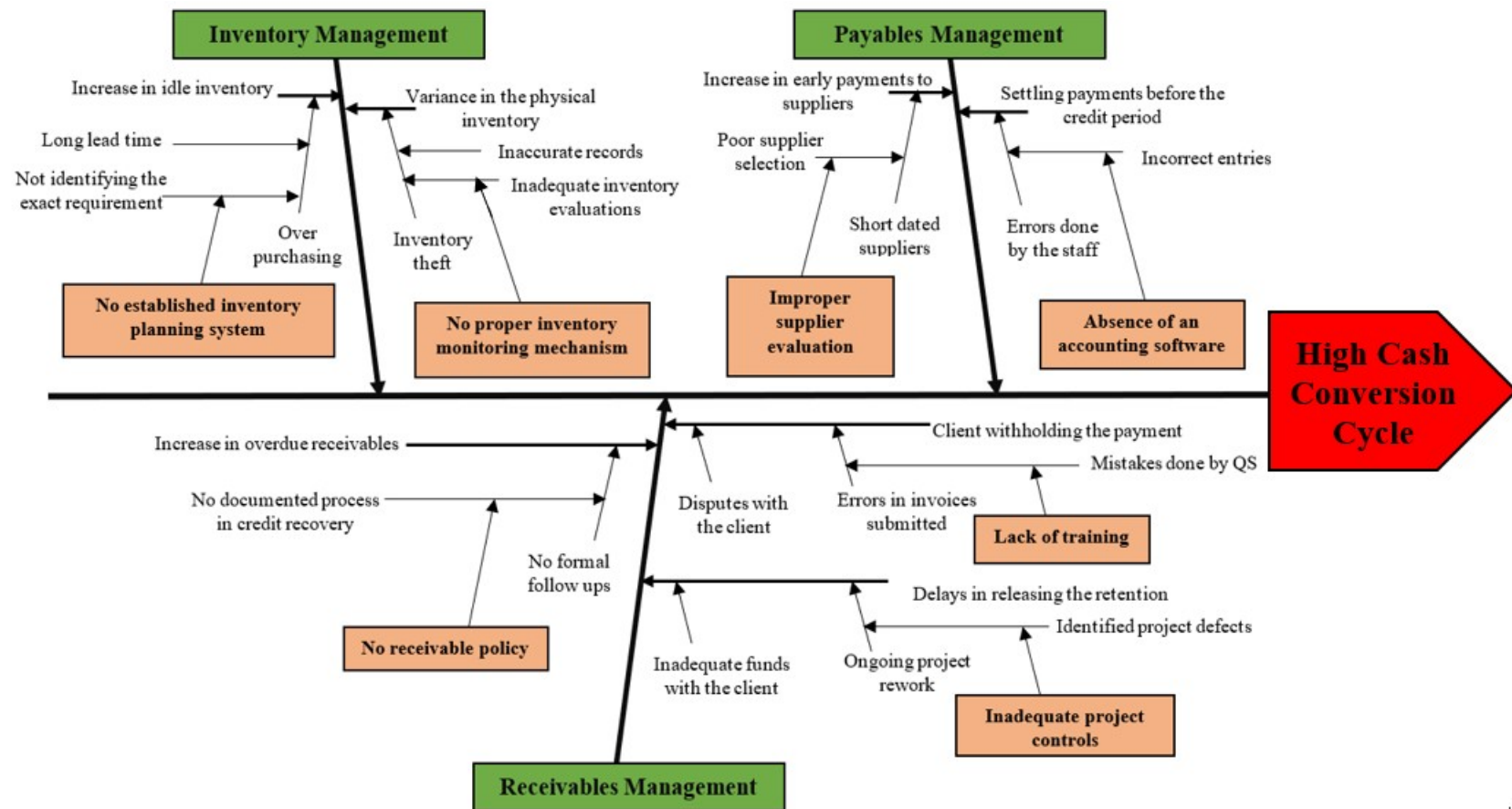


Figure 2.7 Root Cause Analysis

Source: Author.

1.12.1 Inventory Management

One of the associated problems identified was idle inventory in the warehouse. A considerable amount of the closing stock is older than 30 days, and the reason for idle inventory is over-purchasing. Inventories are over-purchased due to long lead times as well as when the required demand is not identified. Therefore, the root cause for this does not have proper inventory planning. Another issue identified with inventory is having a variance of the inventory figures in the records versus the physical inventory. This is due to record inaccuracy or when inventory is missing. These inventories are missing due to theft as no adequate stock evaluations are conducted because there is no proper inventory monitoring mechanism to monitor inventory movement.

1.12.2 Receivable Management

Another associated problem identified was overdue receivables. Overdue receivables are increasing as no formal follow-ups are happening. There is no available guidance provided with documented credit recovery plan and the root cause appear to be not having a receivable policy. In contrast, there are situations when the client withholds the payment as when there are disputes with the clients or when there are errors in the invoices summited. This is because of QS's mistakes, which are happening because of a lack of training. In addition, some clients are not paying the retention when the retention period is over. This is mainly because there is of inadequate funds with the client or due to ongoing rework because of identified defects in the completed projects. The root cause identified for this is inadequate project controls.

1.12.3 Payables Management

The final associated problem identified was an early payment to suppliers. The increase in early payments to suppliers is because of the short credit provided by suppliers. This is because REWC selects suppliers based on narrow criteria, and the root cause can be considered as poor supplier evaluation. Whereas there are situations when the company is settling the payments to suppliers even before the credit period as there are errors in entries made by the staff. This

happens because of manual methods in making the entries; the root cause is the absence of accounting software.

1.13 Summary

This chapter started with an introduction and next covered the company profile. After that, a SWOT analysis was performed to discover the company's position internally and externally. The facts revealed by SWOT were analysed by setting the background and adding flavour to the critical business issue. The selected vital problem of the company was explained with a performance gap and three associated problems using a quantitative aspect. Thereafter root cause analysis was conducted to identify the contributing root causes for the critical problem with the qualitative inputs gained from interviews conducted with key personnel in different divisions.

Chapter 3

LITERATURE REVIEW

1.14 Introduction

A systematic method for discovering, assessing, and synthesizing the body of knowledge already known is a literature review. The chapter which precedes is a review of literature conducted on the main problem and the sub problems identified. The literature review will be conducted utilising ranked academic publications focusing on the CCC, inventory management, receivables management and payables management in the different industries (Annexure V). Aim of the literature review is to assist in developing the study framework while adhering to an applied research methodology. A discussion of the techniques used in this Project will conclude the chapter.

1.15 Theoretical Background to the Problem

1.15.1 Cash Conversion Cycle

Working capital management (WCM) is a vital aspect of business financial management, as the way in which a company manages its working capital can significantly affect its profitability. This is supported by various studies in the academic literature, such as Deloof (2003), Deloof and Jeger (1996), Emery (1984), and Schwartz (1974). The goal of WCM is to optimize the balance of working capital to maximize the value of the firm. This is achieved by implementing efficient management practices, such as expediting collections and delaying payments, which are based on traditional cash conversion cycle theories.

CCC is a commonly used technique for measuring the effectiveness of WCM within a company. As stated by Gentry et al. (1990), CCC is defined as the number of days it takes for a company to convert its investments in working capital into cash through sale of its products. This includes the time it takes for a firm to purchase raw material and convert them in to finished goods, as well as the time it takes to collect payments from customers. The length of

the CCC is influenced by the ICP, RCP and DPO. The CCC can be calculated by adding the ICP, RCP and subtracting the DPO.

Investments in receivables and inventory are commonly used to represent the operational cycle of an organisation. While it has certain limitations, the inclusion of the inventory and receivable in the operating cycle can provide a more accurate assessment of a firm's liquidity position (Richards & Laughlin, 1980). However, the concept of operating cycle alone is not sufficient as a measure of a company's cash flow as it does not consider the temporal dimension of existing obligations. Specifically, it does not consider the time pattern requirement of cash outflow imposed by a company's current liabilities.

It is necessary to consider additional indicators of cash flow related to current obligations, to conduct a more comprehensive analysis of liquidity. Gill (1968) suggests that an increasing POP indicates longer period of working capital financing supplied by trade creditors. Similarly, higher ICP and RCP reflect a huge current asset investment that must be financed over a stretched operating cycle period. Longer payment terms can alleviate potential liquidity management issues, but a longer operating cycle brought on by a high ICP and RCP can exacerbate these issues.

CCC improves the analysis of liquidity by taking POP to consideration which is the time between purchasing goods on credit and paying for them in cash. According to Talonpoika et al. (2014), POP is an important factor in determining the CCC. When inventory is purchased on credit, both the inventory and accounts payable balances increase. As inventory is sold, the inventory account is depleted, but the accounts payable balance remains unchanged until receiving the cash payment. Thus, the CCC is the exact period between cash inflows and withdrawals.

CCC is a metric that indicates the number of days that a business must advance funds while awaiting payments from customers, it can be zero, negative or positive. High ICP and an RCP will increase the CCC, while a low CCC reveals the time a company waits to pay suppliers after receiving revenue in cash. A negative CCC typically indicates well-managed inventory and receivable collection times. As stated by Moss and Stine (1993), a CCC of zero represents a perfect balance between the RCP, ICP and POP.

The duration of CCC for a specific company is influenced by a variety of factors. These include the size of the company, which may affect its capacity to take on debt, the company's capacity to generate internal resources, capital expenditure incurred, growth opportunities in the market, leverage of the company, the age of the company in the specific industry and economic conditions of the country in which the company operates and industry controls. These factors have been studied by various researchers such as Omopariola et al. (2020), Garanina and Olga (2015), Ruyken et al. (2011), Niskanen and Niskanen (2006), and Fazzari and Petersen (1993).

The relationship between an organisation's ability to generate profits using its static resources and its operational effectiveness is of great concern to firms. According to Langford et al. (1993), profitability serves as a benchmark to evaluate the effectiveness of a company's operations, as it illustrates the relationship between earnings and capital. A company is considered successful if its profitability exceeds the Weighted Average Cost of Capital (WACC). Additionally, profitability is directly proportional to efficiency, and vice versa. It also reflects the combined effect of asset, liquidity management, and debt management on the functional outcomes and public acceptability of the company's products or services.

Companies should aim to decrease the number of days in their CCC to enhance profitability. As demonstrated by research conducted by Chang (2018) found a correlation between a shorter CCC and high profitability. This suggests that companies should aim to reduce the number of days in their CCC to improve business performance. Additionally, there is a positive relationship between CCC with ICP and RCP, and a negative relationship with the POP. However, this inverse relationship between accounts payable and profitability is consistent with the idea that less successful businesses tend to delay payments to creditors for longer periods.

In contrast to the previously mentioned studies, some research has found a strong correlation between CCC and profitability. Charitou et al. (2010) found that CCC had a significant impact on returns on investment for all Indonesian companies from 1998 to 2010, and that a high CCC was positively correlated with increased profits. This finding is supported by the study conducted by Gill et al. (2010) also revealed a substantial positive relationship between CCC

and business profitability, suggesting that companies with high profitability tend to pay less attention to effective working capital management.

Similarly, various studies support longer CCC can provide several benefits for a business. Abuzayed (2012) posits that a longer CCC can increase profitability by allowing for the maintenance of larger inventory levels, ensuring continuity of operations during periods of scarcity. Additionally, extending trade credit to customers can increase revenue and ultimately improve profits. Furthermore, shortening the payables payment term can strengthen relationships with suppliers and potentially lead to price reductions for early settlement. However, it is important to note that Ehrhardt and Brigham (2005) suggest that extended CCCs can also be a contributing factor in the bankruptcy of a firm.

The structure of an organisation's supply chain plays a significant role in determining its optimal CCC. The calculation of CCC involves balancing a company's desire to increase profitability and liquidity (Ebben & Johnson, 2011). If the ICP and RCP are too short, the company may miss out on revenue due to stock shortages, or customers may be unable to pay. Conversely, if the POP is extended excessively, the company may lose the opportunity for early payment discounts. Therefore, it is important to carefully consider these factors when determining the appropriate CCC, particularly in industries where the project life cycle is relatively short, such as renovation projects.

Determining the optimal level of CCC for an organisation can be a challenging task. To find the best CCC level, organisations must first strive for transparency and benchmark analysis. Secondly, it is essential to consider the perspectives of all stakeholders in the supply chain, as some organisations may minimize their CCCs by exerting pressure on suppliers. Thirdly, it is important to consider the leverage aspects throughout the entire value chain. This may include reducing ICP, RCP and increasing POP, as a means of improving CCC and moving towards the range of strategic CCC (Olson & Xie, 2009).

1.15.2 Inventory Management

Inventory is a key component in the calculation of cash CCC, which measures the number of days it takes for a company to convert its total inventory of raw materials, work-in-progress,

and finished goods into revenue from finished goods. According to Malik and Anser (2013), a shorter ICP can be achieved using inventory management techniques that require larger inventory levels per dollar of anticipated revenue. Similarly, Sadiq et al. (2018) states high inventory turnover is associated with increased profitability. Additionally, a higher current ratio, which is an indicator of solvency, may also be associated with decisions resulting in longer or shorter liquid holding periods if the business is unable to alter the payment patterns established with creditors.

Finding the appropriate inventory level is crucial as it can impact a company's performance by increasing both opportunity and storage costs. Song and Houtum (2019), conducted a study examining the relationship between a company's inventory management practices and expected financial returns. They emphasized the importance of monitoring and regulating inventory levels for efficient business operations. Their study also found that investing more in inventories reduces a firm's working capital and affects its liquidity, resulting in a decrease in profit for the firm. Furthermore, construction materials make up the majority of inventory, and are the single largest cost factor in every construction project.

Proper inventory management is critical for any business to effectively control and manage its operations. Highlighting the significant consequences of improper handling of these resources, the cost of these inventory range between 50% to 70% of the total cost of construction projects. Excess inventory can lead to financial constraints, rising land costs, deterioration of items in the warehouse, obsolescence, and theft. Businesses maintain these inventories for various reasons, such as a hedge against shortages, anticipated supplier issues or potential unit price increases. However, a lack of materials can cause disruptions in the construction schedule, supply interruptions, and increased cost overruns (Ballou, 2000).

1.15.2.1 Inventory Planning System. Inventory management encompasses both activities scheduling and inventory planning. The primary objectives of inventory planning are to maintain the minimum level of inventory required, plan operational processes, delivery schedules, and purchasing processes, and ensure the availability of supplies, components, and products for planned production and customer deliveries (Sana, 2010). This approach is used in the construction industry to manage the construction delivery processes to ensure that the client receives a facility that meets their expectations and functions as intended; it also includes all strategies, tactics, and methods for achieving this goal.

The success of a business can be influenced by a variety of elements. Some of them are sales, marketing strategies, efficient management of human resources, minimal production expenses, and effective inventory management. According to a study conducted by Mohamad et al. (2016), inadequate inventory planning can be detrimental to a company's viability as it leads to the accumulation of excessive stock, incurs additional costs, and increases the likelihood of damage. The expenses associated with maintaining excess inventory are significant as it results in the unproductive use of warehouse space and failure to generate revenue.

It is crucial for a company to carefully consider the appropriate amount of stock to keep on hand when managing inventory. According to Schultz and Juran (1999), the two main considerations in this decision are ensuring a sufficient safety stock to avoid stockouts and minimizing procurement costs. The level of safety stock required will vary depending on the level of risk the company is willing to accept. More safety will necessitate maintaining a larger inventory, while a lower degree of safety may allow for a smaller inventory. However, as the safety plan expands, it may require more inventory to be maintained.

A business may incur costs associated with idle inventory when the appropriate inventory level is not maintained. These costs depend on the quantity of inventory and how long it remains unsold. According to Abidali and Harris (2006), idle inventory refers to slow-moving materials that hold no value and take up space, thus resulting in an increase in carrying costs as inventory levels rise over time. The carrying cost, also known as the holding cost, includes various expenses such as the capital cost of inventory, storage costs, rent, utilities, security, record

keeping, obsolescence, taxes, insurance, and theft, lost customers, overtime costs, transportation costs, and opportunity costs associated with holding the inventory (Tersine & Tersine, 1990).

In industrial settings, a significant portion of inventory may often be devoted to idle stock. To improve operations and reduce idle inventory, companies must develop an effective inventory management strategy that considers various factors. Given the complexity and significance of inventory management, several challenges have been identified, such as ensuring timely delivery and maintaining the appropriate inventory levels to meet customer and end-user demands. According to Hendricks and Singhal (2009), idle inventory is a primary cause of obsolete inventory, and the level of idle inventory is an indicator of a demand-supply imbalance.

Inventory requirement must be identified to determine the necessary number of materials for operations. Both quantitative and qualitative techniques can be used to assess inventory needs (Plinere & Borisov, 2015). According to Toelle and Terisne (1989) long lead times lead to a build-up of in process inventory. Additionally, Wu et al. (2012) suggest that a higher level of delivery uncertainty will lead to a higher level of inventory impact on performance. Variable manufacturing lead times encourage firms to increase safety stocks or lengthen safety lead times can result in idle inventory.

Properly planning inventory to meet specific demand is essential in avoiding conflicting inventory management goals. Inventory management must find a balance between the costs of maintaining adequate inventory levels and the drawbacks of having excess inventory, while minimizing inventory expenses (Bruccoli et al., 2014). In the construction industry, inventory planning helps businesses determine how much to procure and when to place new orders. By planning inventory effectively, businesses can lower the costs while ensuring that there is enough inventory on hand to complete upcoming tasks.

Effective inventory management is essential for the smooth functioning of any construction company. In recent years, there has been a growing emphasis on the need to design new inventory management processes to improve operational efficiency and reduce costs. A study by Hwang, He, and Zhao (2018) proposed a new inventory management process for

construction companies that incorporates lean principles, just-in-time (JIT) delivery, and advanced technology. The proposed process focuses on reducing inventory levels, improving inventory accuracy, and optimizing inventory replenishment. Implementing an appropriate process flow is one of the key techniques in inventory planning. It is vital to evaluate and adjust the process, as current visualizations of each stage and activity's complexity are not adequate for managing organisational operations (Biazzo, 2002).

Daily project report format is a tool that is widely used in the construction industry to manage inventory. According to Wang et al. (2021), daily reports are essential for construction projects to monitor and track the status of inventory, identify issues, and ensure timely resolution of problems. These reports provide real-time data on inventory movement, which can help managers make informed decisions regarding inventory management.

Moreover, daily reports are useful in identifying discrepancies and errors in inventory records, which can lead to cost overruns and delays. By keeping track of inventory levels and usage, construction companies can minimize inventory holding costs and avoid stockouts. In addition, the use of daily reports can improve communication and coordination between different teams and stakeholders involved in inventory management. The timely sharing of information can help teams work together more efficiently and prevent delays caused by miscommunication (Yang et al., 2007).

The ABC analysis commonly used inventory management technique that categorizes items according to the proportion of revenue they generate. ABC analysis is a fundamental technique in supply chain management, which states that a small percentage of inventory items generate most of the revenue. According to Williams and Patuwo (2004), pareto analysis states that 20% of items account for 80% of sales, while ABC analysis classifies items based on their relative value, with Class A items accounting for most of the revenue, Class B items contributing the next largest portion, and Class C items contributing the least. This method is also referred to as proportional value analysis.

1.15.2.2 Inventory Monitoring Mechanism. Monitoring and measuring inventory in a warehouse are an essential aspect of inventory management. It involves tracking the movement of products and materials through the warehouse and ensuring that inventory levels are accurate and reliable. However, it is often believed that this task is the sole responsibility of a storekeeper and does not require extensive training or expertise. This can lead to inaccurate feedback data and poor inventory management if the monitoring and measurement process is not given adequate attention (Nanaware & Saharkar, 2017).

Inventory monitoring can also aid in identifying the level of accuracy between physical inventory and records. This accuracy is typically evaluated by comparing periodic physical inventory counts to a benchmark. As per a study by Raman and DeHoratius (2008), inventory variance may also be caused by missing or incorrectly counted inventory. Errors in recording replenishments, disbursements, stock levels, stock locations, part identification numbers, and product structures are frequently manifested in inventory variance (Toelle & Tersine, 1989). However, one major cause of inventory variance is employee theft, as found by Rekik et al. (2009) who discovered that over 90% of construction enterprises face some level of theft. Similarly, Smith et al. (2011), also noted that theft can occur due to improper inventory evaluations.

Improper inventory evaluations in building projects are a growing concern for inventory management. Inventory evaluation is crucial to ensure that resources are available at the required time, location, and amount. On-site inventory evaluation issues have drawn attention in the construction sector as supplies are often supplied in bulk without proper identification. It is challenging to trace the whereabouts of supplies on construction sites when needed due to inefficient or inadequate inventory management during the materials receipt process and traditional materials evaluation techniques (Kasim et al., 2012).

Monitoring inventory also involves taking physical location and time into consideration. As suggested by Courtright and Zee (2011), monitoring inventory involves more than just inspecting inventory, it also involves being aware of it to run the warehouse efficiently and provide accurate data for other company management functions such as inventory control, purchasing, processing sales orders, and billing. The better systems keep an eye on both the

process and the inventory. This means that monitoring occurs after each action has been completed, promptly identifying mistakes, and eliminating the need for routine inventory inspections.

It is important for those responsible for inventory management to be informed about proper store practices for inventory monitoring. However, it is also essential to include store policies and procedures in detail on the arrival of inventory and items to be stored according to inventory type (Rosenblatt & Roll, 2007). These store policies and procedures should cover the key steps in assuring the efficient execution of tasks, including receipt of inventory, storage, and the transfer of products out of stores. By having store regulations and procedures in place, warehouse efficiency and accurate inventory levels can be improved.

By implementing a store policy, a construction company can achieve significant benefits in inventory management. According to Rehman, Nasir, and Ahmad (2019), a well-structured store policy can lead to significant improvements in inventory management. A store policy also helps in the effective utilization of space, ensuring that inventory is stored in a manner that minimizes the risk of damage and enables easy retrieval. Additionally, a store policy can help in reducing the time spent searching for inventory, thus increasing productivity. However, it is crucial to develop a policy that is tailored to the specific requirements of the company and its projects. It should consider the company's procurement and supply chain management practices and be flexible enough to adapt to changes in the project requirements.

1.15.3 Receivables Management

Effective management of accounts receivable is crucial for a company's profitability. According to a study by Yao and Yiling (2018), reducing the duration of receiving payments can lead to improved performance. The management of accounts receivable is vital because it accounts for major proportion of a company's assets and plays a crucial role in WCM and corporate strategy. Similarly, Jindal et al. (2017) found that effective receivables management positively impacts working capital and profitability by reducing the collection period. The study also revealed the inverse relationship between CCC and the RCP and the inverse relationship between profitability and the RCP.

Effective management of receivables can lead to improved profitability and liquidity of firms. According to a study by Ali, Hussain, and Rahman (2019), an increase in the receivable collection period has a negative impact on profitability, while a decrease in the collection period can lead to improved liquidity. Moreover, the study suggests that implementing a sound receivable management system, such as creating a credit policy and monitoring customer payment behaviour, can help reduce overdue receivables and improve cash flow.

Proper procedures are followed until final collection as part of managing accounts receivable. According to Mian and Smith (1992), managing receivables involves evaluating the order entry, invoicing, and accounts receivable processes to ensure that the average time lag between completing the transaction and invoicing the receivables can be converted into cash as soon as possible. It is also essential to consider the opportunity cost of retaining receivables, or the opportunity missed by retaining funds in accounts receivable rather than investing it elsewhere.

1.15.3.1 Training and Management. Preparation of progress invoices and claims is a crucial aspect of the construction industry. Progress invoices, which detail the work completed in a building project, are typically created by qualified Qs working for the construction company (Olawum & Ayegun, 2016). However, the process of creating these invoices can be highly dynamic and variable due to various difficulties, such as unexpected demand or variability in personnel skills. Adopting a streamlined strategy to reduce delays in payment for progress invoices can be highly beneficial for the construction company.

In the construction sector, delaying payment for services rendered can have significant effects. Consultants must take steps to approve invoices for payment to clients without delay, but payment may be delayed due to errors made by clients, consultants, or contractors. Research by Zayed and Liu (2014) found that errors in claims submitted by the contractor, such as claims without proper supporting documents or documents submitted without following due process, are major causes of withholding or delaying payment. Qs play a crucial role in ensuring that all necessary documentation is provided to prevent delays in payment to contractors.

During a construction project, disputes between contractor and client frequently emerge. Indeed, disputes have been regarded as being intrinsic to the construction industry. Semple et al. (1994) point out that owing to the highly complex nature of the industry, it is not surprising that the industry is replete with conflicts. Furthermore, the competitive delivery method commonly employed by the construction industry promotes an antagonistic environment, where contractors are often found to be opportunistic in attempting to recover costs, while clients could be uninterested in compensating contractors for legitimate claims.

Payment issues can arise when mandatory documents are missing. Research by Odenigbo et al. (2020) found that contractors' claim submission errors, such as submitting claims without necessary documentation or assessing claims incorrectly, are major causes of payment delays or withholding of funds. Further, errors in progress invoicing are also a major obstacle in the construction industry. Payment may be withheld or delayed due to errors made by quantity surveyors in submitting progress invoices. According to Love and Zhou (2019), unintended departures from proper procedure, which are preventable, are referred to as errors.

There are various types of invoicing errors that can occur. A study by Gunathilaka and Senevirathne (2014) examined the various errors found in building estimating methods and revealed that progress invoice errors were common, including issues with invoice pricing and invoice preparation. Invoice preparation problems discovered included irrelevant preliminary items, incorrect quantities, irrelevant specifications, and insufficient information. These errors can occur due to a lack of experience in job roles.

In general, Qs often encounter difficulties when preparing interim payments, which can lead to delays in the release of funds. Azizi and Yu (2021) identified two types of errors that are typically considered severe: a lack of sufficient supporting documentation and a lack of time for valuation and certification. The lack of supporting documents can result from the contractors' inadequate documentation, while the lack of time for valuation and certification can result from high pressure on the quantity surveyor. Therefore, all stakeholders should work together to provide the best support for Qs.

Developing individual capabilities in businesses is crucial for success. As emphasized by Meyer and Nkado (2001) suggested, Qs body of knowledge includes preliminary cost advice,

project management, procurement and tendering, cost planning, contract documentation, cash flow forecasting, tender evaluation, interim payments, financial reporting, and final accounting, as well as the resolution of disputes. Therefore, identifying the training needs of quantity surveyors and designing a suitable training program is vital for developing individual capabilities.

Training programs are essential for quantity surveyors to enhance their skills and knowledge to improve the quality and efficiency of their work in the construction industry. According to Awe and Oluwunmi (2021), providing training to quantity surveyors can result in improved accuracy and consistency in cost estimation, better project cost control, and reduced construction costs. Quantity surveyors need to have a comprehensive understanding of various construction processes, including cost management, contract administration, and project planning. Therefore, it is crucial to design training programs that focus on enhancing these skills.

1.15.3.2 Receivable Policy. Trade credit can provide many advantages to businesses. However, it also comes with potential negative consequences such as overdue receivables and bad debts. As highlighted by Kasacheva and Udod (2018), these issues can lead to financial difficulties and even insolvency. As a result, businesses implement various policies to mitigate these risks. These can include charges for overdue payments and strict collection programs that may impact sales, credit duration, and bad debt percentages. Therefore, receivable follow-ups are an important aspect of credit management.

Receivable follow ups can include tracking receivables, sending reminders to customers, and providing advice on approaching due dates. One approach to receivable follow up is the preparation of ageing reports. These reports allow lenders to evaluate the quality of the borrower's accounts receivable and take steps to reduce credit risk. As noted by Gallinger and Ifflander (1986), the requirement to submit these reports may encourage managers to increase their efforts to gather accurate accounts receivable information.

A more lenient approach can increase sales but also increase the risk of bad debts and collection costs. As outlined by Frankel et al. (2020), a strict collection program can lead to a shorter average collection time and a lower percentage of bad debts but may also result in lower sales and higher collection costs. The credit period, or the amount of time customers must pay for their purchases, is an important consideration in determining the collection program.

Credit period can increase repeat business and sales, but also comes with a higher risk of bad debt losses. As highlighted by Brigham and Houston (2021), shortening the credit period can negatively impact sales and increase bad debt losses. Therefore, a strategy for accelerating collection is to include discounts for early payments and penalties for late payments in the receivable policy. This can include providing discounts to encourage customers to make prompt payments and charging penalties for late payments (Gentry & De La Garza, 1985)

The necessity of a clear collection policy is evident in receivable collection. Mian and Smith (1992) suggest that companies should consider factors other than credit period and incentives when creating a receivables policy. This can include sending invoices to customers early and reminding them of upcoming payments, introducing more favourable payment schemes, and using bargaining power to negotiate better payment terms and conditions. These considerations can help to identify areas for improvement in the payment process and increase overall efficiency.

1.15.3.3 Project Controls. Renovation projects often have shorter lifecycles compared to new construction projects, but they can still be challenging to complete. Despite the increasing number of renovation projects, they tend to perform worse in terms of cost, time, and quality when compared to new building construction. This is due to the unique risks and uncertainties that are present in renovation projects, as identified by several academics. The design phase of renovation projects can be particularly difficult to manage as clients' requirements are constantly changing (Mitropoulos & Howell, 2002).

Control is an important management task that involves enforcing a set of behaviours that align with organisational operations and desired performance standards. Daft (2014) recognizes control as one of the four main management functions, along with planning, organizing, and

leading. Similarly, Simons (1995), as cited by Herath (2007), defines control as the ability to balance innovative creativity with dependable goal-achievement in pursuit of economic growth. Control activities are the rules, guidelines, and routines that ensure management goals are met.

Most building contracts include a provision for retention, which means that a percentage of the payment is withheld until the project is completed. The main goal of retention is to provide the client with money in case the contractor becomes insolvent (Tianji & Robert, 2001). Additionally, retention can encourage the contractor to complete small unfinished tasks and fix defects once the project is finished. When defects are discovered during the retention period, the release of the retention payment is delayed. This triggers the "defects liability period," which typically lasts between six months and a year.

Defects that arise during the retention period should be resolved by the contractor. The remaining retention is released after the defects are corrected. Many construction companies struggle with delayed retention payments because of rework found in finished projects (Hughes et al., 2010). Rework is a significant contributor to cost and schedule overruns that are common in construction projects. Hwang et al. (2009), indicated that the average direct cost of rework is 5% of the entire construction cost. Therefore, rework impacts cost performance in the construction industry.

To reduce the impact of rework, pre-project and quality management plans should be created with an awareness of its sources. Project managers should focus on limiting rework by creating processes for resolving the causes for defects for those projects where cost tends to be more affected by rework (Jingmond & Agren, 2015). Common defects include those caused by lack of project controls such as structural issues that can cause cracks or collapse, electrical wiring and lighting, plumbing, drainage systems, ventilation, cooling, and heating systems, insulation, and soundproofing, and fire protection and suppression systems. However, Lodhi et al. (2017), suggest clients sometimes tend to delay payments when they lack sufficient financial resources to complete the projects.

Quality Assurance deals in preparation of the quality Plan and formation of organization wide standards. 'Project Quality Plan' can be defined as a set of activities planned at the beginning of

the project that helps achieve quality in the Project being executed. The Purpose of the Project Quality Plan is to define these activities or tasks that intends to deliver products while focussing on achieving customer's quality expectations. These activities or tasks are defined based on the quality standards set by the organization delivering the product. Project Quality Plan identifies which quality Standards are relevant to the project and determines how they can be satisfied (Bhonde & Shaikh, 2015).

A project quality plan is a document that outlines how a company will ensure quality on a specific construction project. It is distinct from a company's quality manual, which only provides general information about the company's quality policies and procedures. To minimize defects, the contract may require the presentation of a quality control plan, or the client may request a project quality plan directly. By providing clear guidelines for the project and processes, competent individuals can take over the project and ensure that it meets program, service, and performance standards (Ashokkumar, 2014).

There have been few studies on techniques for enhancing the practice of project control. Globerson (1994) found that cost control planning should be done utilizing the Work Breakdown Structure (WBS), breakdown of subtasks and smaller project components to ensure task clarity and improved control. This could be done based on the project phase, function, or product. This approach allows for greater clarity and control, and can be tailored to specific project phases, functions, or products. WBS has been shown to be a useful tool in addressing issues that arise during the planning phase and can help to prevent delays.

1.15.4 Payables Management

The relationship between profitability and working capital is well-established. However, the correlation between changes in payables and profitability is less clear. A study by Boisjolya and Conine (2020) found a negative relationship between DPO and profitability. This suggests that extending payment days to hold funds for a longer period can be beneficial for funding a firm's activities, such as investments and acquisitions. However, it could also imply that businesses that are struggling financially may extend payable days because they lack the resources to pay their bills.

Another explanation for the link between payables and profitability is that businesses choose to pay suppliers in advance to take advantage of discounts, rather than holding on to cash in the company. This is supported by research conducted by Moodley et al. (2015) found that firms prefer to pay in advance to gain discounts rather than keeping funds within the company. When liquidity is restricted, firms often need short-term financing to carry out procurement activities. One popular option for this is supplier financing, also known as "trade credit. Findings of Li, Qu, and Liang (2021) that efficient supplier management can improve operational efficiency and reduce costs.

1.15.4.1 Supplier Evaluation Process. The supplier early payment trade credit contract is the most prevalent. It enables clients to settle the supplier for materials procured within a specified window without incurring any charges. However, a penalty fee for paying late encourages early payment. Trade credit is open account financing if there is no early payment incentive and early payment rate of discount is decided solely by the supplier. The wholesale pricing and interest rate of the supplier as well as the outcomes provide new insight on the supplier's justification for its trade credit policies and the predominance of open account financing (Kouvelis & Zhao, 2012).

Suppliers may opt to offer a shorter credit period to customers instead of a longer one. Huang (2003) found that suppliers often determine the length of the credit period based on their analysis of the customer's creditworthiness, the supplier's credit policy, the collection terms, and the relationship between the supplier and customer. By offering a shorter credit period, suppliers can speed up the collection process and potentially earn interest if the customer delays payment beyond the credit period. However, some companies engage in business with suppliers that have short credit periods and high prices, which can be attributed to poor supplier selection.

Poor supplier selection can lead to various supply-related problems. Therefore, a comprehensive evaluation of potential suppliers is necessary before making a final decision. In today's supply chain management practices, several factors beyond cost are considered when evaluating a supplier's performance. This process of determining the effectiveness and efficiency of a supplier's actions is known as supplier assessment. The underlying assumption

is that if the evaluation is successful, it will result in changes to the supplier's behaviour that align with the goals of the company conducting the evaluation (Yoon et al., 2017).

Past research has identified seven key qualities or characteristics that an ideal supplier selection method should possess to be effective. According to Ka-ChiLam et al. (2010), these include the ability to: eliminate multicollinearity among the selection criteria; reduce the dimensionality of the selection data without significant loss of information; automatically assign weights to each criterion; be user-friendly; handle both quantitative and qualitative criteria; reduce the cost of the supplier selection process; handle many alternatives. Based on these seven qualities, the best supplier evaluation process should be designed to effectively select the most suitable supplier.

The introduction of a supplier evaluation process is crucial for a construction company to ensure the efficiency of their procurement operations. It allows the company to evaluate their suppliers' performance based on predetermined criteria, such as on-time delivery, quality of goods or services, and compliance with legal and ethical standards. The benefits of introducing a supplier evaluation process include cost reduction, improved quality control, and risk mitigation. By identifying underperforming suppliers, a construction company can take appropriate measures to improve the relationship with them or find alternative suppliers.

Moreover, it helps to establish long-term relationships with reliable suppliers, leading to increased trust and better communication. A study by Tukamuhabwa (2017) revealed that implementing a supplier evaluation system in a construction company improved supplier performance, reduced procurement costs, and enhanced supply chain efficiency. Therefore, it is imperative for a construction company to introduce a supplier evaluation process to optimize its procurement operations and gain a competitive advantage in the industry.

The use of a supplier evaluation form is required in conducting a proper evaluation on suppliers. According to Safayeni and Purdy (2000), the indirect process assessment technique gives the client much information about the supplier while the supplier compiles the data and creates the documentation. Since this evaluation is frequently well-documented, it lessens some of the customer's doubts regarding the supplier. Substantial supplier documentation can

support decisions made by various organisational units. There is a perception that the supplier can be evaluated based on the indirect assessment of their operations.

Supplier relationship management is a business process that establishes the framework for establishing and maintaining relationships with suppliers. According to Lambert and Schwieterman (2012), this process has become a critical component of business operations due to increased competitive pressures, the need to consider sustainability and risk, the requirement for cost efficiency, and the need to establish close ties with key suppliers whoever can provide the expertise needed to develop and successfully launch new products. By effectively managing relationships with crucial suppliers, a company can reap significant benefits in terms of credit periods.

The integration of operations with suppliers has been demonstrated to contribute to the success of a business. Hald and Chris (2011) argue that forming cross-functional and collaborative relationships with key suppliers enables a company to co-create value, providing additional benefits. Integrating with suppliers can be facilitated by sharing information and performance can be boosted. However, introducing supplier relationship management standards can point organisational personnel toward the proper degree of supplier integration and the need to find a plan specific to each partnership.

1.15.4.2 Accounting Software. In recent years, accountants have increasingly invested in accounting software to improve the efficiency and productivity of their corporate operations. The integration of information technology in accounting has become a strategic tool for gaining a competitive edge in the field. Many studies have been conducted to investigate the impact of computerized accounting software on the accuracy of financial reports (Gogus & Gokhan, 2014). The use of such software can help to eliminate errors that often occur when manually updating accounting entries.

Payment is usually made after the goods have been delivered when paying for suppliers. The purchase contract typically outlines a schedule for payment, which may include the possibility of an early-payment rebate. However, corporations typically have little incentive to make a payment before the final deadline. As a result, the process of receiving an order and making a

payment are typically separate actions. The two most used payment terms are payment within a certain time frame after delivery and payment by a certain day in the month following delivery (Kingsman, 1983).

The accounts department often faces challenges. Some of these challenges occur when dealing with multiple suppliers, various types of goods, multiple deliveries of the same item, returns or errors in orders, and other transactions. According to Brief (1965), manual documentation of these transactions can lead to errors, and in some cases, suppliers may be paid before the credit period has expired. Using an accounting software can help to minimize these errors and improve the coordination and efficiency of accounting procedures.

The manual approach to accounting can be time-consuming and prone to errors. According to Schiff and Szendi (2014), accounting software is a computer-based system that improves internal management and coordination within a business. It is also used to assess financial performance and inform decision-making. Development of more efficient computer programs can help to streamline bookkeeping and account preparation. The challenges of manual accounting procedures have led to the development of computerized systems that can improve efficiency and speed in bookkeeping and financial reporting.

Many providers of accounting software for small businesses promote the user-friendly nature of their products. One of the most widely used accounting software for small business is QuickBooks by Intuit. This software is known for simplifying the accounting process and making it accessible for individuals with little to no accounting knowledge, as reported by Ho et al. (2010). The ease of use of this software is a major selling point for small businesses.

1.16 Study Framework

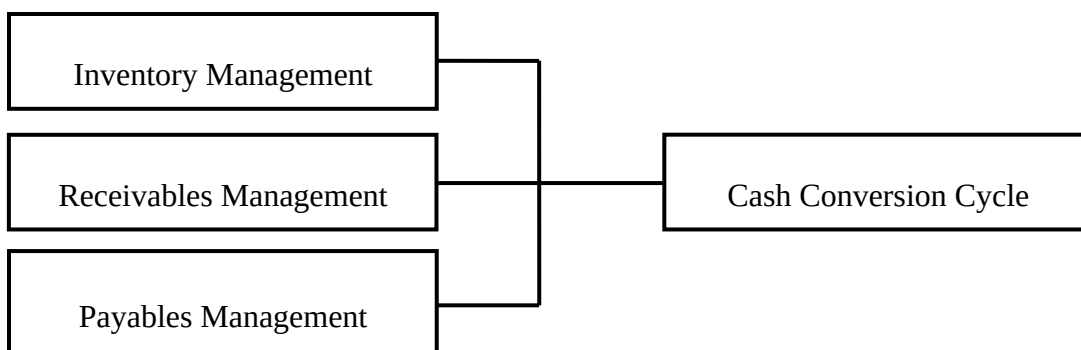


Figure 3.1: Study Framework

Source: Author.

1.17 Description of Techniques

1.17.1 Process Flowcharts

Process flow chart is a useful tool in evaluating and improving the effectiveness and efficiency of a process. They use specific symbols to illustrate the movement of units through the process from start to finish. By analysing the current procedures and identifying the main areas of strengths and weaknesses, it is possible to make operational improvements and increase the overall performance of the process. Additionally, process flow charts allow for the mapping of task ownership and associated tasks, making it easier to secure team participation by identifying task owners (Russell & Taylor, 2019; Lacerda & Leonardo, 2015).

1.17.2 ABC Analysis

In terms of inventory management, organisations can use the ABC classification system to categorize items into three groups: A as very important, B as moderately important, and C as least important. The management pays close attention to the class A items. However, other factors such as lead time, consumption rate, demand, and the criticality of the commodities should also be considered. This system allows for effective inventory control by focusing on the annual sales volume of the most important key items (Bhattacharya & Sen, 2001).

1.17.3 Policy Documents

A policy serves as a guideline for addressing human resource matters within an organization. It reflects the values and expectations of the organization regarding employee conduct and performance. Effective policies provide clear definitions of appropriate and inappropriate workplace behaviour, along with the consequences for non-compliance. A policy typically includes a statement of its objectives and broad guidelines for implementation. It may provide

some discretion in implementation if the reason for such discretion is specified in the policy. Policies are also necessary for situations where diverse interests and preferences may lead to vague or conflicting goals among stakeholders (Edelson, 2010).

1.17.4 Training Need Analysis

Training need assessments are used to identify the current skill gaps among staff members and determine the most appropriate types of training to fill those gaps before starting training programs. The overall training and development plan of an organisation or professional group is influenced by the results of a training needs assessment at the beginning of a cycle. Training success is evaluated using performance evaluation forms, speaking with supervisors, conducting surveys to identify skill shortages, or performing self-evaluations, among other methods. According to Dharmasiri (2019), without a needs analysis, results of the training may be arbitrary and difficult to attribute to the training. Therefore, it is essential to conduct a thorough study of training needs before starting any training program.

1.17.5 Work Breakdown Structure

The Work Breakdown Structure (WBS) is a widely used method for planning and organizing projects in various industries. It breaks down the project into smaller, more manageable components, down to the level of individual work elements. The goal of the WBS is to reflect the entirety of the project's scope and serve as a tool for controlling the project. It is important to ensure that all necessary work and effort is accurately captured within the WBS, as it serves as the foundation for estimating project cost, schedule planning, and managing risk. The WBS also offers a clear and concise overview of the project's goals and outcomes and serves as a basis for managing scope and all additional project management procedures. According to Maylor (2006), many standards that pertain to project management procedures rely on the WBS.

1.17.6 Key Performance Indicators

Key Performance Indicators (KPIs) are numerical values that indicate the extent to which a business is achieving its main goals. These indicators are crucial for measuring performance, as they provide a benchmark for comparing current results to desired outcomes. KPIs make it clear whether the business is on track to meet its objectives. Companies can use KPIs to set goals, develop strategies, and evaluate progress over time (Brown, 2004). Tracking KPIs can also be used to motivate employees by rewarding them for their contributions and holding them accountable for results. As a business grows, it can be common for employees to feel detached from the organisation's overall success. By connecting individuals to specific KPIs, they are more likely to strive for personal achievement and take pride in their work.

1.18 Summary

This chapter included a critical review of the literature concerning the performance gap of the Project. Literature was reviewed concerning identified root causes in Chapter 2. Further, solutions for issues of a similar nature identified by researchers previously were also examined considering the literature. It looked at similar literature related to cross-industry in the context of the Project. A study framework was developed based on the material gathered from numerous reliable sources, including ranked journal articles, conference proceedings and book sections. The chapter concluded by discussing the techniques used for the project and summarizing the key points covered in the literature review.

Chapter 4

THE PROJECT/METHODOLOGY

1.19 Introduction

The focus of this section is to set clear project objectives to develop solutions. This chapter begins by identifying and setting Specific, Measurable, Achievable, Realistic and Time bound (SMART) main objectives of the Project and sub-objectives of the project components. The solutions are provided using management techniques to address the performance gaps identified for each project components in Chapter 2 to achieve the main and sub-objectives. Further, necessary resource allocation, including subsequent costs associated is presented with appropriate justification and the team structure to implement the Project. Finally, the chapter concludes with a summary.

1.20 Field Project Objectives

The project's main objective is to reduce the cash conversion cycle in renovation projects by 38% from current 88% to 54% within 12 months. Further details on component-wise objectives are shown Table 4.1. Computation for objectives is indicated in Annexure VI.

Table 4.2: Project Component, Objectives and Techniques

No.	Project component	Project objectives	Techniques
01	Inventory management	To decrease idle inventory in renovation projects by 90% from current 54% to 5% within six months.	I. Process flowchart II. ABC analysis III. Policy document
02	Receivables management	To decrease overdue receivables in renovation projects by 47% from	I. Training need analysis

		current 38% to 20% in eight months.	II. Receivable policy III. Work breakdown structure
03	Payables management	To decrease early payment to suppliers in renovation projects by 76% from current 41% to 10% in 12 months.	I. Process flowchart II. Key performance indicators

Source: Author.

1.21 The Project: Project Components

The project intends on reducing CCC in renovation projects at REWC. Three components were identified as inventory management, receivables management and payables management in the study framework after conducting the literature review in Chapter 3. Root causes identified in the project components are no inventory planning system, no proper inventory monitoring mechanism, lack of training, absence of a credit policy, inadequate project controls, improper supplier evaluation and no accounting software in use. Therefore, solutions are provided for these root causes under each component compared to the loopholes identified in the current processes. Proposed solutions are further elaborated on in the annexures.

1.21.1 Inventory Management

1.21.1.1 Current Situation of the Project Component. It was identified that REWC has an idle inventory in the warehouse as per the root cause analysis conducted on the associated problems. The closing inventory has inventory purchased before 30 days are considered as idle inventory and inventory purchased within 30 days are not considered as idle inventory. The reason for idle inventory is over-purchasing. There are over purchases as the required demand is not identified for construction work scheduled in the coming weeks. And this is happening because there's no established inventory planning system.



Figure 4.1: Idle Inventory at the Warehouse

Source: Primary data.

In the current situation, the site engineer and the QS schedule the activities that need to be completed within the coming weeks. Then the required inventory for the tasks is compared with Bills of Quantities (BOQ). Then the current inventory is decided with the visual stock levels, and the required inventory list is sent to the head office, which is necessary to complete the upcoming tasks. But sometimes, inventory is requested too early than the scheduled tasks. There are also instances where scheduled tasks are changing. Further, records of current inventory in hand are less likely considered as existing inventory is decided based on visual stock levels.

When the request is sent to the head office, it is received by the procurement executive. Then the procurement executive places the order with an existing list of suppliers, or a new supplier is selected after supplier evaluation. A new supplier is selected if the inventory required is not

currently available with existing suppliers or there is no supplier currently providing the specific inventory type. Then the purchase order is sent to the supplier, and the supplier is paid within the given credit period. The requested inventory list is not double-checked or certified by the project manager or senior QS in the head office.

Variance in the physical inventory versus the inventory figure in records is another problem related to inventory management in renovation projects at REWC. The variance was identified with the qualitative inputs gained through interviews. The actual physical inventory shows a lesser quantity than the quantity shown in the records. This implies that inventory is missing due to theft from the warehouse or the construction site. It has provided an opportunity for theft since no proper stock evaluations are conducted. The root cause for this issue is not having a proper inventory monitoring mechanism. In other words, it explains that the current inventory monitoring mechanism has problems.

When the purchase order is sent to the supplier, inventory is received within the proposed period. When the inventory is received at the construction site, there is an informal inventory counting happening by the storekeeper, who certifies the received quantity by putting a signature on the suppliers' delivery note. Supplier is notified by the head office if any visible damages are seen, or the supplier sends the wrong inventory. Subsequently, the damaged inventory is returned to the supplier if the supplier provides the delivery. On the other hand, inventory in good condition is transferred to the warehouse until the damaged inventory is sent to the supplier.



Figure 4.2: Supplier Delivering Inventory to the Construction Site

Source: Primary data.

Sometimes the inventory is received after 5.00 p.m. when the only person in the construction site is the storekeeper. In such scenarios, unloaded inventory will be kept outside the warehouse or inside the construction site as there is no person to give a hand to transfer the inventory to the warehouse. The next day inventory is transferred to the warehouse when the construction workers arrive with an additional payment. Then the received inventory type and quantity are informed to the site engineer and will be updated in his documents. These documents are verified with the records in the head office as there can be variances due to damaged inventory or wrong inventory sent by the supplier.

Inventory is issued according to the First in First Out (FIFO) method for construction work when the TO or head of subcontractor requests any material. Note of any issued inventory is made in-store keepers documents and later informed to the site engineer. There are instances when the site engineer or the procurement executive notices that there should be adequate previous leftover inventory in the warehouse to perform pending tasks. This situation is highlighted when the site engineer is planning for the inventory requirement, or the procurement executive checks the request for inventory received from the construction site. Physical inventory counts are performed in such situations to identify the variance.

Physical inventory counts are also happening at the end of period, and inventory waste, obsolete inventory and material shrinkages are considered and updated in the wastage book. The balance is considered inventory theft and notified to the head office. After that, the list of inventories remaining in the construction site at the end of the year is updated in the material at site file. Sometimes material at site file is required when preparing the interim invoices or when the project is terminated before the completion due to various reasons such as breach of contract by or as per the decision taken by the client. Figure 4.1 illustrates the current inventory management process.

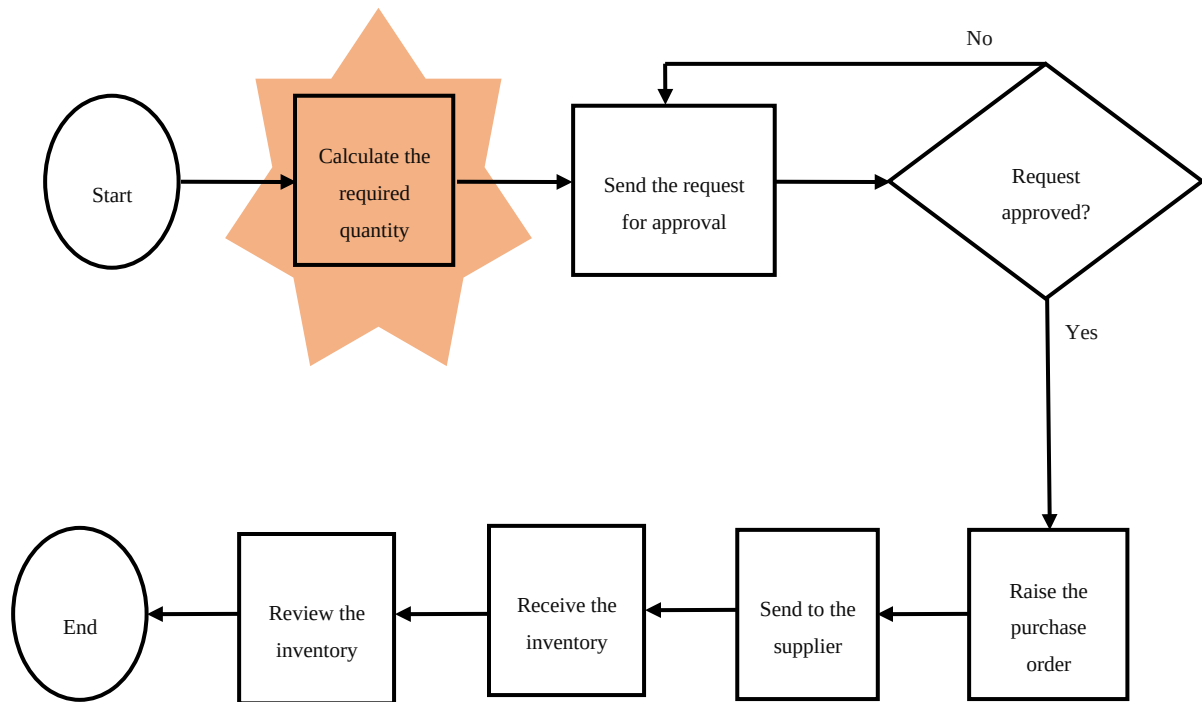


Figure 4.3: Current Inventory Management Process of REWC

Source: Primary data.

1.21.1.2 Solutions Development.

- I. Design a new inventory management process (Annexure VII).
- II. Introduce a daily project report format to identify the inventory requirement (Annexure VIII).
- III. Conduct a workshop to introduce ABC analysis to organise inventory based on importance to the company (Annexure IX).
- IV. Prepare a store policy to follow (Annexure X).

1.21.2 Receivables Management

1.21.2.1 Current Situation of the Project Component. As per the to the root cause analysis conducted on renovation projects at REWC, it was identified that there was an increase in overdue receivables over time. There are receivables within the credit period of 35 days and receivables that have exceeded the 35 days. Therefore, receivables exceeding 35 days after submitting the invoice are considered overdue. In REWC, 35 days is considered the credit period, as mentioned in the ICTAD publication on receiving client payments (ICTAD, 2007). The reason for overdue receivables does not have formal follow-ups due to not having a documented process. Similarly, the root cause for this does not have a receivable policy.

In the current scenario, the client is first advertising the tender (government and semi-government projects) or directly contacting REWC (only in private projects). If the project requires bidding, REWC must pay a tender fee, fill out the tender, and submit it to the client. It is necessary to pay a bid bond to the client through the bank, which is non-refundable even for the contractors who lose the project in bidding. Then the consultant evaluates the best few bids and sends them to the technical committee. After that, the contractor who fulfils the criteria gets the project and will be awarded the letter of acceptance.

REWC is then allowed to commence working on the project after fulfilling all the requirements related to the performance bond, advance bond, signing the contract, insurance and designing the work program. The Benchmark of REWC is to submit an invoice every 30 days to the client for payment. But in the current scenario, there are delays in receivables as the client withholds the payment due to invoice errors. These errors include measurements, and not submitting invoices without proper supporting documents. In such instances client withhold the payment till the corrections are done, and the invoice is resubmitted. After that Interim Payment Certificate (IPC) will be issued with the cheque. Figure 4.2 illustrates the current invoicing process for client payments.

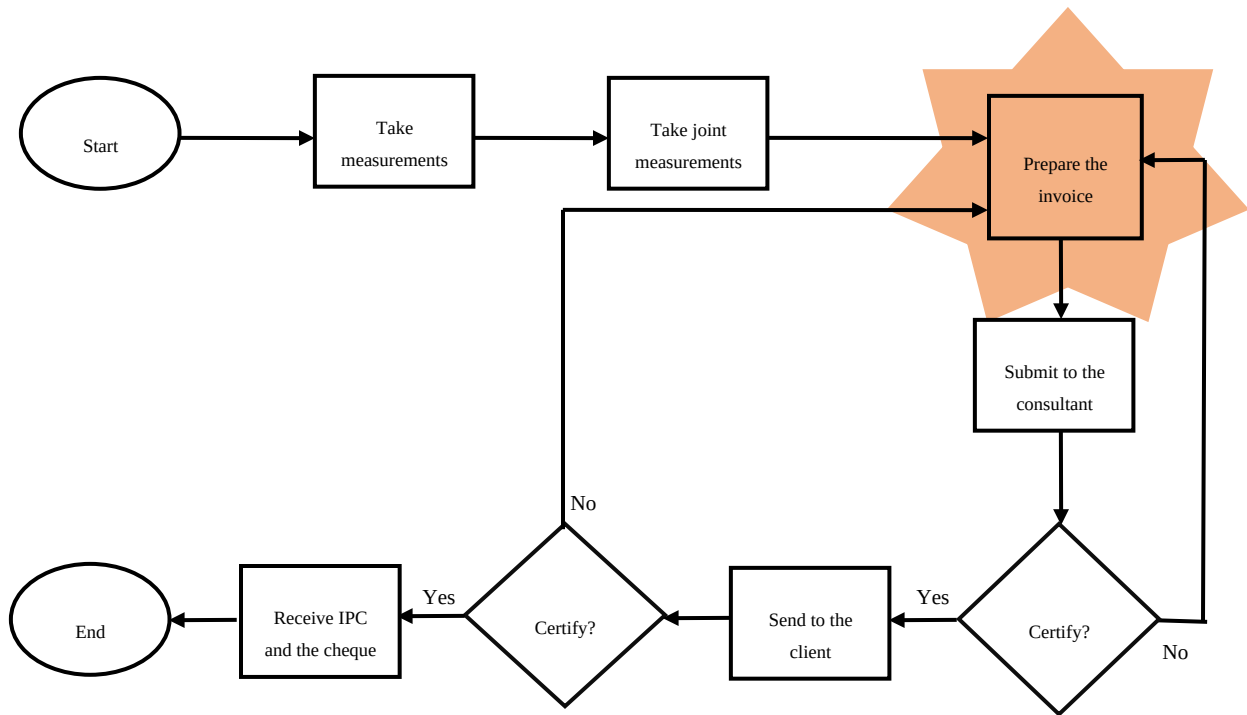


Figure 4.4: Current Invoicing Process for Client Payments

Source: Primary data.

As no formal follow ups happen, there are situations where the client is not making the payment even after exceeding the credit period. The receivable ageing analysis is conducted at the end of the financial period. However, proper formal follow-ups, such as giving reminders through telephone, emails, or mail through the registered post, are not happening properly. This is because there is no introduced process to follow in receivable collection mentioning the credit period, how to do the reminders before exceeding the credit period, and actions to take when the credit period is exceeded.

Subsequently, the project will be concluded within given period. However, the full project value will not be received by the REWC even after handing over the project. This is because of the retention amount held by the client till the retention period is over. The retention period is the warranty period used in the construction industry. Any defects identified within this period should be reworked and certified by the consultant. In some instances, the client can deduct the cost of rework from the retention amount and release the remaining amount to REWC with the

agreement of both parties. In most cases, REWC agrees to do the rework and to release the retention.

In a situation of project defect, the current situation is analysed by the site engineer, project manager, and the TO. Review meetings happen in such cases and then agree to rework or deduct the rework cost from the retention amount with the approval of the managing director and the finance manager. After that, the cost of rework is calculated together with the QS. In most cases, REWC agrees to conduct the rework as it is the responsibility of the REWC to satisfy the client. Enough supervision is happening on the construction site, yet project defects are identified before the end of the retention period due to inadequate project controls.

Either REWC or the consultant can be responsible for these project defects. Reasons for defects can be inappropriate material or brand with low quality recommended by the consultant with low quality or design errors in the provided designs by the consultant. At the same time, errors from REWC are due to poor quality and lack of supervision during the construction work. Therefore, REWC needs to take responsibility for rework only if it's the company's fault. Where the responsibility for the defect is from a consultant, REWC can rework the defects for an extra charge. However, it is necessary to impose adequate project controls to avoid defects due to errors from REWC.

1.21.2.2 Solution(s) Development.

- I. Conduct training for quantity surveyors on taking measurements and invoice preparation practices (Annexure XI).
- II. Develop a receivable policy and conduct a test on the sample policy (Annexure XII).
- III. Design a well-formulated quality plan for renovation projects. (Annexure XIII).
- IV. Introduce work breakdown structure method to identify each task in renovation projects (Annexure XIV).

1.21.3 Payables Management

1.21.3.1 Current Situation of the Project Component. As identified from the root cause analysis, REWC is making the payments to the suppliers before the company average for a credit period of 45 days. This is due to the suppliers providing short credit periods to settle. Short-dated suppliers are selected providing more weightage to criteria such as price, quality, time to deliver, and specific requirements requested by the client. Every criterion is important in selecting suppliers for materials and services. However, in this situation, REWC faces a high CCC due to having shorter payable outstanding period. Therefore, the reason for short-dated suppliers is poor supplier selection due to improper supplier evaluation process.

Firstly, the TO identifies the material requirement and sends the material request form to the head office. The QS and the site engineer must certify this as the material or service requirement should be according to the specifications provided in the BOQ. However, this step is not formally happened in REWC. In some situations, the consultant or resident engineer of the client specifies a brand or certification of a material to be used. The rest of the material, which doesn't have specifications, is requested from the regular suppliers.

After that, the procurement executive calls for a quotation for required materials with relevant specifications. In this situation, several criteria are assessed, providing weightage for each section. After selecting the supplier fulfilling the related criteria are chosen to supply the material. Next, a purchase order is sent to the supplier, and the supplier provides the inventory and the delivery note. Most of the time, materials are directly delivered to the construction site, but in a few instances, materials get delivered to the main stores and the head office. A goods received note is issued when the material is received. After that, the received material is reviewed and sent to the stores.

Subsequently, received materials are updated in the records of the storekeeper, site engineer on the construction site, and accounts staff in the head office. Next, payment is settled to the supplier after reviewing the material. There can be different products, different brands, and wrong orders or damaged materials that need to be returned to the supplier. Once the payment

is settled, a basic supplier evaluation is conducted based on the number of deliveries compared against quantity, delivery, and quality criteria (Annexure XV). After that, the decision to keep or to discontinue the supplier is taken based on the marks earned for each criterion.

However, another problem identified with payables management is REWC settling payments even before the credit period provided by the suppliers. This is happening due to errors made by the staff in recording entries. Currently, REWC is using Ms Excel as there is no accounting software to manage accounting entries. Due to recording entries manually, mistakes and omissions, there are errors in identifying the credit period due to multiple suppliers and multiple deliveries. As a result, REWC is facing problems settling the other prioritized suppliers who need to be paid immediately due to cash purchases or who are over with the credit period. The current supplier evaluation process is illustrated in Figure 4.3.

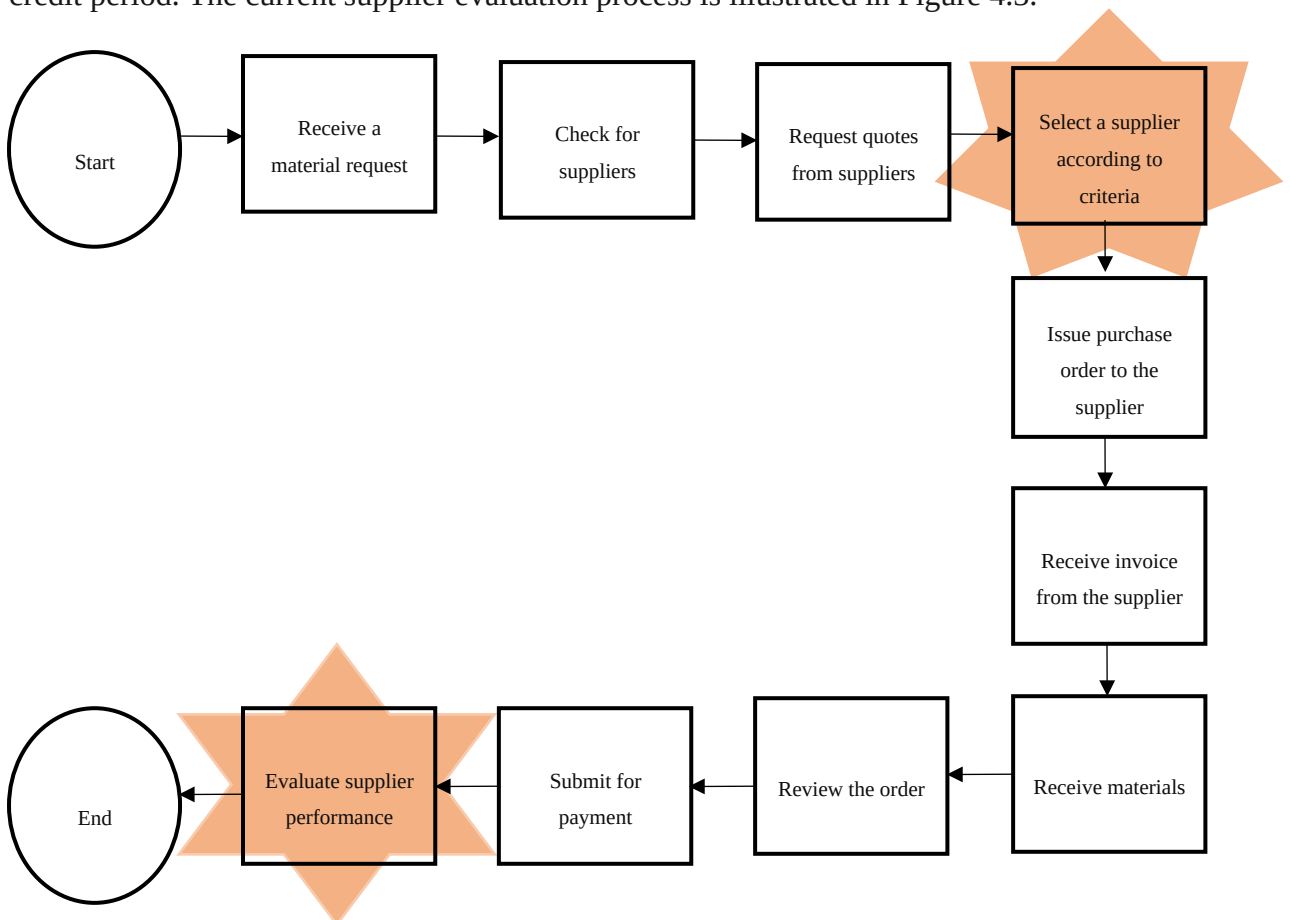


Figure 4.5 Current Supplier Evaluation Process

Source: Primary data.

1.21.3.2 Solution(s) Development.

- I. Introduce an improved supplier evaluation process (Annexure XVI).
- II. Develop well-formulated supplier evaluation criteria sheets (Annexure XVII).
- III. Prepare a set of guidelines for supplier relationship management (Annexure XVIII).
- IV. Introduce QuickBooks as accounting software and conduct a training on using the software (Annexure XIX).

1.22 Resource Allocation

The senior accounts executive, who is also the author, work in close collaboration with the project team to guarantee the successful completion of the Project. Priority was given to internal staff who were chosen based on their previous successful projects and their willingness to take on new responsibilities. There will be no extra cost for resources since the tasks are already part of their current duties. The Table 4.2 shows the resources that will be allocated for each task under the project components and the responsibilities that are expected to be fulfilled.

Table 4.3: Resource Allocation

Project component	Solutions	Allocated resources	Responsibility
Inventory management	Design a new inventory management process.	Site engineer, quantity surveyor and technical officer	Define the upcoming tasks and decide on the required inventory.
		Senior quantity surveyor	Check if the inventory usage is as per the quantity and prices pre-determined in the BOQ.
		Store keeper	Identify the current inventory level in the construction site.
		Procurement executive	Confirm the current inventory in the construction site.
		HR and administration	Design the new inventory

		manager, project manager, Senior procurement executive	management process and communicate the task to the team.
	Introduce a daily project report format.	Project manager	Design the new project report format. Communicate to the involved staff.
		HR and administration manager	Plan the sub-contractors and labours for the upcoming task.
	Conduct a workshop on ABC analysis.	Finance manager, accountant, senior accounts executive	Identify the issues related to inventory planning in the construction sites by conducting informal discussions with the staff. Developing objectives for the workshop. Design the workshop plan and evaluation. Decide a suitable internal facilitator.
		Facilitator	Provide the consent. Prepare the content for the workshop.
		Accounts executive	Identify the costs related to the workshop.
	Prepare a store policy to follow.	Finance manager, accountant, senior accounts executive	Study the store policies followed by construction companies of the same scale.
		Project manager	Decide the stock-level maintenance.

		Site engineer, senior procurement executive, storekeeper	Review of stock levels, obsolete and slow-moving inventory, receiving and rejecting inventory.
		Managing director	Study and approve the new policy.
Receivables management	Conduct training for quantity surveyors on taking measurements and invoice preparation process.	Project manager, senior quantity surveyor, human resources and administration manager	Provide insights and assistance to identify QS's training needs. Provide assistance to design training objectives and training programs. Provide guidance, approval, and required resources to conduct the training program. Prepare training evaluation based on the predetermined criteria.
		Quantity surveyors and trainee quantity surveyors	Provide an honest opinion on the survey.
	Develop a receivable policy and conduct a test on the sample policy.	Finance manager, accountant, senior finance executive, Senior accounts executive	Identify all omissions happening currently in the organisation in the receivable collection process. Provide insights on the criteria that need to be included in the receivable policy.
		Managing director	Check and provide approval for the receivable policy.

		Finance executives, accounts executives	Assist in testing the policy.
	Design a well-formulated quality plan for renovation projects.	Senior quantity surveyor, project manager	Identify the standards and specifications mentioned in the BOQ. Analyse the forms, formats and checklists provided in the ICTAD publication and develop steps to follow.
	Introduce work breakdown structure method to identify each task in renovation projects.	Project manager senior quantity surveyor	Ensure that the full and accurate requirement is captured within the template. Ensure the team understand all the development tasks according to the new template.
Payables Management	Introduce an improved supplier evaluation process.	Finance manager, senior procurement executive	Conduct informal discussions with the staff involved with payables management and develop a new supplier evaluation process.
	Develop a well-formulated supplier evaluation criteria sheet.	Senior procurement executive, senior quantity surveyor	Provide inputs to identify the supplier's performance evaluation criteria. Calculate the weights of each criterion.
	Prepare a set	Finance manager,	Identify the issues associated

	of guidelines for supplier relationship management.	Accountant, senior accounts executive and Senior procurement executive	with suppliers. Study the supplier relationship practices followed in companies of the same scale. Develop the supplier relationship management guidelines.
		HR and administration manager	Communicate the new supplier relationship management guidelines to the staff involved with the process.
		Managing director	Examine and provide approval for the supplier relationship management guidelines.
	Introduce QuickBooks accounting software and conduct a training on using the software.	Finance manager, Accountant, Senior accounts executive	Conduct a questionnaire to identify the need for accounting software. Develop objectives for implementing accounting software.
		Service provider	Provide information on the accounting software and agree to the workshop to train the software.
		Accounts executive	Identify the cost related to implementing the software.
		Accounts executives, finance executives	Get used to the accounting software instead of the manual entries.
		HR and administration	Assist in getting the responses

		manager	from the staff and arranging the training.
--	--	---------	--

Source: Author.

Project team structure to implement the project is illustrated in Figure 4.6 below.

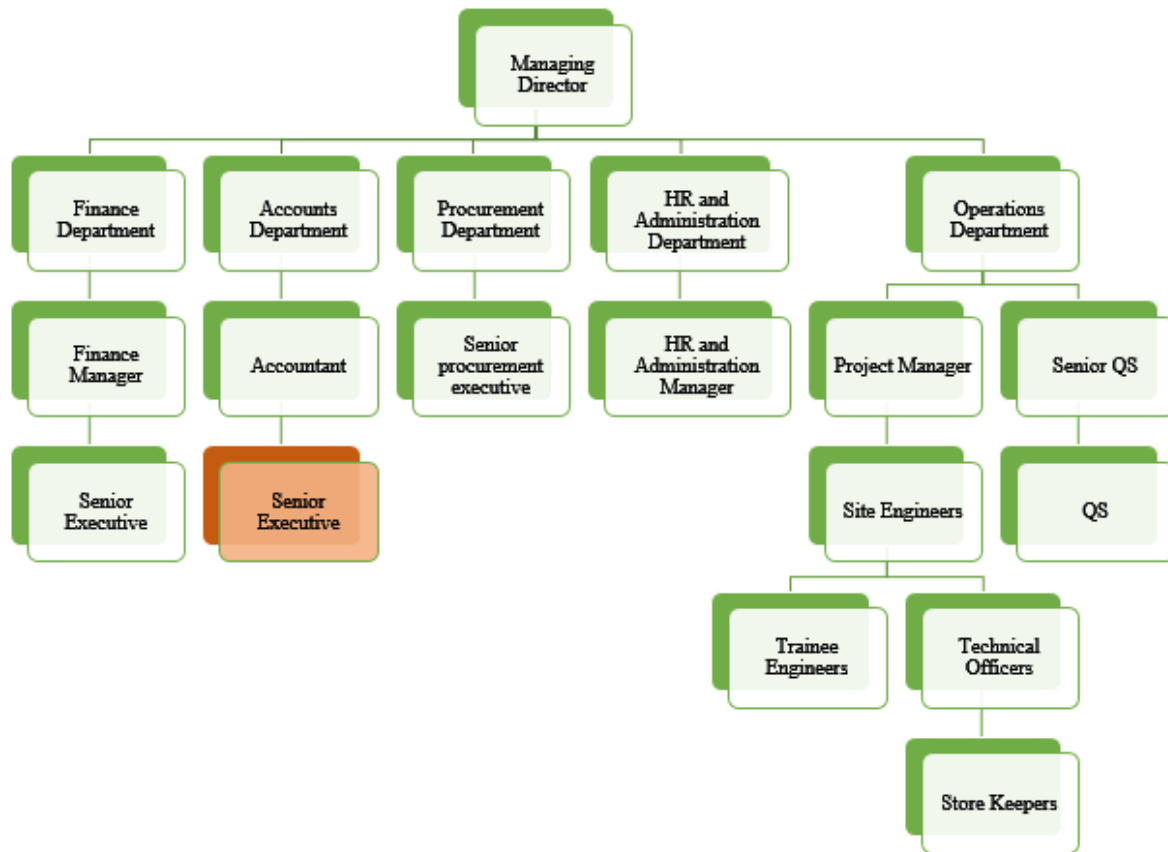


Figure 4.6: Project Team Structure

Source: Author.

1.23Cost Estimates

Table 4.4: Project Components, Proposed Solutions and Budget

No	Project	Proposed solutions	Reference	Cost
----	---------	--------------------	-----------	------

	component			(LKR)
1	Inventory Management	Design a new inventory management process.	Annexure VII	Nil
		Introduce a daily project report format to identify the inventory requirement.	Annexure VIII	Nil
		Conduct a workshop to introduce ABC analysis to organise inventory based on importance to the company.	Annexure IX	15,825
		Prepare a store policy to follow.	Annexure X	135
2	Receivables Management	Conduct a training on taking measurements and invoice preparation practices.	Annexure XI	286,360
		Develop a receivable policy and conduct a test on the sample policy.	Annexure XII	Nil
		Design a well-formulated quality plan for renovation projects.	Annexure XIII	Nil
		Introduce a work breakdown structure method to identify every task in renovation projects.	Annexure XIV	Nil
3	Payables Management	Introduce an improved supplier evaluation process.	Annexure XVI	Nil
		Develop well-formulated supplier evaluation criteria sheets.	Annexure XVII	Nil
		Prepare a set of guidelines for supplier relationship management.	Annexure XVIII	Nil
		Introduce QuickBooks as accounting software and conduct a training on using the software.	Annexure XIX	75,800
Total Cost				378,120

Source: Author.

1.24 Summary

The objective of this chapter was defining the project objectives and presenting the solutions for the problem developed based on the literature discussed in the previous chapter. Overall project objective and component-wise project objectives were defined along with the sub-objectives. The current situation of each component was discussed in detail with the help of flow charts. Solutions were defined along with detailed appendices on implementing them. Then the required resources for each of these proposed solutions were presented. Project team structure and how the staff needs to be joining from different areas of the organizational structure were illustrated. Then the cost of implementing the project solutions was mentioned along with the evidence of the cost calculations.

Chapter 5 FINDINGS

5.1 Introduction

This section of the report delves into the practicality of the proposed Project. It starts by determining the projected tangible and intangible benefits in terms of revenue growth and cost reduction for the first year of the Project. The benefit-cost ratio is then calculated to reveal the net gain that the organisation will gain with the implementation of the Project. Once the feasibility of the Project has been established, the specific outputs and outcomes of the Project that are related to each component will be identified. The chapter will conclude with a comprehensive overview of the content presented in the chapter.

5.2 Benefit-Cost Analysis

Table 5.1 show the estimated financials as benefits to be received from the Project against related expenditures with accompanying annexures.

Table 5.5: Estimated Benefits

No	Detailed benefits	Reference	Cost (LKR)
	<u>Tangible Benefits</u>		
	Reduced idle inventory	Annexure XX(a)	980,889
	Reduced overdue receivables	Annexure XX(b)	2,273,219
	Reduced early settlement for suppliers	Annexure XX(c)	1,335,225
A	Total Tangible Benefits		4,589,333
	<u>Intangible Benefits</u>		
	-		
B	Total Intangible Benefits		0
C	Total Benefits (A+B)		4,589,333

Source: Author.

Benefit-Cost Analysis is used to evaluate the Project's viability by computing the net incremental benefit and benefit/cost ratio, as illustrated in Table 5.2.

Table 5.6: Benefit-Cost Analysis

Item	Value (LKR)
Total Benefits	4,589,333
Total Costs (Annexure XXI)	378,120
Net Incremental Benefits	4,211,213
Benefit/Cost Ratio	12.1

Source: Author.

5.3 Projected Outputs and Outcomes

The outputs and outcomes of the Project are presented in Table 5.3. These outputs and outcomes have been recognized by linking them to the project objectives.

Table 5.7: Projected Outputs and Outcomes

No	Project component	Project objectives	Output	Outcome
1	Cash Conversion Cycle	To reduce the cash conversion cycle in renovation projects by 38% from current 88% to 54% within 12 months	The cash conversion cycle decreased to 54%, from 140 days to 87 days.	Higher profitability. Improved company reputation. Increased credibility of the company.
2	Inventory Management	To decrease idle inventory in renovation projects by 90% from current 54% to 5% within six months.	Idle inventory reduced to 5% which is a reduction of LKR 980,889. Inventory management process. Daily project report format. Store policy.	Less internal communication gaps. Effective control. Enhanced supervision. Improved staff motivation.
3	Receivables Management	To decrease overdue receivables in renovation projects by 47% from current 38% to 20% in eight months.	Overdue receivables reduced to 20%, which is a reduction of LKR 2,273,219. Training plan. Receivable policy. Quality plan.	Enhanced employee motivation. Continuous quality improvements. Improved customer satisfaction.
4	Payables	To decrease early	Early payment to	Improved labour

	Management	payment to suppliers in renovation projects by 76% from current 41% to 10% in 12 months.	suppliers reduced to 10%, which is a decrease of LKR 1,335,225. Supplier evaluation process. Supplier evaluation criteria sheets. Supplier relationship management guidelines.	efficiency. Increased productivity. Enhanced performance of the suppliers.
--	------------	--	---	--

Source: Author.

5.4 Summary

This section of the report began by providing an overview of its content. The goal of the chapter was to demonstrate the practicality of the Project to the organisation. Cost savings were identified to determine the total tangible benefits for the first year. Then, the estimated costs for each solution under the project components were used to compute the Project's benefit-cost ratio. The benefit-cost ratio was estimated to be 12.1:1, which demonstrates the Project's feasibility. Following that, the specific results and outcomes of the Project were outlined in relation to the project objectives, using both percentages and numerical data. Further, qualitative outputs that can be itemized were also presented, linking them to the solutions.

Chapter 6

DISCUSSION OF FINDINGS, RECOMMENDATIONS AND CONCLUSION

6.1 Introduction

The purpose of the chapter is to establish a connection between the literature review and the conclusion of the Project report. The components will be evaluated concerning their status with the literature reviewed in Chapter 3. This demonstrates the author's comprehensive understanding of the concepts and their application in the project context, utilising theoretical references and practical examples. Following the discussion of the components, recommendations will be provided focusing on short-term and medium-term. Finally, the conclusion summarises the key information discussed within the chapter.

6.2 Discussion of Findings

6.2.1 Cash Conversion Cycle

The concept of CCC has its theoretical basis in working capital management, which is an important area of finance. CCC is a widely accepted measure of a company's effectiveness in managing its working capital, and it has been extensively studied in the literature. Empirical research has consistently shown that companies with a low CCC tend to be more efficient and profitable compared to those with a high CCC. For example, a study by Deloof (2003) found that companies with a shorter CCC had higher profitability and liquidity ratios. Similarly, research by Chang (2018) suggests that lowering the number of days in CCC can enhance profitability, and REWC's planned reduction of CCC from 88% to 54% is expected to achieve this goal.

In the case of REWC, its long CCC of 140 days, compared to the industry average of 87 days, has led to a need for external funding from bank loans and overdrafts with high-interest rates. The increase in finance costs has led to a loss of LKR 331,228 in the financial year 2021/22. This highlights the practical implications of having a longer CCC, which is consistent with the

theoretical and empirical evidence that a longer CCC can have a negative impact on a company's financial performance. However, implementing the project to reduce REWC's CCC by 38%, is expected to increase profitability and improve the company's reputation and credibility. Therefore, it is evident that longer CCC affects profitability inversely, and the proposed reduction of CCC aligns with the theoretical principles for effective working capital management.

6.2.2 Inventory Management

Effective inventory management is crucial for business success and profitability. Song and Houtum (2019) emphasize the importance of monitoring and controlling inventory levels to ensure smooth business operations. Their findings suggest that an increase in inventory investment reduces a company's working capital and negatively impacts liquidity, leading to lower profits. These findings are consistent with the inventory turnover theory, which posits that a high inventory turnover is associated with increased profitability (Sadiq et al., 2018).

The idle inventory accounts for a major proportion of the closing inventory due to improper inventory planning, which implies a delay in construction work and project duration. In particular, the average annual idle inventory amounts to LKR 1,077,900, which is equivalent to 54% of the closing inventory, only for renovation projects. However, by implementing the proposed solutions, such as new policies and processes, ABC analysis, and a new project report format, it is expected to reduce idle inventory to 5%, resulting in estimated savings of LKR 980,889 per annum, which is a 90% reduction in idle inventory costs. This reduction in idle inventory is expected to improve the company's working capital management, increase its liquidity, and enhance profitability. Additionally, the project is expected to reduce internal communication gaps, improve effective control, enhance supervision, and improve staff motivation, thereby contributing to the company's overall success.

6.2.3 Receivables Management

In their research, Yao and Yiling (2018), discovered a strong connection between liquidity and profitability indicators, implying that shortening the collection time for receivables can lead to

improved performance. They also emphasized the significance of managing accounts receivable, as it constitutes a substantial portion of a firm's assets and is a vital aspect of working capital management and corporate strategy. It is vital to keep in mind that small changes in accounts receivable can have a significant impact on working capital. Furthermore, managers should take steps to avoid potential lost sales due to stricter credit terms.

According to a study by Ali, Hussain, and Rahman (2019), effective management of receivables can lead to improved profitability and liquidity of firms. The study found that an increase in the receivable collection period has a negative impact on profitability, while a decrease in the collection period can lead to improved liquidity. Moreover, the study suggests that implementing a sound receivable management system, such as creating a credit policy and monitoring customer payment behaviour, can help reduce overdue receivables and improve cash flow.

This is particularly relevant for firms facing overdue receivables, which can account for a significant portion of their closing receivables, as in the case of REWC. At REWC, overdue receivables represent a significant portion of the closing receivables, averaging LKR 4,836,637 or 38% of the total. With the implementation of a receivable policy, the project is expected to reduce overdue receivables from 38% to 20%, which is a decrease of 47% in eight months. This will result in a reduction of LKR 2,273,219 per annum and improve employee motivation, continuous quality improvements, and customer satisfaction.

6.2.4 Payables Management

Accounts payable is an essential component of working capital management, and an increase in the payable days can have a significant impact on the company's profitability. According to a study by Cheng, Liu, and Liu (2021), payable outstanding period has a significant negative relationship with the firm's profitability, implying that a longer payable period tends to reduce the company's profitability. Furthermore, their study also revealed that firms with lower payable outstanding period have higher risk and are more prone to financial distress.

In the case of REWC, the root cause analysis identified that the company pays its suppliers earlier than the average settlement period due to the short credit period provided by suppliers. Out of all the payments made to suppliers in a year, 41% accounts for early payments made due to short credit periods provided, amounting to an average of LKR 1,756,875. By implementing a proper supplier evaluation process and developing supplier relationship management, the project aims to reduce early payments to suppliers from 41% to 10%, a decrease of 76%. This reduction will result in savings of LKR 1,335,225 and will help alleviate pressure on the long CCC. Moreover, the project intends to gain qualitative outcomes such as improved labour efficiency, increased productivity, and enhanced performance of the suppliers, which are in line with the findings of Li, Qu, and Liang (2021) that efficient supplier management can improve operational efficiency and reduce costs.

6.3 Recommendations

6.3.1 Cash Conversion Cycle

Short term: It is suggested that individual meetings should be held with the individuals responsible for carrying out these tasks before putting any plans into action related to inventory management, receivable management, and payable management. This will assist in setting goals and addressing any questions or concerns they have. These meetings are likely to increase their motivation and lead to successful implementation as the success of these plans is closely linked to the performance of those assigned to them.

Short term: Implementing proposed solutions necessitates allocating operational funding from the annual corporate budget, carefully considering projected outputs and outcomes. Approval from the management, along with the provision of authority to the project task owners, is required for the successfully execute the of project tasks. Further, the successful execution of the recommended solutions necessitates the involvement of a team with a diverse set of functional expertise. The senior management should ensure the availability of the proposed human resources. Additionally, evaluations should be conducted for accounting software purchases and training programs to ensure their effectiveness and efficiency.

Medium term: Reducing the CCC requires collective efforts from all the employees of a company. Therefore, it is essential to create awareness among them about the importance of reducing CCC and how their actions can impact the company's financial health. This can be achieved through regular training sessions and workshops, where employees can be informed about the significance of managing payables, receivables, and inventory efficiently. Moreover, the management can also introduce incentive schemes to encourage employees to contribute to reducing the CCC. Such incentives can include bonuses for employees who find innovative ways to reduce the CCC or cost-saving initiatives. By involving employees in the CCC reduction process and creating awareness about it, a company can not only achieve its financial objectives but also foster a sense of ownership and commitment among its employees towards the company's success.

Short term/ Medium term: To get the expected outcomes, it is essential for managers to continually monitor and track the progress made towards achieving set objectives and ensure that the company is on track to meet its goals. Monitoring can be done using key performance indicators (KPIs), which provide valuable data on the performance of the organization. Managers should regularly review KPIs to identify areas that require improvement and take corrective actions. Regular progress reports can be presented to the management team to provide an overview of the company's performance. By monitoring internal progress and taking corrective actions where necessary REWC enhance overall performance with the implemented project.

6.3.2 Inventory Management

Medium term: It is important for management to establish new procedures during the training program and stress the significance of adhering to them. This can be facilitated by using a responsibilities matrix, which clearly defines the expectations for everyone. Even if some individuals do not fully understand the entire process, they can begin by fulfilling their specific responsibilities. It is suggested to initiate these new processes with one renovation project and then fully adopt them across all projects within one year. The senior staff of each department should closely monitor the usage of these newly introduced processes and standards by their

subordinates. The success of these solutions is contingent upon the active involvement of the senior staff in motivating and supporting their team members.

Short term: It is also recommended to consider the installation of CCTV cameras to deter and prevent inventory theft. Not only that CCTV cameras serve as a visual deterrent for would-be thieves, but they also provide valuable evidence in the event of a theft. Additionally, being able to monitor inventory and real-time construction activity can aid in identifying any potential security vulnerabilities, allowing for proactive measures to be taken to address them. Furthermore, the CCTV footage can be used as evidence in a case if employees of REWC are involved with the missing inventory. In addition, it can be used to claim insurance to recover lost inventory.

6.3.3 Receivables Management

Medium term: In this project, an assessment was performed to identify the training needs of the QSSs. It is crucial to continue assessing the training needs of all employees on an ongoing basis. The company's leadership should also allocate sufficient funding for employee training. Areas for improvement in employee skills and knowledge can be identified and addressed through training programs by conducting periodic needs assessments. It is important to encourage employee participation in these training programs. The effectiveness of the training programs should be evaluated both before and after the training is provided. Finally, it is important to track employee performance as they apply their new skills and knowledge in their roles.

Short term: Preparing invoices for client payments is a critical process as it directly affects the company's cash flow. To streamline this process, it is recommended to establish a standardized invoicing procedure that includes all relevant details, such as project name, invoice number, payment terms, and due date. Additionally, regular communication with clients regarding invoicing and payment terms can help ensure timely payments and avoid potential disputes. By implementing a streamlined invoicing process, construction companies can improve their cash flow, enhance client satisfaction, and minimize administrative errors.

Short term: A committee to monitor receivable follow-ups is recommended for REWC to ensure the introduced receivable policy is feasible. The committee should monitor the

efficiency and effectiveness of receivable collection compared to previous processes. Additionally, they must assist task owners in performing the new duties until they become comfortable with their duties and the new policy. Further, having a dedicated committee in charge of receivable follow-ups allows REWC to keep a close eye on their client's payment history and manage their risk.

Short term: It is recommended that REWC place a strong emphasis on correct supervision to avoid project defects and ensure completing the project successfully within the time and budgeted cost. The supervisor's role is crucial in identifying and addressing issues early on, enforcing safety procedures and regulations, and maintaining the quality of the work. Having a dedicated supervisor on site, REWC can ensure that the work is being carried out correctly and efficiently and that any defects or errors are caught and addressed promptly. Furthermore, proper supervision can help control the budget, avoid cost overruns, and release the retention payment without any issues.

6.3.4 Payables Management

Short term to medium term: Periodically updating supplier evaluation criteria is crucial for REWC to ensure that suppliers are selected based on the client-provided material specifications. The construction industry is constantly evolving, with new technologies and materials being developed and regulations being implemented. Not regularly updating the evaluation criteria makes it risky to select suppliers who may not be compliant with industry standards. Additionally, by regularly reviewing and updating the evaluation criteria, a company can identify new suppliers that may be able to supply better products or services at a minimum cost.

Short term to medium term: It is strongly recommended that REWC prioritize better recruitment practices when hiring staff for their accounting department to avoid errors in accounting. This includes thoroughly evaluating candidates' qualifications, skills, and experience and conducting background checks to ensure their integrity and trustworthiness. Furthermore, recruiting responsible staff with a strong understanding of accounting principles, laws and regulations, and the software and tools used for accounting entries will help the

organisation avoid errors and ensure the accuracy and integrity of records. This is crucial to avoid errors currently happening in making payments to suppliers.

6.4 Conclusions

The beginning of this project involved introducing the background and importance of the performance gap, as well as identifying related issues. This was followed by an examination of the underlying causes. Next, a literature review was performed, and developed the study framework subsequently. Identified project components were set with objectives, and after that current situation of the project components was discussed, and solutions were proposed. The next chapter revealed that a benefit-cost ratio of 12.1 can be achieved by implementing the Project. Finally, findings from the current situation of the project components were discussed against findings from the literature accompanied by midterm and short-term recommendations.

REFERENCES

- Abdurrahman, A. A., Hanifah, H. R., & Fauzi, M. H. (2021). The impact of supplier evaluation and selection on supply chain performance in SMEs. *Journal of Engineering and Applied Sciences*, 16(2), 263-270.
- Abuzayed, B. (2012). Working capital management and firms' performance in emerging markets: the case of Jordan. *International Journal of Managerial Finance*, 8(2), 155-179. doi: <https://doi.org/10.1108/17439131211216620>.
- Access Engineering PLC. (2017). Annual Report. Colombo: Access Engineering PLC.
- Access Engineering PLC. (2018). Annual Report. Colombo: Access Engineering PLC.
- Access Engineering PLC. (2020). Annual Report. Colombo: Access Engineering PLC.
- Access Engineering PLC. (2022). Annual Report. Colombo: Access Engineering PLC.
- Access Engineering PLC. (2021). Annual Report. Colombo: Access Engineering PLC.
- Advocata. (2019, 11 18). Empowering Our SMEs. Retrieved from Advocata: <https://www.advocata.org/commentary-archives/2019/18/11/empowering-our-smes>
- Ali, M., Hussain, A., & Rahman, A. (2019). The impact of payables management on firm profitability: Evidence from the textile sector of Pakistan. *Global Business Review*, 20(3), 642-655.
- Ashokkumar, D. (2014). Study of Quality Management in Construction Industry. *International Conference on Engineering Technology and Science*, 3(1), 36-43.
- Atkinson, A. R. (1999). The role of human error in construction defects. *Structural Survey*, 17(4), 231-236. <https://doi.org/10.1108/02630809910303006>
- Awe, O. O., & Oluwunmi, A. O. (2021). The effect of supplier evaluation on purchasing performance: A case study of Nigeria manufacturing companies. *European Journal of Business and Management Research*, 6(3), 1-11.

- Azizi, N. S., & Yu, T. Z. (2021). Preliminary Study of Changes in Work Practices of Quantity Surveyors in Building Information Modelling (BIM) Projects. *International Journal of Sustainable Construction Engineering and Technology*, 12(1), 140-152.
- Ballou, R. H. (2000). Evaluating Inventory Management Performance using a Turnover Curve. *International Journal of Physical Distribution & Logistics Management*, 30(1), 72-85. doi: <http://dx.doi.org/10.1108/09600030010307993>
- Bhattacharya, C., & Sen, S. (2001). Doing Good Always Lead to Doing Better? Consumer Reactions to Corporate Social Responsibility. *Journal of Marketing Research*, 38(2), 225-243. <https://doi.org/10.1509/jmkr.38.2.225.1883>.
- Biazzo, S. (2002). Process Mapping Techniques and Organisational Analysis. *Business Process Management Journal*, 8(1), 42-52. <http://dx.doi.org/10.1108/14637150210418629>
- Boisjolya, R. P., & Conine , T. E. (2020). Working capital management: Financial and valuation impacts. *Journal of Business Research*, 108, 1-8. <https://doi.org/10.1016/j.jbusres.2019.09.025>
- Brief, R. P. (1965). Nineteenth Century Accounting Error. *Journal of Accounting Research*, 3(1), 12-31. <http://dx.doi.org/10.2307/2490048>
- Brigham, E. F., & Houston, J. F. (2021). *Fundamentals of financial management*. Cengage Learning.
- Brown, D. H. (2004). *Language Assessment: Principles and Classroom Practices*. Longman.
- Bruccoleri, M., Cannella, S., & La Porta, G. (2014). Inventory record inaccuracy in supply chains: the role of worker's behaviour. *International Journal of Physical Distribution and Logistics Management*, 44(10), 796-819. <http://dx.doi.org/10.1108/IJPDLM-09-2013-0240>
- Central Bank of Sri Lanka. (2021). *Annual Report 2020*. Colombo: Central Bank of Sri Lanka.
- Central Bank of Sri Lanka. (2021). *Annual Report 2021*. Central Bank of Sri Lanka.

- Chang, C.-C. (2018). Cash Conversion Cycle and Corporate Performance: Global. *International Review of Economics and Finance*, 56(1), 568-581. <https://doi.org/10.1016/j.iref.2017.12.014>
- Charitou, M. S., Elfani, M., & Lois, P. (2010). The Effect of Working Capital Management on Firms Profitability: Empirical Evidence from An Emerging Market. *Journal of Business and Economics Research*, 8(12), 63-68. <https://doi.org/10.19030/jber.v8i12.782>
- Courtright, E. M., & Zee, J. V. (2011). The Database for Inventory, Monitoring, and Assessment (DIMA). *Rangelands*, 33(4), 21-26. <https://doi.org/10.2111/1551-501X-33.4.21>
- Daft, R. L. (2014). The Leadership Experience. *Journal of Education for Business*, 89(8), 457-458. <https://doi.org/10.1080/08832323.2014.921593>
- Deloof, M. (2003). Does Working Capital Management Affect Profitability of Belgian Firms? *Journal of Business Finance & Accounting*, 30(3). <https://doi.org/10.1111/1468-5957.00008>
- Deloof, M., & Jeger, M. (1996). Trade Credit, Product Quality, and Intragroup Trade: Some European Evidence. *Financial Management*, 25(3), 945-968. <https://doi.org/10.2307/3665806>
- Dharmasiri, A. S. (2019). HRM for Managers. Postgraduate Institute of Management.
- Ebben, J. J., & Johnson, A. C. (2011). Cash Conversion Cycle Management in Small Firms Relationships with Liquidity, Invested Capital, and Firm Performance. *Journal of Small Business & Entrepreneurship*, 24(3), 381-396. <https://doi.org/10.1080/08276331.2011.10593545>
- Edelson, N. M. (2010). Can standard operating procedures be motivating? Reconciling process variability issues and behavioural outcomes. *Total Quality Management and Business Excellence*, 16(2), 231-241. <https://doi.org/10.1080/14783360500054236>

- Ehrhardt, M. C., & Brigham, E. F. (2005). *Financial Management Theory and Practice*. Thomson Washington DC.
- Emery, G. (1984). A Pure Financial Explanation for Trade Credit. *Journal of Financial and Quantitative Analysis*, 9(3), 271-285. <https://doi.org/10.2307/3666022>
- Fazzari, S. M., & Petersen, B. C. (1993). Working Capital and Fixed Investment: New Evidence on Financing Constraints. *The RAND Journal of Economics*, 24(3), 328-342. <https://doi.org/10.2307/2555961>
- Frankel, R., Hwan, K. B., Tao, M., & Xiumin, M. (2020). Bank Monitoring and Financial Reporting Quality: The Case of Accounts Receivable Based Loans. *Contemporary Accounting Research*, 1911-3846. <https://doi.org/10.1111/1911-3846.12595>
- Gallinger, G. W., & Ifflander, J. A. (1986). Monitoring Accounts Receivable Using Variance Analysis. *Financial Management*, 15(4), 69-76. <https://doi.org/10.2307/3665782>
- Garanina, T., & Olga, P. (2015). Liquidity, cash conversion cycle and financial performance: case of Russian companies. *Investment Management and Financial Innovations*, 12(1), 90-100.
- Gentry, J. A., Vaidyanathan, R., & Lee, H. (1990). A Weighted Cash Conversion Cycle. *Financial Management*, 19(1), 90-99. <https://doi.org/10.2307/3666040>
- Gentry, J. A., & De La Garza, J. M. (1985). A Generalized Model for Monitoring Accounts Receivable. *Financial Management*, 14(4), 28-38. <https://doi.org/10.2307/3665334>
- Gill, P. (1968). *Systems Management Techniques for Builders and Contractor*. McGraw-Hill. <https://doi.org/10.1002/qj.49709440214>
- Globerson, S. (1994). Impact of various work-breakdown structures on project conceptualization. *International Journal of Project Management*, 12(3), 165-171. [https://doi.org/10.1016/0263-7863\(94\)90032-9](https://doi.org/10.1016/0263-7863(94)90032-9)
- Gogus, C. G., & Gokhan, O. (2014). The Roles of Technology Acceptance Model. *Academy of Information and Management Sciences Journal*, 17(1), 1-25.

- Gunathilaka, A., & Senevirathne, I. P. (2014). Identification of Errors that are being made in Sri Lankan Construction Industry. The 3rd World Construction Symposium 2014. Colombo: Department of Building Economics, University of Moratuwa, Sri Lanka. Retrieved from <https://ciobwcs.com/downloads/WCS2014-Proceedings.pdf>
- Hald, K. S., & Chris, E. (2011). Supplier evaluation processes: the shaping and reshaping of supplier performance. *International Journal of Operations & Production Management*, 31(8), 888-910. <https://doi.org/10.1108/01443571111153085>
- Harrington, T. C., Lambert, D. M., & Vance, M. P. (1990). Implementing an Effective Inventory Management System. *International Journal of Physical Distribution & Logistics Management*, 20(9), 17-23. <http://dx.doi.org/10.1108/EUM00000000000376>
- Hendricks, K. B., & Singhal, V. R. (2009). Demand-Supply Mismatches and Stock Market Reaction. *Stock Mark Manufacturing & Service Operations Management*, 11(3), 509-524. <http://dx.doi.org/10.1287/msom.1080.0237>
- Herath, S. (2007). A framework for management control research. *Journal of Management Development*, 26(9), 895-915. <http://dx.doi.org/10.1108/02621710710819366>
- Ho, W., Xu, X., & Dey, P. K. (2010). Multi-criteria decision-making approaches for supplier evaluation and selection: A literature review. *European Journal of Operational Research*, 1(1), 16-24. <https://doi.org/10.1016/j.ejor.2009.05.009>
- Huang, Y. (2003). Optimal retailer's ordering policies in the EOQ model under trade credit financing. *Journal of Operational Research Society*, 54(9), 1011-1015. <https://doi.org/10.1057/palgrave.jors.2601588>
- Hughes, W., Hillebrandt, P., & Murdoch, J. (2010). The impact of contract duration on the cost of. *Construction Management and Economics*, 8(1), 11-14. <https://doi.org/10.1080/014461900370906>
- Hwang, B.-G., Thomas, S. R., Haas, C. T., & Caldas, C. H. (2009). Measuring the Impact of Rework on Construction. *Journal of Construction Engineering and Management*, 135(3), 187-198. [https://doi.org/10.1061/\(ASCE\)0733-9364\(2009\)135:3\(187\)](https://doi.org/10.1061/(ASCE)0733-9364(2009)135:3(187))

- ICTAD. (2007). Standard Bidding Document Procurement of Works. ICTAD.
- Jindal, D., Jain, S., & Vartika. (2017). Effect of Receivables Management on Profitability: A Study of. *International Journal of Applied Sciences and Management*, 2(2), 246-255.
- Jingmond, M., & Agren, R. (2015). Unravelling causes of defects in construction. *Construction Innovation*, 15(2), 198-218. <https://doi.org/10.1108/CI-04-2014-0025>
- Ka-ChiLam, Tao, R., & KitLam, M.-K. (2010). A material supplier selection model for property developers using Fuzzy Principal Component Analysis. *Automation in Construction*, 19(5), 608-618. <https://doi.org/10.1016/j.autcon.2010.02.007>
- Kasacheva, O. V., & Udod, V. A. (2018). Overdue Accounts Receivable: Quality Analysis for Prevention and Regulation. *Digest Finance*, 23(3), 254-260. <https://doi.org/10.24891/df.23.3.254>
- Kasim, N., Liwan, S. R., Shamsuddin, A., Zainal, R., & Kamaruddin, N. C. (2012). Improving on site material tracking for inventory management in construction projects. Melaka.
- Kingsman, B. G. (1983). The Effect of Payment Rules on Ordering and Stockholding Purchasing. *Journal of Operational Research Society*, 34(11), 1085- 1098. <https://doi.org/10.1057/jors.1983.240>
- Kouvelis, P., & Zhao, W. (2012). Financing the Newsvendor: Supplier vs. Bank, and the Structure of Optimal Trade Credit Contracts. *Operations Research*, 60(3), 566-580. <https://doi.org/10.1287/opre.1120.1040>
- Lacerda, R. T., & Leonardo, E. (2015). Research Opportunities in Business Process Management and Performance Measurement from a Constructivist View. 23(1), 18-30. <https://doi.org/10.1002/kpm.1495>
- Lambert, D. M., & Schwieterman, M. M. (2012). Supplier relationship management as a macro business process Supplier relationship management as a macro business process. *Supply Chain Management. An International Journal*, 17(3), 337-352. <https://doi.org/10.1108/13598541211227153>

- Langford, D., Iyagba, R., & Komba, D. (1993). Prediction of solvency in construction companies. *Construction Management and Economics*, 11, 317-325. <https://doi.org/10.1080/014461993000000036>
- Lee, C. C., & Hogg, K. (2009). Early career training of quantity surveying professionals. The Construction and Building Research Conference of the Royal Institution of Chartered Surveyors. Cape Town: Northumbria University, Newcastle Upon Tyne. Retrieved from <https://nrl.northumbria.ac.uk/id/eprint/2341/>
- Li, J., Qu, X., & Liang, J. (2021). The impact of accounts payable on firm performance: Evidence from Chinese listed firms. *International Journal of Managerial Finance*, 17(1), 24-43.
- Love, P., & Zhou, J. (2019). Project controls for electrical, instrumentation and control systems: Enabling role of digital system information modelling. *Automation in Construction*, 103(1), 202-212. <http://dx.doi.org/10.1016/j.autcon.2019.03.010>
- Malik, Q. A., & Anser, R. (2013). Cash Conversion Cycle and Firms' Profitability – A Study of Listed Manufacturing Companies of Pakistan. *Journal of Business and Management*, 8(2), 83-87.
- Maylor, H., Brady, T., Davies, T. C., & Hodgson, D. E. (2006). From precertification to programmification. *International Journal of Project-Management*, 24(8), 663-634. <http://dx.doi.org/10.1016/j.ijproman.2006.09.014>
- Meschnig, G., & Kaufmann, L. (2015). Consensus on supplier selection objectives in cross-functional sourcing teams: Antecedents and outcomes. *International Journal of Physical Distribution and Logistics Management*, 45(8), 774 - 793. <https://doi.org/10.1108/IJPDLM-06-2014-0129>
- Meyer, T., & Nkado, R. (2001). Competencies of professional quantity surveyors: A South African perspective. *Construction Management and Economics*, 19(1), 481-491. <https://doi.org/10.1080/01446193.2001.9709624>

- Mian, S. L., & Smith, C. W. (1992). Accounts Receivable Management Policy: Theory and Evidence. *The Journal of Finance*, 47(1), 169-200. <https://doi.org/10.1111/j.1540-6261.1992.tb03982.x>
- Mitropoulos, P., & Howell, G. A. (2002). Renovation Projects: Design Process Problems and Improvement Mechanisms. *Journal of Management in Engineering*, 18(4), 179-185.:// [http://dx.doi.org/10.1061/\(ASCE\)0742-597X\(2002\)18:4\(179\)](http://dx.doi.org/10.1061/(ASCE)0742-597X(2002)18:4(179)
- Mohamad, S. J., Suraidi, N. N., Rahman, N. A., & Suhaimi, R. D. (2016). A Study on Relationship between Inventory Management and Company Performance: A Case Study of Textile Chain Store. *Journal of Advanced Management Science*, 4(4), 299-304. <http://dx.doi.org/10.12720/joams.4.4.299-304>
- Moodley, T., Ward, M., & Muller, C. (2015). The relationship between the management of payables and the return to investors. *South African Journal of Accounting Research*, 31(1), 35-43. <https://doi.org/10.1080/10291954.2015.1105555>
- Moss, J. D., & Stine, B. (1993). Cash Conversion Cycle and Firm Size: A study of retail firms. *Managerial Finance*, 19, 25-34.
- Nanaware, M. R., & Saharkar, U. R. (2017). Application of Inventory Control Technique in Construction. *International Journal of Engineering Research and General Science*, 5(4), 49-54.
- Niskanen, J., & Niskanen, M. (2006). The Determinants of Corporate Trade Credit Policies in a Bank-dominated Financial Environment: The Case of Finnish Small Firms. *European Financial Management*, 12(1), 81-102. <https://doi.org/10.1111/j.1354-7798.2006.00311.x>
- Odenigbo, O., Odusami, K. T., Okolie, K. C., & Vincent, O. C. (2020). Causes of Delayed Payment in Construction Project in Nigeria. *European Journal of Engineering Research and Science*, 5(9), 1049-1053. doi: <https://doi.org/10.24018/ejeng.2020.5.9.2066>
- Olawale, Y., & Sun, M. (2014). Construction project control in the UK: Current practice, existing problems, and recommendations for future improvement. *International Journal*

- of Project Management, 33(3), 623–637.
<https://doi.org/10.1016/j.ijproman.2014.10.003>
- Olawum, T. O., & Ayegun, O. A. (2016). Are Quantity Surveyors Competent to Value for Civil Engineering Works? Evaluating QSs' Competencies and Militating Factors. *Journal of Education and Practice*, 7(16), 9-18.
- Olson, D. L., & Xie, M. (2009). A comparison of coordinated supply chain inventory management systems. *International Journal of Services and Operations Management*, 6(1), 73-88. <https://doi.org/10.1504/IJSOM.2010.029494>
- Omopariola, E. D., Abimbola, W., Edwards, D. J., & Thwala, W. D. (2020). Contractors' perceptions of the effects of cash flow on construction project. *Journal of Engineering, Design and Technology*, 18(2), 308-325. <http://dx.doi.org/10.1108/JEDT-04-2019-0099>
- Parada, J., & Rutgers, V. (2021, July). Deloitte. Retrieved from Global Powers of Construction 2021:<https://www2.deloitte.com/global/en/pages/energy-and-resources/articles/deloitte-global-powers-of-construction.html>
- Plinere, D., & Borisov, A. (2015). Case Study on Inventory Management Improvement. *Information Technology and Management Science*, 18, 91-96. <http://dx.doi.org/10.1515/itms-2015-0014>
- Postgraduate Institute of Management. (2021). PIM Presentation Guidelines. Postgraduate Institute of Management.
- Raman, A., & DeHoratius, N. (2008). Inventory Record Inaccuracy: An Empirical Analysis. *Management Science*, 54(4), 627–641. <https://doi.org/10.1287/mnsc.1070.0789>
- Razeek, T. (2022, April 30). Ceylon Today. Retrieved from Construction industry on the verge of collapse: <https://ceylontoday.lk/2022/04/30/construction-industry-on-the-verge-of-collapse/>

- Rekik, Y., Sahin, E., & Dallery, Y. (2009). Inventory inaccuracy in retail stores due to theft: An analysis of the benefits of RFID. *International Journal of Production Economics*, 118(1), 189–198. <https://doi.org/10.1016/j.ijpe.2008.08.048>
- Richards, V. D., & Laughlin, E. j. (1980). A Cash Conversion Cycle Approach to Liquidity Analysis. *Financial Management*, 9(1), 32-38. <https://doi.org/10.2307/3665310>
- Roberts, L. D. (2015). Exploring ethical issues associated with using online surveys in educational research. *Educational Research and Evaluation*, 212(2), 95-108. <https://doi.org/10.1080/13803611.2015.1024421>
- Rosenblatt, M. J., & Roll, Y. (2007). Warehouse design with storage policy considerations. *International Journal of Production Research*, 22(5), 809-821. <http://dx.doi.org/10.1080/00207548408942501>
- Rehman, S., Nasir, M., & Ahmad, A. (2019). Supplier evaluation and selection criteria in supply chain management: A systematic literature review. *Business and Economic Review*, 11(1), 55-80.
- Russell, R. S., & Taylor, B. W. (2014). *Operations and Supply Chain Management* (8th ed.). New Delhi: Wiley India (Pvt) Ltd.
- Ruyken, P. T., Wagner, S. M., & Jonke, R. (2011). What is the right cash conversion cycle for your supply chain? *International Journal of Services and Operations Management*, 10(1), 13-29.
- Sadiqu, A., Ahmad, N., & Umar, M. (2018). Inventory turnover and profitability: Evidence from Pakistan. *International Journal of Scientific and Research Publications*, 8(10), 617-623.
- Safayeni, F., & Purdy, L. (2000). Strategies for supplier evaluation: a framework for potential advantages and limitations. *IEEE Transactions on Engineering Management*, 47(4), 435 - 443. <http://dx.doi.org/10.1109/17.895339>
- Saleek, J. G. (2005). *Accounts Receivable Management Best Practices*. John Wiley and Sons Inc.

- Sambasivan, M., & Soon, Y. W. (2007). Causes and effects of delays in Malaysian construction industry. *International Journal of Project Management*, 25(5), 517-526. <http://dx.doi.org/10.1016/j.ijproman.2006.11.00>
- Sana, S. S. (2010). A Production Inventory Model of Imperfect Quality Products in a Tree Layer Supply Chain. *Decision Support Systems*, 50, 539-547. <http://doi:10.1016/j.dss.2010.11.012>
- Schiff, A., & Szendi, J. (2014). Helping Small Business Entrepreneurs Avoid Critical Mistakes in QuickBooks Accounting Software. *The Entrepreneurial Executive*, 19(1), 169-181.
- Schultz, K. L., & Juran, D. C. (1999). The Effects of Low Inventory on the Development of Productivity Norms. *Management Science*, 45(12), 1664-1678. <http://dx.doi.org/10.1287/mnsc.45.12.1664>
- Schwartz, R. (1974). An Economic Model of Trade Credit. *Journal of Financial and Quantitative Analysis*, 9(4), 643-657. <https://doi.org/10.2307/1882390>
- Semple, A., Hartman, F. T., & Jergeas, G. F. (1994). The Construction Dispute Problem: Trends and Causes. *Journal of Management in Engineering*, 10(3), 39-44. [https://doi.org/10.1061/\(ASCE\)9742-597X\(1994\)10:3\(39\)](https://doi.org/10.1061/(ASCE)9742-597X(1994)10:3(39))
- Smith, A. D., Smith, A. A., & Baker, D. L. (2011). Inventory management shrinkage and employee anti-theft approaches. *International Journal Electronic Finance*, 5(3), 210-234. <http://dx.doi.org/doi:10.1504/ijef.2011.041337>
- Song, J.-S., & Houtum, G.-J. v. (2019). Capacity and Inventory Management: Review, Trends, and Projections. *Manufacturing and Service Operations Management*, 1–11. <https://doi.org/10.1287/msom.2019.0798>
- Talonpoika, A.-M., Monto, S., Pirttila, M., & Karri, T. (2014). Modifying the cash conversion cycle: revealing concealed advance payments. *International Journal of Productivity and Performance Management*, 63(3), 341-353. <http://dx.doi.org/10.1108/IJPPM-12-2012-0130>

- Tersine, R. J., & Tersine, M. G. (1990). Inventory Reduction: Preventive and Corrective Strategies. *The International Journal of Logistics Management*, 1(2), 17-24. <https://doi.org/10.1108/09574099010804563>
- Teruel, P. J. G., & Solano, P. M. G. (2007). Analysing the relationship between accounts receivable and profitability: Evidence from Spanish SMEs. *Small Business Economics*, 29(4), 415-427. <https://doi.org/10.1007/s11187-006-9028-8>
- Tianji, X., & Robert, L. (2001). Risk assessment on contractor's pricing strategies. *Construction Management and Economics*, 19(1), 77-84. <http://dx.doi.org/10.1108/ECAM-08-2012-0082>
- Toelle, R. A., & Tersine, R. J. (1989). Excess Inventory: Financial Asset or Operational Liability. *Production and Inventory Management Journal*, 30(4), 32-33.
- Tukamuhabwa, B. R. (2017). The impact of supplier evaluation on procurement performance in the construction industry. *Journal of Construction Business and Management*, 1(1), 1-13. <https://doi.org/10.1177/1074318417739018>
- Wafa, S. A., Uddin, M. N., Uddin, M. M., & Islam, M. S. (2021). A framework of supplier evaluation criteria for the construction industry: A case study in Bangladesh. *Journal of Construction in Developing Countries*, 26(2), 23-41.
- Wang, X., Li, Q., Li, Y., Li, Z., & Song, Y. (2021). The application of digital technologies in construction industry: A review. *Journal of Cleaner Production*, 314, 128012. <https://doi.org/10.1016/j.jclepro.2021.128012>
- Williams, C. L., & Patuwo, E. B. (2004). Analysis of the effect of various unit costs on the optimal incoming quantity in a perishable inventory model. *European Journal of Operational Research*, 156, 140-147. [http://dx.doi.org/10.1016/S0377-2217\(02\)00919-0](http://dx.doi.org/10.1016/S0377-2217(02)00919-0)
- Wu, D., Chen, J. C., & Olson, D. L. (2012). An empirical study of the impact of delivery lead time variability on inventory levels. *International Journal of Production Research*, 50(2), 439-452.

- Yang, Y. C., Park, C. J., Kim, J. H., & Kim, J. J. (2007). Management of Daily Progress in a Construction Projects of Multiple Apartment Buildings. *Journal of Construction Engineering and Management*, 133(3), 242-253. [http://dx.doi.org/10.1061/\(ASCE\)0733-9364\(2007\)133:3\(242\)](http://dx.doi.org/10.1061/(ASCE)0733-9364(2007)133:3(242))
- Yao, H., & Yiling, D. (2018). Managerial incentives and accounts receivable management policy. *Managerial Finance*, 44(7), 865-884. <https://doi.org/10.1108/MF-05-2017-0148>
- Yoon, J., Talluri, S., Yildiz, H., & Ho, W. (2017). Models for supplier selection and risk mitigation: a Holistic Approach. *International Journal of Production Research*, 56(10), 3636-3661. <https://doi.org/10.1080/00207543.2017.1403056>
- Zhou, L., Zhang, H., Li, J., & Ma, X. (2017). Evaluating suppliers in a supply chain with multiple criteria in uncertain contexts. *Sustainability*, 9(10), 1-23.

APPENDICES

Annexure I: Summary of the Income Statements

Table A1.1: Summary of the Income Statements

	2017	2018	2019	2020	2021

Revenue	27,300,757	27,725,972	26,086,301	24,591,014	26,889,841
COS	11,327,400	11,849,228	12,403,285	12,292,538	14,914,009
Gross Profit	15,973,357	15,876,744	13,683,016	12,298,476	11,975,832
Admin Expenses	6,045,580	6,178,700	6,567,494	5,795,084	6,103,491
Distribution Expenses	292,985	312,538	357,566	172,643	260,174
Bank Guarantee Charges	198,422	224,586	251,536	299,328	333,153
Insurance Guarantee Charges	6,100	6,464	7,240	7,920	7,269
Security Bond Charges	6,725	7,093	7,944	9,453	11,911
Lease Interest	200,500	207,176	232,037	276,124	347,916
Loan Interest	2,312,500	2,544,870	2,667,055	2,938,395	3,128,778
Overdraft Interest	1,624,500	1,680,195	1,787,419	1,908,228	2,114,368
Total Finance Expenses	4,348,747	4,670,384	4,953,231	5,439,448	5,943,395
Net Profit	5,286,045	4,715,122	1,804,725	891,301	(331,228)

Source: Primary data.

Annexure II: Organisational Structure

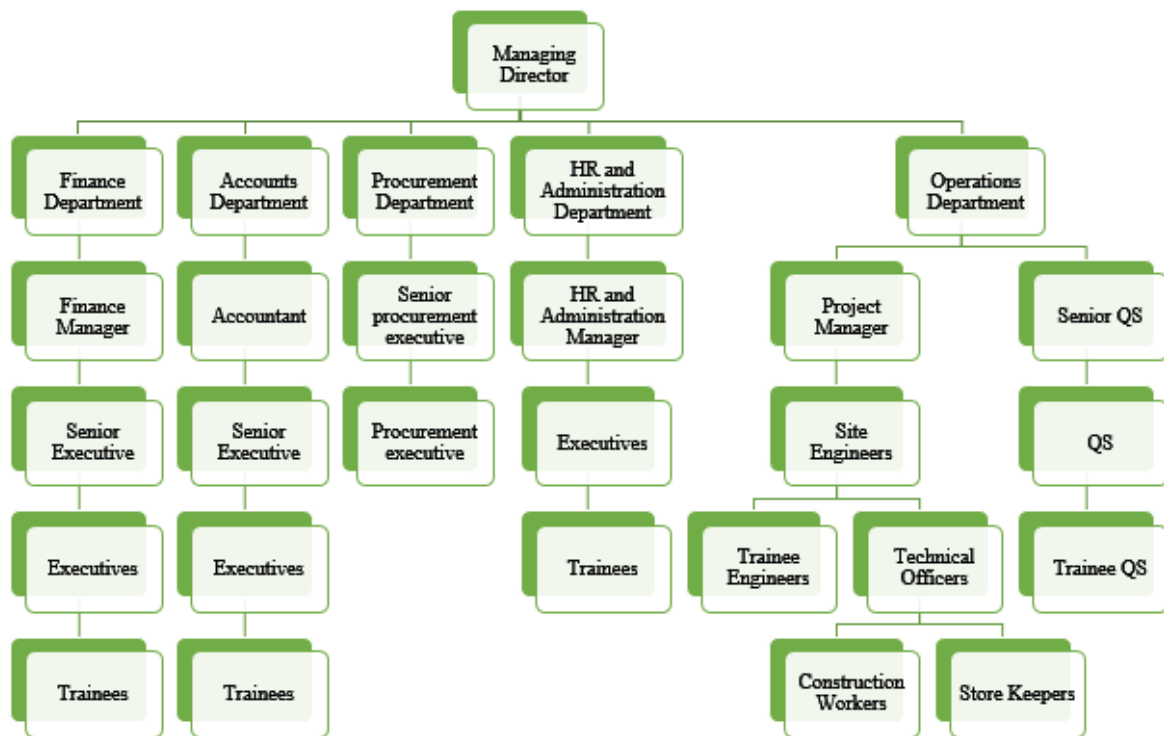


Figure AII.1: Organisational Structure

Source: Primary data.

Annexure III: Cash Conversion Cycle Calculation

Table AIII.1: Cash Conversion Cycle Calculation for R.E. Weerakoon Constructions

Elements	2017/18	2018/19	2019/20	2020/21	2021/22	Company Average
Inventory Days						
Opening Inventory	1,385,884	1,551,783	1,802,373	2,096,139	2,015,937	
Closing Inventory	1,551,783	1,802,373	2,096,139	2,015,937	2,378,588	
Average Inventory	1,468,834	1,677,078	1,949,256	2,056,038	2,197,263	
COS	11,327,400	11,849,228	12,403,285	12,292,538	14,914,009	
	47	52	57	61	54	54
Receivable Days						
Opening Receivables	11,140,900	11,342,281	12,025,748	12,904,254	13,036,408	
Closing Receivables	11,342,281	12,025,748	12,904,254	13,036,408	13,168,239	
Average Receivables	11,241,591	11,684,015	12,465,001	12,970,331	13,102,324	
Revenue	27,300,757	27,725,972	26,086,301	24,591,014	26,889,841	
	150	154	174	193	178	170
Payable Days						
Opening Payables	2,654,001	2,494,137	2,616,587	2,944,199	3,363,164	
Closing Payables	2,494,137	2,616,587	2,944,199	3,363,164	3,589,136	
Average Payables	2,574,069	2,555,362	2,780,393	3,153,682	3,476,150	
COS	11,327,400	11,849,228	12,403,285	12,292,538	14,914,009	
	83	79	82	94	85	84
Cash Conversion Cycle						
	115	127	150	160	147	140

Source: Company data.

Table AIII.2: Cash Conversion Cycle Calculation for Access Engineering PLC 9

Elements	2018/2019	2018/2019	2019/2020	2020/2021	2021/2022	C. Average
Inventory Days						
Opening inventory	839,846,530	966,233,113	1,617,677,510	1,439,247,027	2,282,986,302	36
Closing inventory	966,233,113	1,617,677,510	1,439,247,027	2,282,986,302	2,071,828,710	
Average inventory	903,039,822	1,291,955,312	1,528,462,269	1,861,116,665	2,177,407,506	
COS	11,382,941,278	15,126,447,918	16,183,403,341	15,622,640,045	18,944,612,742	
	29	31	34	43	42	
Receivable Days						
Opening receivables	4,297,018,213	5,772,400,376	7,490,671,978	8,743,200,936	10,007,501,087	154
Closing receivables	5,772,400,376	7,490,671,978	8,743,200,936	10,007,501,087	10,170,484,200	
Average Receivables	5,034,709,295	6,631,536,177	8,116,936,457	9,375,351,012	10,088,992,644	
Credit sales	14,786,897,901	17,937,033,665	19,356,105,575	17,917,039,628	21,852,864,677	
	124	135	153	191	169	
Payable Days						
Opening payables	3,610,047,268	5,547,064,752	3,331,035,416	3,469,940,107	4,012,683,917	104
Closing payables	5,547,064,752	3,331,035,416	3,469,940,107	4,012,683,917	6,354,783,373	
Average Payables	4,578,556,010	4,439,050,084	3,400,487,762	3,741,312,012	5,183,733,645	
COS	11,382,941,278	15,126,447,918	16,183,403,341	15,622,640,045	18,944,612,742	
	147	107	77	87	100	
Cash Conversion Cycle						
	6	59	111	147	111	87

Source: Access Engineering PLC (2022, 2021, 2020, 2019, 2018).

Annexure IV: Summary of the Interviews Conducted with Key Stakeholders

Table AIV.1: Summary of the Interviews Conducted with the Managing Director

No	Question	Mr R.E. Weerakoon
1	How many years since this business started?	I started the business in 1998 as a government building construction company, so we have been in the industry for over 20 years.
2	What are the main issues faced by REWC currently?	There are several issues we are facing now. But the most significant one is difficulties in managing working capital. Due to this, we have to seek finance from banks and other financial institutes.
3	What do you think is the reason to seek finance from external parties?	We have delays in receiving the payment from clients. Clients delay the payments even after completing the project stages and submitting the invoices. Sometimes they take more time to release the retention than the contract retention period.
4	What do you think is the reason for delays in collecting the receivables?	Even though we do a receivable ageing analysis, no proper follow-up is done with telephone calls, emails or by post. And also, sometimes there are delays in certifying the invoices by the client, and ongoing project rework sometimes delays the retention payment.
5	Do you have any suggestions to overcome this situation?	I there's a requirement of training for employees. Introducing new processes also required as a standard method will assist in overcoming most of the problems here.

Source: Primary data.

Table AIV.2: Summary of the Interviews Conducted with the Accountant

No	Question	Miss. P.A.N. Priyangani
1	How many years have you been working at REWC?	I joined the company in 2002, so it's been 20 years since I joined it.
2	What is the main issue faced by REWC currently?	Currently, we have a serious problem with working capital. Also, the finance cost is rising because of external funding. We have earned a loss for renovation projects in the financial year 2021/22. Even though there is a gross profit, it becomes a net loss after deducting the finance expenses.
3	What is the main reason for the working capital problems faced by REWC?	There are many reasons for working capital problems, from processes to people to everything. This is because of the gap from when we buy inventory, convert it into receivables and then collect cash, which is higher than the time we have to settle the payables. This creates cash flow problems. We have a high cash conversion cycle. This gap is the reason for working capital problems
4	Do you have any suggestions to overcome this situation?	We need processes and documented procedures to minimize errors happening from our end.

Source: Primary data.

Table AIV.3: Summary of the Interviews Conducted with the Site Engineer

No	Question	Mr. Gihan Mahindaratne
1	How many years have you been working at REWC?	I have been working here for six years now.
2	What are the main issues with the inventory at the construction site?	There's a lot of remaining inventory in the construction site stores at the end of the period. Inventory sit idle for long periods.
3	What is the reason for the high closing inventory?	Not all the closing inventory is idle inventory. There is some inventory purchased within 30 days. But there are also inventory older than 30 days and some older than six months. The reason for this is not identifying the required demand. The amount of inventory required for the next task is not compared with the current inventory. There's no flow from when the inventory required for the next task is identified, it is not double-checked with the BOQ and no proper approval process.
4	Do you have any suggestions to overcome this situation?	I suggest introducing inventory planning through an inventory analysis based on the value of the consumption and new process in purchasing. Also, there should be awareness of idle inventory's financial impact.

Source: Primary data.

Table AIV.4: Summary of the Interviews Conducted with the Store Keeper

No	Question	Mr K. Dharmasena
1	How many years have you been working at REWC?	I have worked for the company for more than eight years. But have been assigned as the store keeper for this project for the last six months.
2	What are the main issues with the inventory at the construction site?	There is a variance in the physical inventory compared with the inventory value in the records. Even after deducting the obsolete, damaged inventory and recording errors, there is still a variance. These inventories are missing due to inventory theft.
3	What are the reasons for inventory theft?	No proper inventory count happens when the inventory is delivered to the construction site by the supplier. Only the amount delivered and the quantity ordered is the amount that we know. All the inventory is locked inside, but sometimes I must go to the other construction sites. In such situations, I have to hand over the warehouse keys to the head of labour, who needs inventory for construction work. Inventories might be missing in such situations, but there's no one to take responsibility for.
4	Do you have any suggestions to overcome this situation?	CCTV cameras are not covering the area enough. Also, if there are periodic inventory counts, missing inventory can be identified earlier, and the person responsible for the act can be found. Other than that, if there are good policies and procedures from when inventory is received to inventory issues, it would be much easier to avoid such situations.

Source: Primary data.

Table AIV.5: Summary of the Interviews Conducted with the Quantity Surveyor

No	Question	Miss Gayanjalie Peiris
1	How many years have you been working at REWC?	I have been working here since 2017.
2	What are the main issues with receivables in renovation projects?	The target of our company is to submit an invoice once a month. When we submit an invoice, it is certified by the client's engineer within 14 days and should be sent to the client within the next 21 days. Therefore, the client should pay the contractor within 35 days. However, this period exceeds when there are errors in the bills submitted, as the client withholds the payment till, we correct the errors and resubmit the invoices.
3	What are the reasons for errors in invoices?	There are often errors like missing documents to support the invoices. Sometimes even though the measurements are correct, the value of some completed work is omitted, and we have to wait till the next invoice to include the measurement and convert the work in progress to a receivable.
4	Do you have any suggestions to overcome this situation?	I think proper training should be introduced to the QS as some are properly qualified yet inexperienced. And also, if a senior QS checks the invoices before submitting would help to overcome the errors. Other than a checklist to avoid missing supporting documents will help to get the payment from the client without any delays.

Source: Primary data.

Table AIV.6: Summary of the Interviews Conducted with the Finance Manager

No	Question	Mr K.P.J. Dalpadathu
1	How many years have you been working at REWC?	I have been working here for ten years now.
2	What are the main issues with receivables in renovation projects?	Normally the project period of renovation projects is much lesser when compared to the project period of a new building construction project or extension building project. But the clients are taking longer periods than the provided credit period. Because of this, we currently have overdue receivables older than six months. This is incorrect, as most of the clients of renovation projects are private and semi-government clients because their payments are not problematic as payments from government projects.
3	What are the reasons for overdue receivables?	The main reason is not following clients formally. We currently do a receivable ageing analysis, but there are no formal follow-ups. This is because there's no specific document for credit recovery.
4	Do you have any suggestions to overcome this situation?	If the reminders are made before the end of provided credit period is useful in collecting the money soon. If a credit policy is introduced, there will be a standard process for recovering receivables as well, as there will be responsible personnel for the same with mentioned periods, and actions need to be taken if the payments are not paid within the credit period.

Source: Primary data.

Table AIV.7: Summary of the Interviews Conducted with the Project Manager

No	Question	Mr. S.H.B. Egodawela
1	How many years have you been working at REWC?	I have been working here for over 15 years.
2	What are the main issues with receivables in renovation projects?	As the accountant and the QS have provided information related to receivables, let me give you an insight from the project management angle. There are situations where we had to put much effort into releasing the retention. I have to say there are errors from the clients, but those are things beyond our control. However, we can rectify the errors which are happening from our end.
3	What are the reasons for delays in releasing the retention from our end?	When a retention period is given, we are bound to rectify the defects occurring. It's like a warranty; if there are no defects, we can release the retention. But when there are defects within the period, we must rework from our pockets and release the retention amount. It is most of the time 10% of the contract value. The reason for defects is poor project planning, as there are no adequate project controls.
4	Do you have any suggestions to overcome this situation?	I think it is better to introduce a WBS with a checklist, so every task is clearly understood and not missed. Introducing quality measures, check-ups, and proper documentation communicated to the project team would help minimize project defects.

Source: Primary data.

Table AIV.8: Summary of the Interviews Conducted with the Senior Procurement Executive

No	Question	Mrs Achini Gunasinghe
1	How many years have you been working at REWC?	I have been working here since 2017, over five years now.
2	What are the main issues with payables in renovation projects?	Payables mostly consist of suppliers of the inventory to the construction sites. Most of these suppliers provide short credit, so settling within such a short period is difficult. But we somehow get funds externally and settle as there are penalties for late payments.
3	What are the reasons for an Early payment to suppliers?	We cannot change their credit periods. Now we have several new suppliers with short payment periods, and as per my understanding, there are loopholes in supplier evaluation. Due to that, we have poor supplier selection. We have not built proper long-term relationships with suppliers other than with very few.
4	Do you have any suggestions to overcome this situation?	The current supplier evaluation process has considered very limited supplier evaluation criteria such as price and quality and has omitted important criteria like the credit period. Therefore, a new supplier evaluation process with broad criteria should be introduced. Further, we used to have suppliers with much longer credit periods. I think it is time to consider building long-term relationships with suppliers again.

Source: Primary data.

Table AIV.9: Summary of the Interviews Conducted with the Accounts Executive

No	Question	Mrs Dinusha Lakmali
1	How many years have you been working at REWC?	I have been working here for three years now.
2	What are the main issues with payables in renovation projects?	In the past, we have taken time exceeding the credit period to settle the payments. Due to this, later suppliers-imposed penalties to avoid late payments. And some of the suppliers are not providing us with inventory anymore. Therefore, we try our best to settle early for payables to maintain supplier relationships. But in most cases, we settle early because of the wrong entries, mixing the credit period, mistakenly paying for the wrong supplier, and so on.
3	What do you think is the reason for these errors made in entries?	With time we have to change. Even though the company is a private limited company, it is a family-owned business. There are no proper processes, and the company is still using traditional methods. We still use manual methods in making the entries. So, there will be errors for sure. Most entries are made in Microsoft Excel, which takes more time and effort.
4	Do you have any suggestions to overcome this situation?	I think we should use new accounting software to manage payables. If we get an accounting software like QuickBooks, it would be user-friendly, less costly and help avoid errors like this in the future.

Source: Primary data.

Annexure V: Ranked Journals Used

Table AV.1: Ranked Journals Used

Name of the journal	Rank	Number of articles used
RAND Journal of Economics	A*	1
Decision Support Systems	A*	1
International Journal of Production Economics	A*	1
Journal of Construction Engineering and Management	A*	2
Management Science	A*	2
European Journal of Operational Research	A*	1
Journal of Accounting Research	A*	1
Automation in Construction	A*	1
Supply Chain Management: An International Journal	A	1
International Journal of Operations and Production Management	A	1
international Journal of Project Management	A	1
International Review of Economics and Finance	A	1
Financial Management	A	7
Construction Management and Economics	A	3
Engineering, Construction and Architectural Management	A	1
Manufacturing and Service Operations Management	A	2
International Journal of Production Research	A	3
International Journal of Physical Distribution and Logistics Management	A	1
Business Process Management Journal	B	1
International. Journal of Electronic Finance	B	1
Managerial Finance	B	1
International Journal of Productivity and Performance Management	B	1
Journal of Multinational Financial Management	B	1
Journal of Small Business and Entrepreneurship	C	1
International Journal of Services and Operations Management	C	2
Investment Management and Financial Innovations	C	1
Journal of Business and Management	C	1
International Journal of Economics and Financial Issues	C	1
Total		41

Source: Author.

Annexure VI: Project Objective Setting

Table AVI.1: Project Objective Calculation

Project Components	Company Average	Company Benchmark	Performance Gap
Cash conversion cycle			38%
Idle inventory	54%	5%	90%
Overdue receivables	38%	20%	47%
Early payment to payables	41%	10%	76%

Source: Company data.

*160 days is the highest CCC in renovation projects at REWC in the past five years.

Annexure VII: Proposed Inventory Management Process

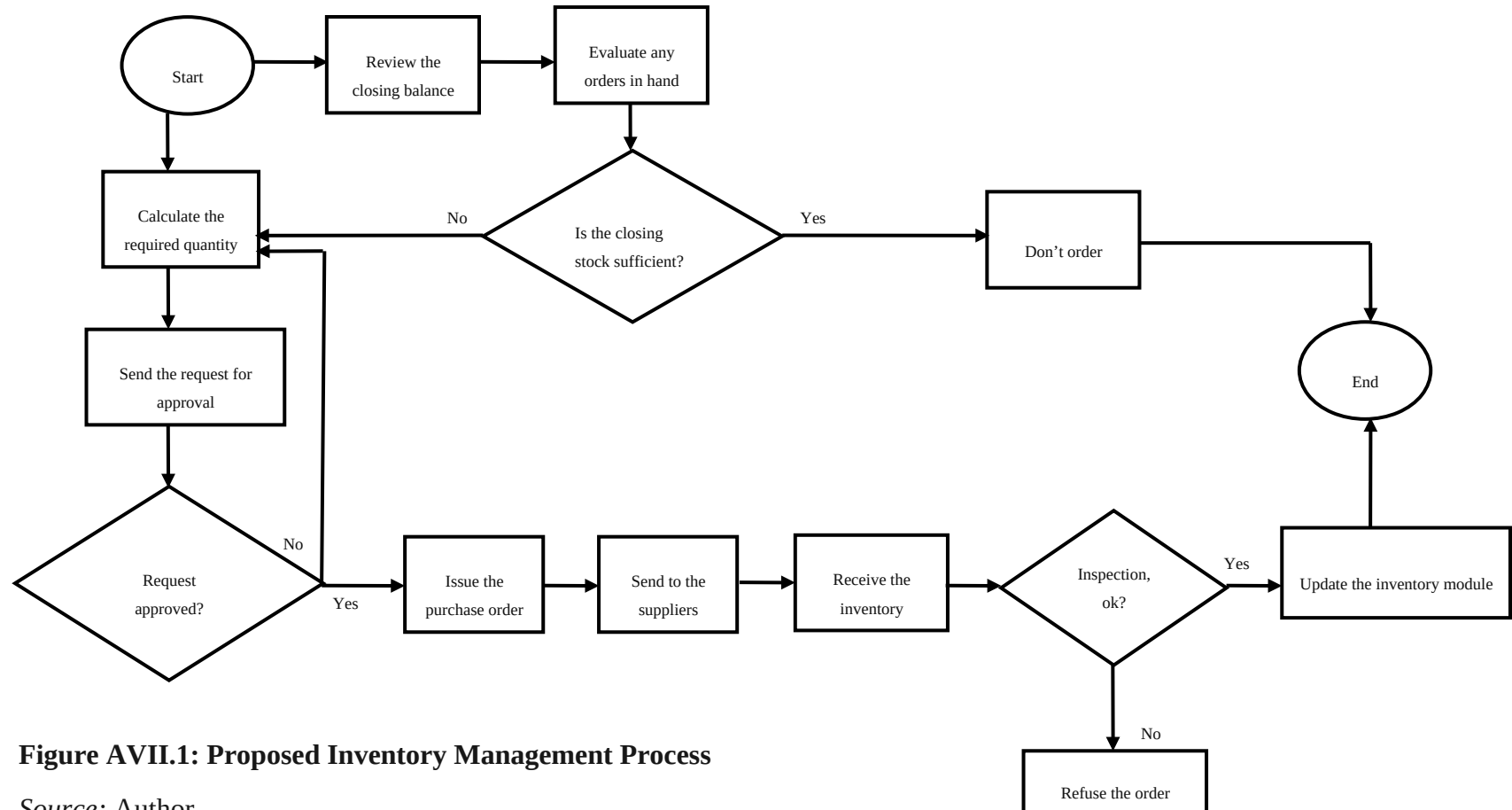


Figure AVII.1: Proposed Inventory Management Process

Source: Author.

Annexure VIII: Proposed Daily Project Report Format

REWC		<u>DAILY PROJECT REPORT</u>																			
PROJECT INFORMATION																					
Project /Section																					
Contract No																					
Date																					
Subcontractor		Time Duration		Site Engineer	Q.s.	T.O.	Supervisors	Trainee	Masons	Carpenters	Bar benders	Plumbers	Welders	painters	Unskilled	Store Keeper	Driver	Personnel			
		From	To																		
Site Staff		8.00	5.00	1	1	1	1	2													
Company															2	1	1				
Sub Name:									4	2	2	0	0	0	5						
1 Manju																					
2																					
Yesterday Planned	Location	Work Plan											Unit		Qty						
1	Stair Case (1 st Flight,2 nd Flight)	Sides of risers in 1 st Flight and 2 nd Flight											m		22						

2							
Today Planned	Location	Work Plan		Unit	Qty		
1	Stair Case (1 st Flight, 2 nd Flight)	Sides of risers in 1 st Flight and 2 nd Flight		m	22		
2							
Tomorrow Planned	Location	Work Plan		Unit	Qty		
1	2 nd -floor Slab	Curing Work					
2							
Material Stock (IN)				Material Stock (OUT)			
Material		QTY		Material		QTY	
Cement Bags		50 Bags		Plywood Sheets		5 Nos	
						Balance	
						50+10=60	
						30-5=25	
Labour Expenses (Budgeted)							
Contractor		Contract/sub		Company		Total Amount	
Name		Skill	U/Skill	Skill	U/Skill		
01) Manju		8×1500	5×1000	0×2000	3×1500	21,500.00	
02)							
Prepared by (Name and Designation) Signature							

Figure AVIII.1: Proposed Daily Project Report

Source: Author.

Annexure IX: Proposed Workshop to Introduce ABC Analysis

a) Identify the need for ABC analysis

The resources that a company uses for production or sales are known as inventory. The coordination of inventory including its availability, management, utilization, and acquisition is called inventory control. The objective of inventory control is to ensure the appropriate inventory is present at the right location, at the right time, and in the correct quantity. This requires a series of interrelated activities that align with an organisation's manufacturing operations. For the construction project, the project creates an inventory management system. The project is required to outline the issues businesses are having with their inventory needs and to offer effective inventory management. The following problems occurred in R.E.Weerakoon Constructions related to inventory.

- i. Not identifying the required demand resulting over purchasing.
- ii. Over purchasing resulting idle inventory.
- iii. Wrong store practices result in inventory theft.

b) The objective of the Workshop

To save cost and time and free up management to concentrate on the materials with the highest value.

c) Overview of the Workshop

The author suggests conducting a one-day workshop to address the identified problems related to inventory management. A suitable and qualified internal facilitator is expected to conduct the workshop for the relevant construction sites and head office employees. To enhance the workshop's effectiveness, it is expected to conduct a multimedia presentation on ABC analysis, discussions and group activities to achieve its objectives. Senior employees will be appointed as mentors and will provide the learnings gained from the workshop to other employees, such as storekeepers and security guards in Sinhala.

Table AIX.1: Summary of the Proposed Workshop

Delivery Method	Classroom
Workshop Duration	06 hours
Workshop Time	9.00 a.m. – 4.00 p.m.
Facilitator Type	Internal
Workshop Room Set Up	Conference
Maximum number of participants	15 participants
Workshop Fee	Company sponsored
Workshop supporting materials	PowerPoint presentation, White board, projector, pens, A4 sheets, plastic file.

Source: Author.

Table IX.2: Schedule for the Workshop

Time	Media	Description
9.00-9.05	-	Introduction to the objective of the workshop
9.05-9.15	-	Raise questions on the current situation of the issues related to inventory management.
9.15-9.17	PPT	Contents of the workshop
9.17-9.20	PPT	Introduction to inventory
9.20-9.25	PPT	Need for inventory Management
9.25-9.35	PPT	What is ABC Analysis?
9.35-9.40	PPT	Importance of ABC Analysis?
9.40-10.00	PPT	Steps in implementing ABC Analysis
10.00-10.15	Tutorial	Example calculation
10.15-10.20	PPT	Advantages
10.20-10.25	PPT	Limitations
10.25-10.30	PPT	Conclusion
10.30-10.45	PPT	Questions
10.45-11.00	-	Tea Break

11.00-12.30	Group Activity	Providing an opportunity to conduct an ABC analysis on inventory for a specific renovation project. This task shows a comprehension of the information acquired during the workshop. I. Explanation II. Instructions for the activity III. Allow time to ask question and clarify problems IV. An opportunity to explain the analysis conducted on each construction site and what details it implies on the inventory.
12.30-12.45	-	Summary of the key learning areas and Questions and Answers session.
12.45-1.00	-	Wrap up Review the objectives of the workshop. Review what was learned and the supporting documents provided. Appointing mentors to distribute the learnings to other employees. Closing remarks. Thank learners. Provide and collect feedback forms.
1.00-2.00	-	Lunch

Source: Author.

d) Recommended Facilitator Profile

The facilitator should have the following credentials or equal,

- i. An experienced and qualified employee in the accounting field.
- ii. Experience in inventory management.
- iii. Knowledge of the current situation of the R.E. Weerakoon Constructions.
- iv. Capacity to communicate and conduct a workshop effectively.

e) Feedback Form

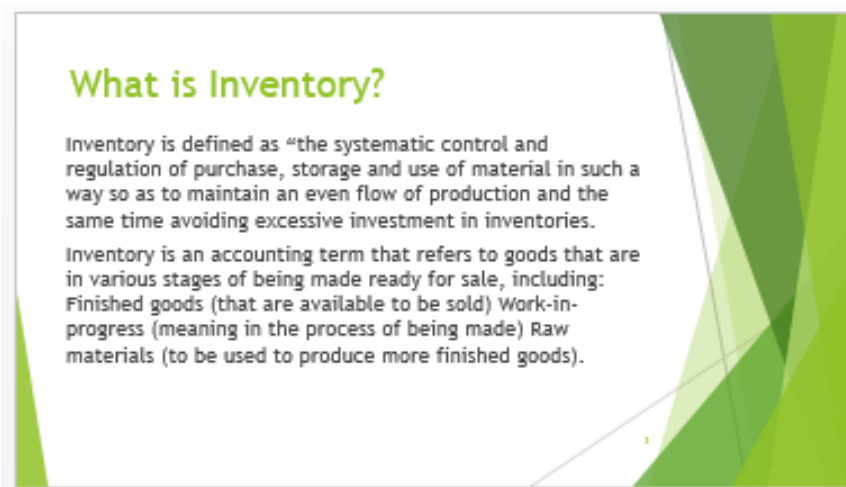
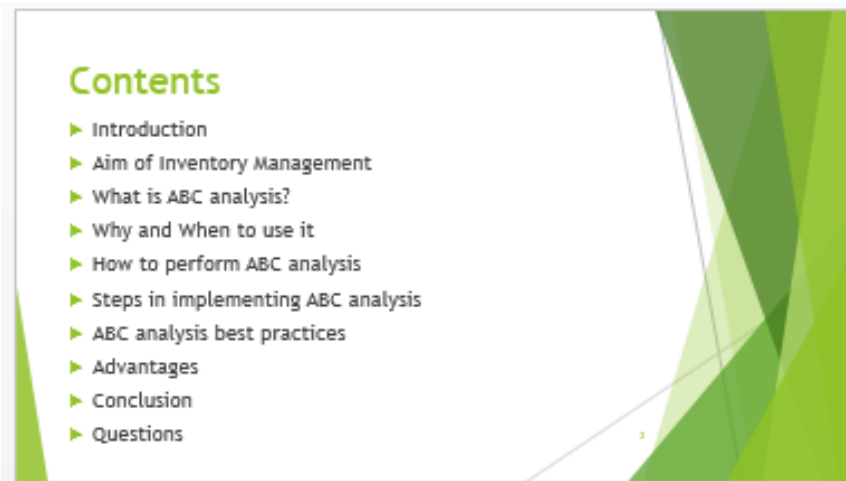
Feedback Form					
Date					
Workshop	Introducing ABC Analysis to manage inventories at R.E. Weerakoon Constructions				
Name (Optional)					
Facilitator					
Please help us to improve our offerings and service by indicating your impression of the items listed below. 5 Strongly Agree 4 Agree 3 Neutral 2 Disagree 1 Strongly Agree					
Curriculum	1	2	3	4	5
1. The subject matter was informative					
2. Workshop materials were organized and easy to follow					
3. The workshop objectives were communicated and achieved					
Facilitator					
1. The facilitator was knowledgeable					
2. The audio-visual aids were effectively utilized to improve the instruction.					
3. Class participation and group interaction were encouraged.					

4. The facilitator did a perfect job of teaching the material.					
5. Adequate time was provided for attendee questions					
6. The facilitator showed respect for the participants					
Procedure and Communication					
1. I knew about the class in advance					
2. I was allowed adequate time for breaks					
3. Temperature, lighting, and the seats of the classroom were arranged well.					
Overall Impression					
1. The knowledge gained will support in my job and the accomplishment of my individual goals.					
2. Others will benefit from my knowledge					
3. The workshop was worthwhile					
What is the most interesting thing about this workshop?					
What is the least interesting thing about this workshop?					
What other areas who you recommend conducting workshops on?					
Any additional comments?					

Figure IX.1: Evaluation Form

Source: Author.

f) Proposed Contents for the PowerPoint Presentation



Aim of Inventory Management

- ▶ The right quality
- ▶ The right quantity of supplies
- ▶ The right source
- ▶ At the right time
- ▶ At the right place
- ▶ For the right cost

What is ABC Analysis?

- ▶ ABC means Always Better Control
- ▶ This is based on cost criteria. It helps to exercise selective control when confronted with large number of items, rationalizes the number of orders, number of items and reduce the inventory.
- ▶ ABC analysis divides an inventory into three categories—"A items" with very tight control and accurate records, "B items" with less tightly controlled and good records, and "C items" with the simplest controls possible and minimal records.

Category 'A' Items

- ▶ The most important items of the company.
- ▶ Accounts only for 10% to 25% of the total inventory items.
- ▶ Accounts only for 70% to 80% of the annual consumption value of the company which is also the highest value.
- ▶ Have very tight inventory control.

Category 'B' Items

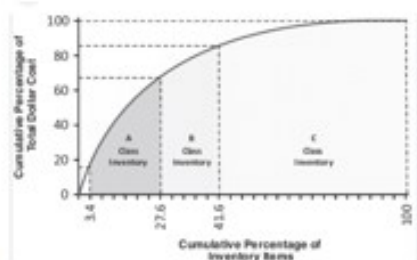
- ▶ Less important items than 'A' items but more important than the 'C' items of the company.
- ▶ Accounts for 25% to 30% of the total inventory items.
- ▶ Accounts for 15% to 25% of the annual consumption value of the company.
- ▶ Have intermediate inventory control.

Category 'C' Items

- ▶ Marginally important items of the company.
- ▶ Accounts only for 45% to 50% of the total inventory items.
- ▶ Accounts for 5% to 10% of the annual consumption value of the company.
- ▶ Have low inventory control.

Philosophy Behind

- ▶ 20% effort
- ▶ 80% results



When and Why to Use It?

- ▶ The ABC analysis can be also be useful in several steps of the sourcing value chain. Demand identification, to gather the annual spend on the site. Opportunity assessment to identify the leveraging opportunities.
- ▶ The ABC tool is used to identify the vital few from the trivial many according to defined set of criteria.
- ▶ Different decisions may be taken from the result of ABC analysis.

ABC Classification of Inventory Items

- ▶ A items: 70% of the annual consumption of inventory is covered by only 10% of the items in the inventory, deserve highest attention.
- ▶ B items: 20% of the items covering 20% of the inventory investment (less stringent control).
- ▶ C items: Rest 10% of the inventory items (very little control)

Particulars	A Class Items	B Class Items	C Class Items
Control	High	Intermediate	Low
Requirement	Low	Intermediate	High
Check	Tight	Intermediate	Low
Safety Check	High	Low	Rare

How to Perform ABC Analysis

- ▶ Identify the Objective
- ▶ Collect Data
- ▶ Sort by Decreasing Order of Impact
- ▶ Calculate the Revenue Impact
- ▶ Sort Items in to Buy Classes
- ▶ Analyze classes

Steps for Implementing ABC Analysis

- ▶ Classify the items of inventories determining the expected use in its units and the price per unit for each item.
- ▶ Determine the total value for each item by multiplying the expected units by its unit price.
- ▶ Rank the items in accordance with the total value, giving first rank to the items with highest total value and so on.
- ▶ Compute the ratio (percentages) of number of units of each item to total units of all items and the ratio of total value of each item to total value of all items.
- ▶ Combine items on the basis of their relative value to form three categories A,B, and C.

13

Steps for Implementing ABC Analysis

- ▶ Classify the items of inventories determining the expected use in its units and the price per unit for each item.

No	Item	Annual Demand	Cost per Unit
1			
2			
3			
4			
5			
6			
7			

14

Steps for Implementing ABC Analysis

- ▶ Determine the total value for each item by multiplying the expected units by its unit price and rank the items in accordance with the total value, giving first rank to the items with highest total value and so on.

No	Item	Annual Demand	Cost per Unit	Annual Cost
1				
2				
3				
4				
5				
6				
7				

15

Steps for Implementing ABC Analysis

- ▶ Compute the ratio (percentages) of number of units of each item to total units of all items and the ratio of total value of each item to total value of all items.

No	Item	Annual Demand	Cost per Unit	Annual Cost	Annual usage %
1					
2					
3					
4					
5					
6					
7					

Steps for Implementing ABC Analysis

- ▶ Combine items on the basis of their relative value to form three categories A,B, and C.

No	Item	Annual Demand	Cost per Unit	Annual Cost	Annual Usage %	Category
1						
2						
3						
4						
5						
6						
7						

ABC Analysis Best Practices

- ▶ Keep Classification Simple
- ▶ Assign Service and Labor Levels at the same time
- ▶ Segment KPIs by time
- ▶ Establish perform reviews
- ▶ Manage across locations
- ▶ Count inventory in transit
- ▶ Reclassify purposefully
- ▶ Consider sales and inventory in Tandem
- ▶ Leverage technology and resulting data.



Figure IX.2: Proposed Contents for the PowerPoint Presentation

Source: Author.

g) Example Calculation

Table IX.3: ABC Analysis Conducted on Project Renovation of the Roof at Youth Centre Maharagama

No	Item	Annual Demand	Cost per Unit	Annual Cost	Annual Usage %	Cumulative Annual %	Category
1	Box Bars Ms(Mild steel) 6" x 3"	415	17,825	7,3 97,375	35.58%	35.58%	A
2	BNA Cold Mix 1Ton	534	12,000	6,4 08,000	30.82%	66.40%	A
3	GI Pipes 4"	95	21,275	2,0 21,125	9.72%	76.12%	A
4	GI Pipes 8"	24	55,200	1,3 24,800	6.37%	82.49%	B
5	Plywood	200	5,880	1,1 76,000	5.66%	88.15%	B
6	Bricks	20000	38	7 60,000	3.66%	91.81%	B
7	Cement	100	3,000	3 00,000	1.44%	93.25%	C
8	Orange X5 13A Twin S/Socket Outlet	100	2,570	2 57,000	1.24%	94.49%	C
9	ACL Telephone Wire (Cable) 1 Pair Bobbin 500m	10	15,250	1 52,500	0.73%	95.22%	C
10	Sand	8	15,200	1 21,600	0.58%	95.80%	C
11	BNA Bitumen Dam Proof Coat	250	435	1 08,750	0.52%	96.32%	C
12	Cement blocks	1000	105	1 05,000	0.51%	96.83%	C
13	National PVC Pressure Pipes PE - 90MM(3") - 280MM(10")	3	35,646	1 06,938	0.51%	97.34%	C
14	National PVC Pressure Pipes PE - 90MM(3") - 280MM(10")	2	43,146	86,292	0.42%	97.76%	C
15	Filler 20ltr	50	1,379	68,950	0.33%	98.09%	C
16	Red Soil (1 Cube)	5	12,000	60,000	0.29%	98.38%	C
17	Orange X5 2 Way 4 Gang 10A Switch	18	2,420	43,560	0.21%	98.59%	C
18	Lanwa - Square Tube (30MM x 30MM) GI	23	1,699	39,077	0.19%	98.78%	C
19	Safety Nets	20	1,859	37,180	0.18%	98.96%	C
20	Waterproofing Brilliant White	10	3,535	35,350	0.17%	99.13%	C

21	Polycon Corrugated Conduit 15M	10	2,505	25,050	0.12%	99.25%	C
22	Binding	25	850	21,250	0.10%	99.35%	C
23	6" x 4" Metal [Cube 1]	1	20,000	20,000	0.10%	99.45%	C
24	Orange X5 2 Way 4 Gang 10A Switch	8	2,360	18,880	0.09%	99.54%	C
25	GI Pipes 3/4"	5	2,300	11,500	0.06%	99.60%	C
26	Orange X5 2 Way 4 Gang 10A Switch	7	1,840	12,880	0.06%	99.66%	C
27	Anton Clamp Saddle (63 x 20)	6	1,273	7,638	0.04%	99.70%	C
28	Alkali Paint 20L	8	967	7,736	0.04%	99.74%	C
29	Toa Spray - Heat Resistant Silicon Spray	5	1,680	8,400	0.04%	99.78%	C
30	Anton Trunking (50 x 12.5 mm)	5	1,822	9,110	0.04%	99.82%	C
31	Lanka Tile Grout 1kg	30	254	7,620	0.04%	99.86%	C
32	Anton Clamp Saddle (50 x 20)	6	1,138	6,828	0.03%	99.89%	C
33	Nut and Bolts	553	10	5,464	0.03%	99.92%	C
34	Anchor Bolt (Brass) - 10mm x 75	95	60	5,700	0.03%	99.95%	C
35	Binding Wire	5	850	4,350	0.02%	99.97%	C
36	8" Amano Cladding Sheets	7	400	2,800	0.01%	99.98%	C
37	Hand Axe	1	1,405	1,405	0.01%	99.99%	C
38	Drywall Screw 8 100g	16	130	2,080	0.01%	100.00%	C
39	Anton Socket (25 MM x 20 MM)	15	95	1,418	0.01%	100.01%	C
40	Water Seal Padlow Cement 1kg	5	190	950	0.00%	100.01%	C
41	Seamless Diamond Disc	1	967	967	0.00%	100.01%	C

Source: Company data.

Annexure X: Proposed Store Policy

1. Introduction

1.1 Scope

- 1.1.1** Brief store policies used by R.E. Weerakoon Constructions will be revised by introducing new policies and procedure manuals to manage the inventory adequately.
- 1.1.2** The contents of this manual are confidential and are only for the benefit of employees of R.E. Weerakoon Constructions.
- 1.1.3** This manual will begin on the date determined and approved by the Managing Director, and that date will also be used for any upcoming revisions.

1.2 Mission, Objectives and role of the manual

- 1.2.1** Describe the major store policies and procedures regarding the physical count and verification of inventory, as well as the transfer of obsolete and slow-moving inventory.
- 1.2.2** Instruct the employees in charge of handling and documenting inventory transactions.
- 1.2.3** Ensure that the rules established for the stores' section are applied consistently and uniformly.

2. Stock Level Maintenance

- 2.1** To guarantee that stock is always available, stock levels are tracked, maintained, and stock requests are made:
 - i. The stock controller or designated official must print out a weekly report detailing stock items outside the minimum and maximum levels.
 - ii. The report alerts the stock controller to the items that have run out and need to be reordered.

3. Review of stock levels

- 3.1** It is essential for construction sites to regularly assess their inventory based on data such as the amount of stock distributed, the level of service achieved, the total worth of the inventory held, and the rate of inventory turnover, in order to efficiently manage the store's operations.
- 3.2** To maintain a precise minimum, maximum, and reorder levels and guarantee the availability of stock/materials at all times for the end-users to meet their established goals or targets, stock-level evaluations are undertaken with strategic business units once a year.

4. Obsolete and Slow/ Non-Moving Inventory

- 4.1** Inventory must be examined every three months using reports from the inventory records to spot any slow-moving inventory.
- 4.2** The storekeeper and site engineers must classify inventory as slow-moving and obsolete. This classification is based on the item's condition and shelf life.
- 4.3** Before being entered into the books of accounts, the accountant and manager of finance must assess and approve the provision for obsolete and slow-moving inventory under R.E. Weerakoon Constructions.
- 4.4** The site engineer of the relevant site must request the inventory write-offs, and finance manager of R.E. Weerakoon Constructions must authorize it.
- 4.5** If one of the following circumstances occurs, inventory must be written off from inventory and accounting records:
- I. Products are damaged.
 - II. Products can't be sold as scrap.
 - III. The item is no longer needed for its intended use and cannot be transferred to another.

5. Receiving the Inventory

- 5.1** A store receipt voucher must be prepared for the reception of inventory at the stores. When receiving an inventory or item, inventory records need to be updated immediately.
- 5.2** After receiving the authorized request, the main warehouse will distribute inventory products to stores in the construction sites.
- 5.3** Store inventory is to be separated into long-term and expense (short-term).
- 5.4** The inventory system must be updated if there is a change in the inventory, including receiving goods returned from other construction sites' stores.
- 5.5** All high-value inventory goods must be protected against loss from theft by insurance.
- 5.6** Inventory cannot be inspected when received, the store keeper must store them in a convenient but safe location for later inspection and stamping the inventory.

6. Rejecting Inventory

- 6.1** Suppose the item is not as described in the purchase order terms, contract's conditions, and specifications. In that case, it must be rejected by the receiving personnel and sent back to the supplier immediately. There are several circumstances in which inventory can and must be rejected, including but not limited to the following:
 - i. Incomplete or insufficient documentation, such as certifications or test results.
 - ii. An excessive or inadequate quantity of inventory.
 - iii. Complete or partial non-compliance with the requirements or quality standard.
 - iv. Delivery delays.
 - v. A deficiency in testing results or certificates.
 - vi. Improper documentation
 - vii. Damaged inventory.

- 6.2 If a rejection or return occurs after a store receipts voucher has been issued, the receiving staff must issue a "return to supplier" form and send it to the procurement department.
- 6.3 The receiving staff must take a photo of the inventory in cases of significant visual damage, especially expensive inventory, before submitting it to the procurement executive.
- 6.4 Rejected inventory must be marked as quarantined and transported to a designated area separated from the regular storage and receiving areas if the inspection occurs after the inventory has been received, i.e., after receipt or during use.
- 6.5 If, as determined by the finance manager, the supplier fails to collect the rejected inventory within a given timeframe, the company reserves the right to sell, auction, or dispose of the inventory.

7. Physical Count Verification

- 7.1 A sample-based physical count and verification of each inventory at the stores must be done every three months. Such counts must cover all inventory at the end of the year.
- 7.2 During inventory counts, the person taking the count should not be responsible for safeguarding those inventories.
- 7.3 At least one accounting team member must be on the count team.
- 7.4 During the physical count, no inventory level of products may be received or transferred. During the physical verification phase, pre-verified inventory stocks for the three consumption halls were set aside based on consumption patterns. The relevant stores must be notified seven days before the physical inventory count.
- 7.5 If an inventory count is complete, the results must be compared to the inventory accounting records. Any discrepancy should be reported to the finance division. Any proven discrepancies must be reported to Finance and Accounts department.

7.6 Before finalizing any adjustment entries, any revisions necessary to inventory records must be approved in accordance with authorised personnel at R.E. Weerakoon Constructions.

7.7 A surprise physical count of inventories must be performed based on the inventory values.

8. Inventory Transfer

8.1 All inventory transfers will be recorded in inventory records the same day the inventory is moved.

8.2 Transportation expenses for intra-company inventory transfers will not raise the value of that inventory.

8.3 All inventory transfers must receive the appropriate authorities' approval by R.E. Weerakoon Constructions authorization.

Annexure XI: Proposed Training for Quantity Surveyors

a) Training Need Analysis

A brief survey was conducted among 12 quantity surveyors to identify the exact training requirements needed to cover the skill gap shown by quantity surveyors. This also covers the priority areas to focus on in providing the training. Responses from the participants were collected and analysed. The responses showed the importance of interim valuations, preparation of interim valuations, processing of variations and extra works, management of claims, prepare the final accounts. The questions and the brief of responses are as follows:

Name: _____

Years of experience?

- ☐ Less than 1 year
- ☐ 1-2 years
- ☐ 2-5 Years
- ☐ More than 5 years

What are the most important areas for a quantity surveyor out of the following? (Rate from 1 to 5, 1 = less likely and 5 = Mostly)

- ☐ Preparation of interim valuations
- ☐ Processing of variations and extra works
- ☐ Management of claims
- ☐ Preparation of final accounts
- ☐ Progress monitoring
- ☐ Documentation and record keeping

What areas would you like to have trainings the most?

- ☐ Preparation of interim valuations
- ☐ Processing of variations and extra works
- ☐ Management of claims
- ☐ Preparation of final accounts
- ☐ Progress monitoring
- ☐ Documentation and record keeping

Figure AXI.1: Questionnaire for Training Need Analysis

Source: Author.

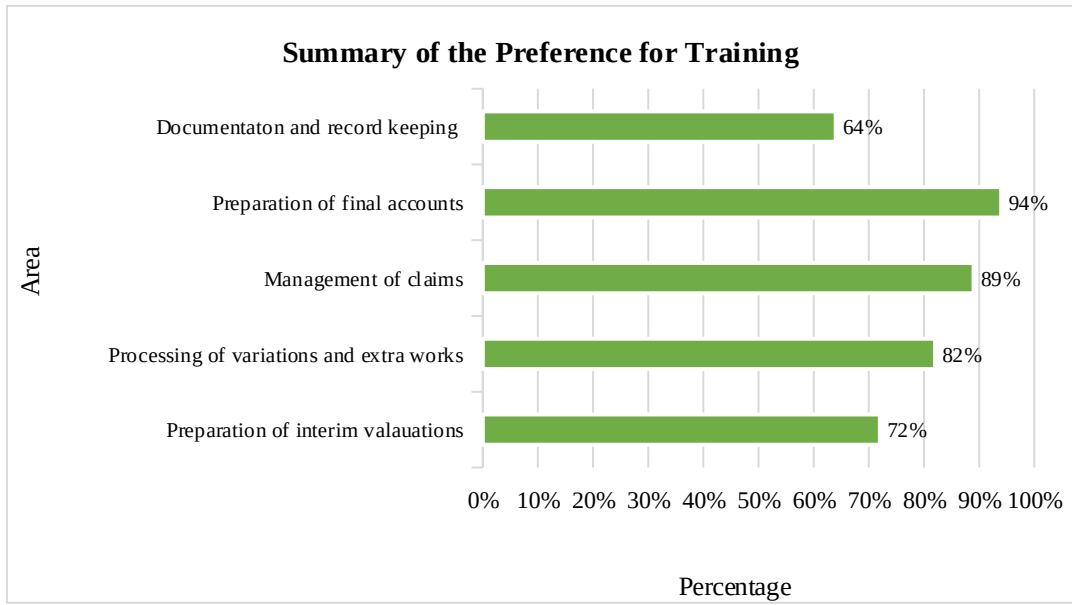


Figure AXI.2: Summary of the Preference for Training

Source: Author.

b) Training Objectives

By the completion of the training program, the attendees will have the ability.

- i. To process variation of extra work by valuing, negotiating, analysing, incorporating and reporting.
- ii. To manage claims by studying the conditions, assessing extension of time, and negotiating claims.
- iii. To prepare final accounts by taking measurements, scrutinizing the final bills, scrutinizing the variations and assessing the price fluctuations.
- iv. To prepare interim valuations by taking on-site measurements, preparing interim bills, assessing material at the site and preparing summary accounts.
- v. To document and record keeping by preparing site status reports, preparing various correspondence, and assisting in documentation and record keeping.

c) Training Program Design

Training Program Overview

The author suggests a two-day training program to address the competency gap. Quantity Surveyors are expected to get classroom training from a qualified trainer, including lectures, group activities, case studies, and discussions. The accomplishment of the predetermined training objectives is supposed to gauge the training program's efficacy. The mentors will be in charge of closely monitoring and coaching their team members. This is done to ensure that even after training, the personnel won't continue using the same old methods. This strategy is anticipated to be used consistently until all quantity surveyors are comfortable with the areas in which they showed weaknesses previously. Following are the necessary data for the training program.

Table AXI.1: Overview of the Training Program

Training Type	Off-the-job training
Delivery Method	Classroom training
Training Duration	14 hours (2days*7 hours + Break)
Training Tie	9.00 a.m. to 5.00 p.m.
Trainer Type	External
Training Room Setup	Classroom
Maximum Number of Participants	12 Participants
Training Fee	Rs. 250,000/- sponsored by the company
Training Supporting Materials	PPT, printed material, papers and pens, projector

Source: Author.

Outline of the Program Contents

1. Interim Valuations (2 hours)
 - a. On-site measurements
 - b. Preparation of interim bills
 - c. Preparation of price fluctuation claims (Application of price fluctuation formulae etc.)
 - d. Assessment of material at a site
 - e. Preparation of a summary of accounts
2. Processing Variations and Extra Work (3 hours)
 - a. Valuing variations
 - b. Negotiation of variations
 - c. Retrieve information such as records, valuation orders, and confirm verbal communications.
 - d. Variation analysis
 - e. Reporting on the impact of variations
 - f. comparative cost studies for variations
3. Management of Claims (3 hours)
 - a. Examining the contract conditions and all other contractual provisions for claims
 - b. Evaluating time extension
 - c. Negotiation claims
4. Preparation of Final Accounts (3 hours)
 - a. Conducting individual and joint measurements
 - b. Understanding variations, additional work, and daywork accounts.
 - c. Finalizing price fluctuations and other contractual claims.
 - d. Preparing final bills, including preliminary expenses, prime costs, and provisional sums.
5. Documentation and Record Keeping (1.5 hours)
 - a. Preparation of site status report
 - b. Preparation of various correspondence

- c. Assist in site documentation and record keeping
- 6. Revision (1.5 hours)

d) Evaluation of the Training Program

The actual evaluation of the training programs will be conducted using the evaluation criteria listed below.

Training Evaluation

Participants would be requested to provide their honest opinion on a training evaluation form to assess how they "Feel" about the experience. Before the next batch of trainees receives instruction, the training program will be revised if the feedback suggests that modifications to the course materials, method of delivery, or facilitator are necessary. The training assessment form is provided below.

Date:	Training Name:				
Name (Optional):	Trainer:				
1= Strongly Disagree 2= Disagree 3= Neutral 4= Agree 5= Strongly agree					
Tick the most appropriate box	1	2	3	4	5
Overall rating of the quality of the training is satisfactory.					
The presenter clearly stated the objectives.					
Overall rating of the presenter is satisfactory.					
Participation and interaction were encouraged.					
The topics covered were relevant to me.					
The content was organized and easy to follow.					
The material distributed was helpful.					
This training experience would be helpful in my work.					
The trainer was knowledgeable about the topics.					
The trainer was well prepared.					
The training objectives were met.					
The time allocated for the training was sufficient.					
The location and other facilities were satisfactory.					
What was the most interesting thing you learnt?					
What was the least interesting thing you learnt?					
What would have made session more effective?					
How the knowledge and skills learnt in training would be applied in your job? (mention at least one item)					

Figure: AXI.3: Training Evaluation Form

Source: Author.

Annexure XII: Proposed Receivable Policy

a) Developing a Receivable Policy

1.0 Objective

The implementation of a credit management policy for R.E. Weerakoon Constructions has been strategically formulated to ensure strict adherence to the contractual obligations associated with credit-based customer relationships, thereby promoting the sustainability of payment recovery. This policy is instrumental in reinforcing the company's liquidity position and expanding its client segment, thus fostering long-term financial resilience.

2.0 Scope

Business Type: Construction Projects

Primary Operations: Renovation of buildings, encompassing both commercial and residential properties

Geographical Scope: Nationwide presence, serving diverse locations within Sri Lanka

3.0 Policy

Upon meeting the predefined eligibility criteria, clients who satisfy the threshold requirements set by the company will be eligible for credit-based facilities. The credit limit and advance payment requirements for each client are specified in Table XIV.1. It is mandatory for all clients to provide an advance payment before any order can be processed, based on their respective section.

In order to facilitate an immediate transition to cash-based operations, the advance payments received from clients must possess high liquidity. However, for high-value projects, the deposit should be in the form of a bank guarantee, with the amount and duration of the guarantee contingent upon the settlement procedures of the respective clients.

Effective collaboration between all relevant stakeholders is vital, and prior scheduled meetings should be conducted to negotiate and establish a formal agreement that strictly adheres to the specified criteria. Under certain circumstances, the maximum credit limit for a particular client category may be augmented.

All payment obligations must be fulfilled within 35 days from the date of invoicing. The finance team assumes responsibility for diligently monitoring client behaviour during order processing and conducting annual due diligence investigations to evaluate their creditworthiness.

Table AXII.1: Credit Limit and Advance Payments

Creditworthiness	Past settlement pattern	Advance payment
Low	More than 90 days	30%
Moderate	Settled 35-90 days	20%
High	Settled with 35 days	15%

Source: Author.

*If the client's current ratio is less than 1:1, the range also applies to the "low" category.

4.0 Key Roles and Responsibilities

- i. Commencing on the 35th day, the finance team assumes the responsibility of monitoring client payments in a diligent manner.
- ii. The finance manager is entrusted with the task of issuing monthly statements to clients who have outstanding dues to the organization.
- iii. Credit approvals and order escalation protocols are outlined as follows:
- iv. For amounts ranging between Rs. 1,000,001 and Rs. 2,500,000, the authority lies with the Finance Manager.
- v. For amounts exceeding Rs. 2,500,001, the approval rests with the Managing Director.
- vi. In order to provide management with performance updates, the finance manager must generate a periodic report on the client's performance during the first week of each consecutive month.

- vii. Prior to processing any order, it is imperative to input the client's essential information into the system.
- viii. The Finance team holds the responsibility of regularly notifying the operations department about any outstanding balances, while the operations department is responsible for reminding clients before the expiration of the 56-day grace period.

5.0 Commercial Prospection – Client

The clients of R.E. Weerakoon Constructions are diverse, and the following behavioural outcomes are based on their past performance:

Government Bodies (High Risks) – 45% coverage

The company's current client group accounts for 45% of its business revenue; however, historical data indicates a significantly lower creditworthiness due to delays associated with treasury-related matters. In order to mitigate potential cash flow challenges, it is imperative that the new policy aims to decrease engagement with this client group by a minimum of 50%.

Residential (Moderate Risk) – 35% coverage

The company derives 30% of its revenue from the residential project's customer segment. However, an analysis of historical data reveals a moderate to high credit risk associated with exceeding the prescribed 35-day credit period for this particular segment. To proactively address potential cash flow challenges, it is advisable to implement a reduction of 25% in the engagement with this category.

Mercantile companies (Lower Risk) -20% coverage

Driven by its profit-oriented approach, the company has conducted 20% of its business with this specific customer sector, which has demonstrated a comparatively lower credit risk according to historical data. In order to bolster cash inflows, it is recommended that the new policy focuses on augmenting coverage within this consumer demographic by 40%.

6.0 Quotations/ Bid (Commercial Conditions)

All quotations or bids must adhere to the following criteria:

- i. **Payment Methods:** Acceptable forms of payment include cheques and electronic slip transfers.
- ii. **Guarantee Requirement:** A bank guarantee should be provided as the preferred form of guarantee.
- iii. **Credit Rating Consideration:** Moderate to higher credit ratings are preferred, and any rating falling below the specified threshold must be duly considered for management purposes.

7.0 New Clients

All new clients seeking engagement with R.E. Weerakoon Constructions are subject to comprehensive screenings to assess their creditworthiness. To facilitate this evaluation, the following documents will be thoroughly examined:

- i. **Customer's Purpose of Business:** A clear understanding of the customer's intended business activities.
- ii. **Customer's Structure:** Verification and confirmation of the customer's status as a public sector entity, if applicable.
- iii. **Past Settlement Track Record:** Assessment of the customer's payment history and settlement patterns with other suppliers.
- iv. **Contractual Agreement/Expectation Points:** Review and confirmation of the contractual agreement and mutually agreed-upon expectation points, which require appropriate sign-off.
- v. **Customer Profile:** Essential customer information including name, address, and nature of the business.
- vi. **Copy of Certificate of Incorporation:** Submission of the certificate of incorporation for clients engaged in commercial activities.
- vii. **Authorized Instructor Details:** Documentation of authorized individuals responsible for liaising with R.E. Weerakoon Constructions.
- viii. **Financial Information:** Provision of relevant financial information pertaining to the customer's financial standing and stability.

8.0 Payment Terms

All company-wide transactions with clients necessitate the approval of payment terms. The designated method of payment is through cheques or electronic slip transfers, which are deemed irrevocable. Payment information is conveyed to clients by means of invoicing a bill.

To provide additional assurance, guarantees in the form of bank guarantees or documentary credits are required. These mechanisms serve to enhance financial security and mitigate potential risks.

9.0 Credit Limits

Government Sector:

The government sector constitutes 45% of the annual sales (inclusive of VAT). Stringent due diligence and comprehensive financial analysis must be conducted, with approvals being contingent upon the managing director's decision.

Residential Sector:

The residential sector accounts for 35% of the annual sales (inclusive of VAT). Thorough analysis, encompassing dynamic assessment and financial analysis, must be conducted, and approvals are subject to the managing director's discretion.

Mercantile Sector:

The mercantile sector represents 20% of the annual sales (inclusive of VAT). Existing clients in this sector must undergo due diligence analysis, and approvals rely on the managing director's evaluation.

In the event of a credit limit falling below expectations, and if the business desires to retain the client, a recommendation supported by an explanation can be submitted along with additional documentation. In such cases, the consent of the directors or management is required. The supplementary document may include the following:

- i. Recommended credit limit.
- ii. Client's revenue for the past six months.

- iii. Historical repayment behaviour, considering payments made within 60 days and other relevant factors.

Simultaneously, exceeding the prescribed credit limits should be evaluated through more comprehensive financial analysis, taking into account ratios and cash flows. The following types of clients are not eligible for credit facilities:

- a) Clients on credit hold in accounts receivables.
- b) Industries with high potential financial risks.
- c) Retail clients (individuals making personal purchases).
- d) Clients with incomplete profiles.

The credit limitations may be adjusted periodically based on the repayment performance of R.E. Weerakoon Constructions. The managing director retains the authority to modify credit limits in response to requests from the operations manager.

10.0 Reporting

Table AXII.2: Risk Analysis Category

Credit Limit	Types of Reports
Below and Equal Rs.1,000,000	Ageing reports
Between Rs 1,000,001 – Rs 2,500,000	Ageing reports
Between Rs 2,500,001 – Rs 5000,000	Ageing reports and reasons
Above Rs 5,000,001	Full reports Financial documents Recommendations

Source: Author.

11.0 Friendly Collection

To effectively communicate with relevant clients who have surpassed their credit limits in terms of value or days, the following methods of communication can be employed:

- i. Prepare a debtor's aging report on a monthly basis, providing a comprehensive overview of outstanding debts.
- ii. Assign a relationship officer to categorize and address the overdue payments based on customer segments.
- iii. Initiate follow-ups for overdue payments that have exceeded the designated period.
- iv. Contact the customer directly via phone calls to remind them about the pending payment.
- v. Communicate with the customer through email, with the relationship officer taking the lead in sending the correspondence.
- vi. The finance manager should send a formal letter to the customer, emphasizing the urgency of payment.
- vii. Issue a reminder to the customer, requesting confirmation of the outstanding balance along with valid reasons for the delay.
- viii. If necessary, the managing director should personally send a letter to the customer, underscoring the importance of resolving the payment issue.
- ix. In the event that other measures have proven ineffective, it may be appropriate to recommend taking legal action as a course of action.

12.0 Late Payment Penalties

The company maintains a firm stance against clients who surpass their credit limits in both monetary value and duration. As a consequence, a penalty rate is enforced annually for clients listed as late payers.

The amount of the late payment penalty is determined by the outstanding receivables, inclusive of VAT. The annual rate ranges between 8% and 15%, with the final decision on the rate falling under the purview of the managing director. The penalty clause must be explicitly included under the following headings:

- i. Sales Quotes: The penalty clause should be incorporated in the sales quotes provided to clients.
- ii. Business Agreements: The penalty clause is an integral part of the business agreements between the company and its clients.

- iii. **Billing Invoices:** The penalty clause should be clearly stated in the billing invoices sent to clients.

By implementing this penalty structure, the company aims to uphold strict adherence to credit limits and foster timely payment practices among its clients.

13.0 Litigation

- i. To mitigate financial risks associated with recoveries, the company has established several methods to be employed as necessary:
- ii. **Formal Letter of Notice:** The manager of the finance department may issue a formal letter of notice to clients with outstanding payments, emphasizing the need for timely settlement.
- iii. **Credit Recovery Insurance Policy:** The company may opt to utilize a credit recovery insurance policy, providing an additional layer of protection against potential losses.

Suggestion of Lawsuits: In more severe cases, the managing director may recommend pursuing legal action to recover outstanding debts.

***Note:** It is imperative to retain all relevant information in electronic format for a minimum of five years to ensure compliance and record-keeping.

Steps in Implementing the Receivable Policy:

- i. **Root Study:** Conduct a thorough analysis of the underlying factors contributing to receivables and potential risks.
- ii. **Consultation with Finance and Accounts Team:** Engage in discussions and gather insights from the finance and accounts team to ensure their input is incorporated into the policy design.
- iii. **Policy Design:** Develop a comprehensive and robust receivable policy based on the findings and input obtained during the study and discussions.
- iv. **Sample Validation:** Validate the policy through testing and review of sample cases to ensure its effectiveness and alignment with the company's objectives.

- v. Re-corrections: Make necessary adjustments and refinements to the policy based on feedback and identified areas for improvement.
- vi. Formal Announcement: Officially communicate the finalized receivable policy to all relevant stakeholders through a formal announcement, outlining the key provisions and implementation timeline.
- vii. Training: Provide training sessions to the employees involved in receivable management, ensuring they are well-versed in the policy's requirements and procedures.
- viii. Implementation: Execute the receivable policy across the organization, actively monitoring its implementation and making any necessary adjustments along the way to ensure its effectiveness and adherence.

b) Testing the Sample Receivable Policy

Step 1: Identify the Target Audience with Impact

- i. Finance and Accounts Department
- ii. Operations Department
- iii. Managing Director

Step 2: Establish Test Objectives

- i. Provide a detailed description of the overall structure and expectations of the test.
- ii. Explain each test result and its correlation to the policy design.
- iii. Review the results to determine if any changes or adjustments are required.

Step 3: Create a Policy Testing Environment

- i. Select a test unit (e.g., a steering committee) to oversee the testing process.
- ii. Develop test prototype scenarios to simulate real-world situations.
- iii. Create dummy stakeholder profiles including client information and transaction details.
- iv. Prepare digital documents as evidence of the testing process.

Step 4: Consider Target Markets for Title Holders Testing

Super users, individuals who have performed exceptionally well during training, may be selected to participate in testing the policy.

Step 5: Assign Responsibilities and Roles

Table AXII.3: Roles, Responsibilities and Tasks

Roles	Responsibilities	Risks
Finance and Accounts Executive	Implement the test (data entry)	Moderate
Finance and Accounts Department Manager	Approvals and the analysis of test	Moderate
Operations Manager	Approvals and the analysis of test	Moderate
Managing Director	Funding and facilitating resources	High

--	--	--

Source: Author.

Step 6: Test Real-World Scenarios (Case Building)

Utilize previous instances or stories to evaluate the impact of the new policy on key stakeholders. Identify and rectify any issues or loopholes from the previous practices. The steering committee should document the results in a sample results table.

Table AXII.4: Roles, Responsibilities and Tasks

Situation	Old practice results	New policy results	Pass/ Fail

Source: Author.

Step 7: Document Change Requirements with Fact-Based Examples

Table AXII.5: Documents Change Requirements

Documents	Description
Service quality	Error rate tolerance should be 99.5%
Risk	Diversifies stakeholder behavior
Service transition	Linked to diversified group of activities
Service design	Specification with nil error
Skills and knowledge	All stakeholders are having the KSA to work on the process

Source: Author.

C) Staff Training on New Procedures

This training plan aims to determine the appropriate training strategies and exercises necessary to achieve the desired learning outcomes.

Step 1: Identify the Training's Intended Audience

Finance division staff, including the finance manager and Senior managers

Step 2: Training Objectives

- i. Ensure that all affected employees receive updated training to enhance company advancement.
- ii. Ensure that personnel receive the appropriate level of training for the new procedures.
- iii. Maintain an operational efficiency of 90% during the training session.

Step 3: Areas of Coverage

- i. Decision-making case studies.
- ii. Updated procedural guidelines.

Step 4: Training Risks

- i. Unforeseen competency-related issues.
- ii. Lack of timely replacements for training staff.
- iii. Lack of comprehension regarding the new procedures.

Step 5: Assess the Current Capabilities of Each Audience

Table AXII.6: Capacity of the Audience

Position	size	Experience	Training priority
Executive	8	1-3years	High
Managerial	2	8 years or above	Moderate

Source: Author.

Step 6: Training Approaches

- i. Job-based training over a period of 6 months.
- ii. Weekly sessions with discussions and question-and-answer sessions.
- iii. Volunteer participation in the project committee.

Step 7: Training Schedule

Day 1: Introduction and orientation - Internal Mail

Day 2: On-the-job training under the guidance of senior managers

Day 3: Mobile session

Step 8: Roles and Responsibilities in Training

Table AXII.7: Roles and Responsibilities in Training

Role	Profile	Focus
Training lead	Responsible for completing the training program within the scope	Senior managers
Staff	Responsible to take all the materials needed for implementation	Continuous training
Management	Responsible to sponsor all the materials to conduct the training	Decision making

Source: Author.

Step 9: Training Infrastructure

- Provision of laptops for staff members
- Hands-on training sessions

Step 10: Training Materials and Curriculum

Table AXII.8: Training Materials and Curriculum

Role	Profile	Developer	Reviewer	Decision
On the job training	Hand on training	Policy makers	Senior managers	Management
Questioning and answer session	Feedback and improvements	Senior managers	Management	Management
Participation in project committee	Voice of engagement	Steering committee		

Source: Author.

Step 11: Develop a Training Checklist

Table AXII.8: Training Checklist

Criteria	Pass	Fail
Payment recovery is speed		
Payments are recovered within the deadlines		
Performance		
behaviors		

Source: Author

Annexure XIII: Proposed Quality Plan



R.E Weerakoon Constructions (Pvt) Ltd.

Project: Renovation of Bungalow at Ethanamadala, kalutara North

Version: 01

QUALITY PLAN FOR A RENOVATION PROJECT

Section 1

PURPOSE OF THE PROJECT QUALITY MANAGEMENT PLAN

The purpose of the Project Quality Management Plan is to establish a comprehensive framework for ensuring that the home renovation project at the construction company meets the desired quality standards. This document will outline the specific quality objectives that are to be achieved during the project, as well as the strategies and procedures that will be used to achieve these objectives.

The Project Quality Management Plan will be tailored to fit the needs of this project and will cover all aspects of the project life cycle that are relevant to ensuring quality outcomes. This includes planning, design, construction, and final inspection and testing.

1.1 Project Name

Renovation of Bungalow at Ethanamadala, Kalutara North.

1.2 Overview of the Project

The Renovation of Bungalow project involves the refurbishment and remodelling of an existing residential property to improve its functionality, aesthetics, and overall value. This project typically consists of a series of interrelated activities, such as demolition, construction, installation of new fixtures, finishes, and systems, and final inspections and testing.

1.3 Scope of the Project

- i. **Assessment and planning:** The project begins with an assessment of the existing property, including its structural condition, design, and layout. This assessment helps to identify areas that require renovation and develop a plan for the project, which includes timelines, budgets, and resource allocation.
- ii. **Demolition and preparation:** Once the plan is finalized, the existing structures, fixtures, and finishes that need to be replaced or removed are demolished. The site is then prepared for the construction and installation phases of the project.
- iii. **Construction and installation:** The construction and installation phase involve the actual renovation work, which includes the installation of new walls, floors, ceilings, windows, doors, plumbing, electrical systems, and other features. This phase also includes the installation of new finishes such as paint, tiles, and countertops.
- iv. **Final inspections and testing:** Once the renovation work is completed, final inspections and testing are carried out to ensure that the renovated property meets all safety, building code, and quality standards. This includes inspections of electrical, plumbing, and HVAC systems, as well as tests for structural integrity and fire safety.
- v. **The renovation of a bungalow project requires careful planning, effective project management, and collaboration among all stakeholders involved in the project. The successful completion of this project can result in a transformed living space that enhances the comfort, functionality, and value of the property.**

1.4 Objectives of the Project

The overall quality objectives for this project include:

- i. Meeting or exceeding the expectations of the client in terms of the quality of workmanship, materials used, and overall finished product
- ii. Adhering to all applicable codes, standards, and regulations for construction and renovation projects.
- iii. Ensuring that all work is completed safely and without harm to workers, occupants of the home, or the environment.

- iv. Minimizing the risk of defects, rework, and delays that can compromise the quality and timeliness of the project.
- v. Maintaining effective communication and collaboration among all stakeholders involved in the project to ensure that quality objectives are met throughout the project life cycle.

Section 2

PROJECT QUALITY MANAGEMENT

2.1 Organisation, Responsibilities, and Interfaces

Table AXIII.1: Organisation, Responsibilities, and Interfaces

Designation	Job Role	Quality Responsibility
The project manager	The project manager is responsible for overall project quality management.	This includes ensuring that the project quality management plan is in place, communicating the quality objectives and expectations to the team, and monitoring the progress of the project against the established quality standards. The project manager is also responsible for conducting project reviews, coordinating audits, and ensuring that corrective actions are taken when necessary.
Operations Manager	The quality manager is responsible for the development and implementation of the project quality management plan.	This includes identifying the quality objectives, developing quality metrics, and ensuring that all project activities are carried out

		in accordance with the established quality standards. The quality manager is also responsible for auditing work products and processes, identifying non-conformities, and working with the team to develop corrective and preventive actions.
Project Team Members	The project team members are responsible for implementing the project quality management plan.	This includes adhering to the established quality standards, documenting project activities, and participating in project reviews. They are also responsible for auditing their own work products to ensure that they meet the established quality standards.
Mentors or Coaches	Mentors or coaches are responsible for providing guidance and support to project team members to ensure that they have the necessary skills and knowledge to implement the project quality management plan.	This includes providing training, coaching, and mentoring to team members to help them improve their work quality and productivity.
Auditors	Auditors are responsible for conducting audits of work products and processes to ensure that they meet the established quality standards.	This includes reviewing project documentation, interviewing team members, and observing project activities to identify non-conformities and recommend

		corrective and preventive actions.
--	--	------------------------------------

Source: Author.

2.2 Tools, Environment, and Interfaces

Table AXIII.2: Tools, Environment, and Interfaces

Tool	Description
Checklists	Checklists are used to ensure that all the required tasks and activities have been completed as per the established quality standards. The data elements include the list of tasks and activities, quality standards or metrics, and status of completion.
Control charts	Control charts are used to monitor the performance of a process over time. The data elements include the process performance data, time, upper and lower control limits, and the mean value.
Histograms	Histograms are used to display the distribution of a set of data. The data elements include the data values, frequency, and the range of values.
Pareto charts	Pareto charts are used to identify the most significant causes of a problem or issue. The data elements include the causes or issues, frequency, and the percentage contribution of each cause or issue.
Fishbone diagrams	Fishbone diagrams are used to identify the root causes of a problem or issue. The data elements include the problem or issue, potential causes, and the relationships between the causes and the problem.
Scatter diagrams	Scatter diagrams are used to identify the relationship between two variables. The data elements include the variables, the data points, and the correlation coefficient.

Statistical sampling	Statistical sampling is used to estimate the quality of a population based on a sample. The data elements include the sample size, sampling method, sampling frame, and the quality metric being measured.
Flowcharts	Flowcharts are used to visualize a process and identify areas of improvement. The data elements include the process steps, decision points, inputs, outputs, and the flow of information.

Source: Author.

Section 3

PROJECT QUALITY MANAGEMENT

The following sections define how this project will apply each of these practice groups to define, monitor and control quality standards.

1.1 Quality Planning

Table AXIII.3: Quality Planning

Quality Standard	Comment
ISO 9001:2015	Quality Management System Standard: This standard provides a framework for managing quality and ensuring customer satisfaction. To satisfy this standard, the project team should develop a project quality management plan that outlines quality objectives, metrics, and measures for the project.
Building codes and regulations	Building codes and regulations are specific to the location where the renovation project is taking place. The project team should ensure that all renovation work complies with the relevant codes and regulations.
Product functionality	The project team should define the functional requirements for

requirements	the renovated home and ensure that the renovation work meets these requirements. Examples of quality metrics and measures for product functionality include usability, reliability, and performance.
Project management performance	The project team should define quality metrics and measures for project management performance, such as meeting project milestones, staying within budget, and managing project risks.
Documentation	The project team should ensure that all project documentation is complete, accurate, and up to date. Quality metrics and measures for documentation include completeness, accuracy, and timeliness.
Testing	The project team should define quality metrics and measures for testing the renovated home, such as the number of defects found during testing and the time required to fix defects.
Other	In addition to identifying the relevant quality standards, the project team should also define acceptance criteria for project deliverables and product performance. Acceptance criteria should be specific, measurable, and achievable. Examples of acceptance criteria for a home renovation project include: i. Walls are painted with a minimum of two coats of paint and there are no visible brush or roller marks. ii. Plumbing fixtures are installed in accordance with the manufacturer's instructions and there are no leaks. iii. Electrical wiring is installed in accordance with the relevant codes and regulations and all outlets and switches function correctly. iv. Flooring is installed without gaps, bulges, or other

	defects.
--	----------

Source: Author.

1.2 Quality Assurance (QA)

Table AXIII.4: Quality Assurance

Action	Comment
Conduct regular quality audits	Quality audits are an effective way to determine if the project is conforming to the defined quality standards. Audits should be conducted at regular intervals throughout the project life cycle, and metrics should be established to measure the effectiveness of the audit process. Metrics for quality audits may include the number of audits conducted, the percentage of non-conformities identified and corrected, and the time taken to close out non-conformities.
Monitor project performance	The project team should continuously monitor project performance to ensure that quality objectives are being met. Metrics for project performance may include meeting project milestones, staying within budget, and managing project risks.
Review and update the project quality management plan	The project team should periodically review and update the project quality management plan to ensure that it remains relevant and effective. Metrics for the quality management plan may include the number of updates made, the number of non-conformities identified and addressed through plan updates, and the effectiveness of the plan in achieving project quality objectives.
Conduct testing and inspections	Testing and inspections should be conducted at various stages of the project to ensure that the product or service being delivered

	meets the defined quality standards. Metrics for testing and inspections may include the number of defects found, the time taken to fix defects, and the effectiveness of the testing and inspection processes.
Document and communicate quality performance	The project team should document all quality-related activities and communicate them to relevant stakeholders. Metrics for documenting and communicating quality performance may include the number of quality-related documents produced, the accuracy of the documentation, and the effectiveness of communication efforts in ensuring that stakeholders are aware of project quality performance.

Source: Author.

1.3 Quality Control (QC)

Table AXIII.5: Quality Control

Action	Comment
Conduct regular quality audits	Regular quality audits will be conducted to ensure that project activities are conforming to the defined quality standards. The quality audits will be conducted by the Quality Assurance (QA) team or an independent auditor, and non-conformities will be tracked, reported, and corrected through the Corrective Action Preventive Action (CAPA) process.
Monitor project performance	Project performance will be monitored throughout the project life cycle to ensure that quality objectives are being met. The project manager will be responsible for monitoring project performance, and metrics such as project milestones, budget, and

	risk management will be used to measure project performance.
Conduct testing and inspections	Testing and inspections will be conducted at various stages of the project to ensure that the product or service being delivered meets the defined quality standards. The Quality Control (QC) team will be responsible for conducting testing and inspections, and any defects found will be reported and corrected through the CAPA process.
Review and update the Project Quality Management Plan	The Project Quality Management Plan will be periodically reviewed and updated to ensure that it remains relevant and effective. The project manager will be responsible for reviewing and updating the Plan, and any changes will be communicated to the project team and stakeholders.
Document and communicate quality performance	All quality-related activities will be documented, and the performance will be communicated to relevant stakeholders. The project manager will be responsible for documenting and communicating quality performance, and metrics such as the number of quality-related documents produced, and the accuracy of the documentation will be used to measure quality performance.
Other	To determine compliance with the defined quality standards, the QA team will conduct regular audits and inspections to ensure that project activities are conforming to the defined quality standards. Metrics such as the number of non-conformities identified and corrected, and the time taken to close out non-conformities will be used to measure compliance with the quality standards.

Source: Author.

"We, the undersigned, have reviewed the Project Quality Management Plan and agree with the approach it presents. Changes to this Plan will be coordinated with and approved by the undersigned or their designated representatives."

Project Manager

Quality Manager

Date

Annexure XIV: Proposed Work Breakdown Structure Method

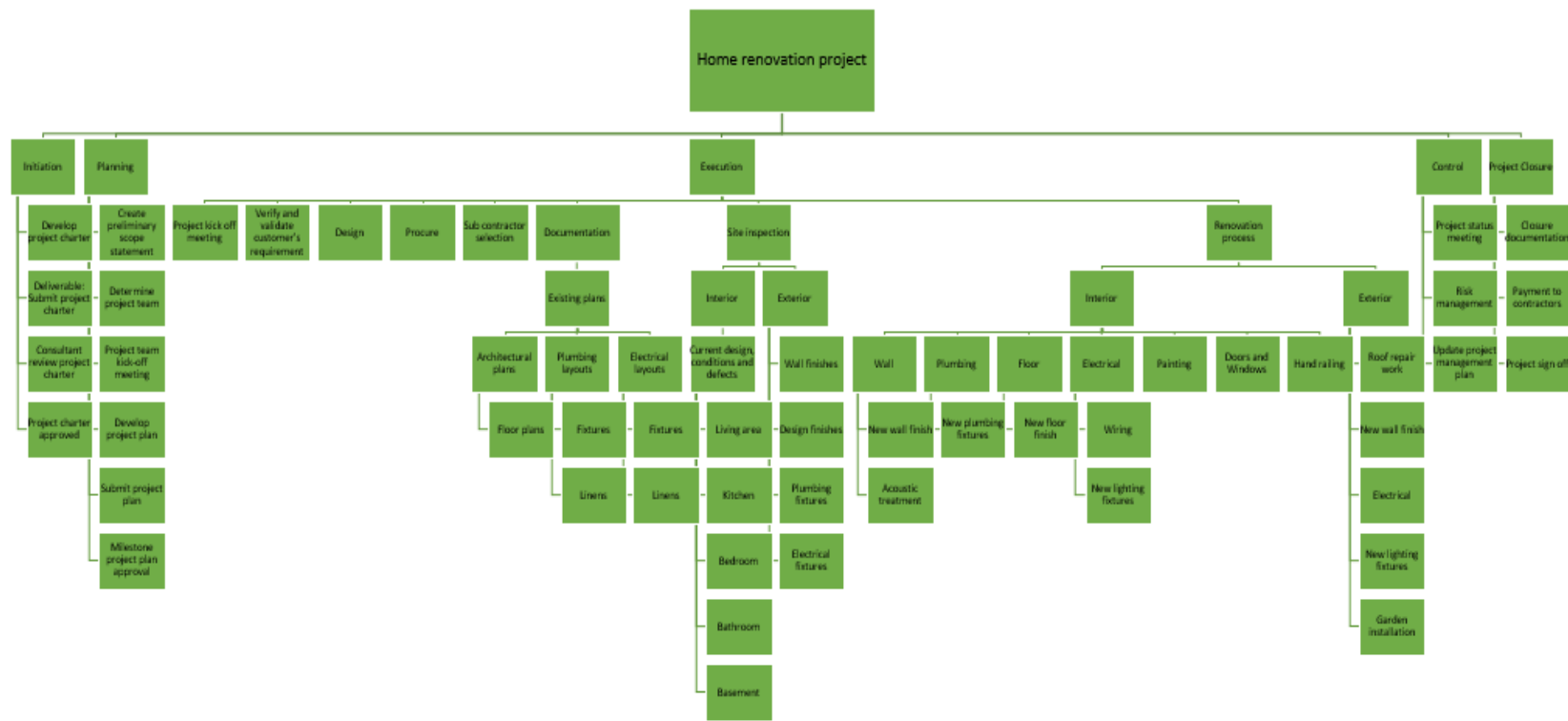


Figure AXIV: Work Breakdown Structure for a Home Renovation Project

Source: Author.

1. Home renovation project
 - 1.1. Initiation
 - 1.1.1. Develop project charter
 - 1.1.2. Deliverable: Submit project charter
 - 1.1.3. Consultant review project charter
 - 1.1.4. Project charter approved
 - 1.2. Planning
 - 1.2.1. Create a preliminary scope statement
 - 1.2.2. Determine project team
 - 1.2.3. Project team kick-off meeting
 - 1.2.4. Develop project plan
 - 1.2.5. Submit project plan
 - 1.2.6. Milestone project plan approval
 - 1.3. Execution
 - 1.3.1. Project kick-off meeting
 - 1.3.2. Verify and validate customer's requirement
 - 1.3.3. Design
 - 1.3.4. Procure
 - 1.3.5. Subcontractor selection
 - 1.3.6. Documentation
 - 1.3.6.1. Existing plans
 - 1.3.6.1.1. Architectural plans
 - 1.3.6.1.1.1. Floor plans
 - 1.3.6.1.2. Plumbing layouts
 - 1.3.6.1.2.1. Fixtures
 - 1.3.6.1.2.2. Linens
 - 1.3.6.1.3. Electrical layouts
 - 1.3.6.1.3.1. Fixtures
 - 1.3.6.1.3.2. Linens
 - 1.3.7. Site inspection

1.3.7.1. Interior

1.3.7.1.1. Current design, conditions and defects

1.3.7.1.1.1. Living area

1.3.7.1.1.2. Kitchen

1.3.7.1.1.3. Bedroom

1.3.7.1.1.4. Bathroom

1.3.7.1.1.5. Basement

1.3.7.2. Exterior

1.3.7.2.1. Wall finishes

1.3.7.2.2. Design finishes

1.3.7.2.3. Plumbing fixtures

1.3.7.2.4. Electrical fixtures

1.3.8. Renovation process

1.3.8.1. Interior

1.3.8.1.1. Wall

1.3.8.1.1.1. New wall finish

1.3.8.1.1.2. Acoustic treatment

1.3.8.1.2. Plumbing

1.3.8.1.2.1. New plumbing fixtures

1.3.8.1.3. Floor

1.3.8.1.3.1. New floor finish

1.3.8.1.4. Electrical

1.3.8.1.4.1. Wiring

1.3.8.1.4.2. New lighting fixtures

1.3.8.1.5. Painting

1.3.8.1.6. Doors and Windows

1.3.8.1.7. Handrailing

1.3.8.2. Exterior

1.3.8.2.1. Roof repair work

1.3.8.2.2. New wall finish

- 1.3.8.2.3. Electrical
- 1.3.8.2.4. New lighting fixtures
- 1.3.8.2.5. Garden installation

1.4. Control

- 1.4.1. Project status meeting
- 1.4.2. Risk management
- 1.4.3. Update project management plan

1.5. Project Closure

- 1.5.1. Closure documentation
- 1.5.2. Payment to contractors
- 1.5.3. Project sign off

Annexure XV: Proposed Supplier Evaluation Process

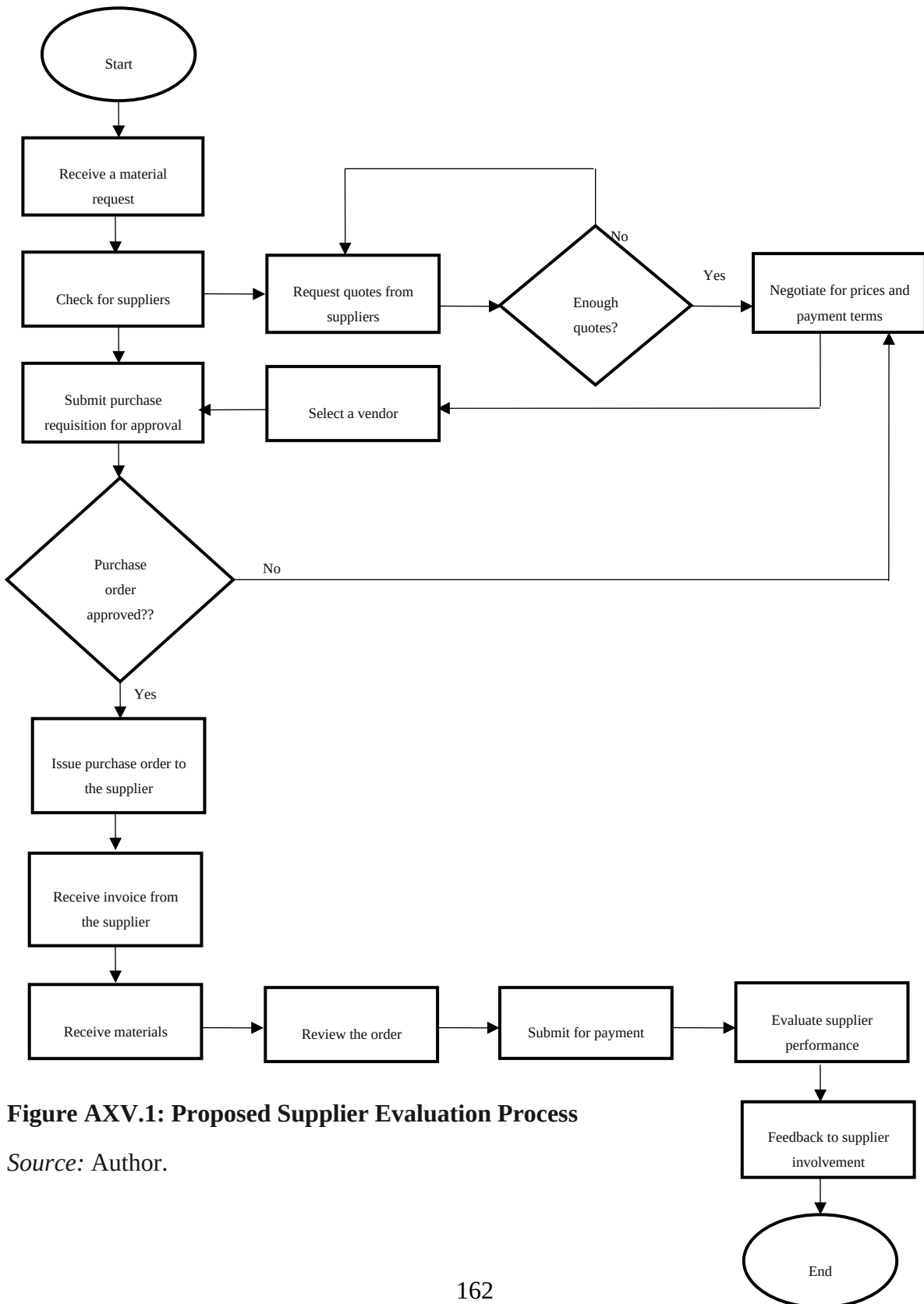


Figure AXV.1: Proposed Supplier Evaluation Process

Source: Author.

Annexure XVI: Current Supplier Evaluation Form

a) Materials

Supplier Evaluation – Materials										Issue 01 Date of Issue 2015 – 06 –01 Page 1 of 1	
Year- Period- Jan to June / July to Dec											
Location -											
Item	Name of Supplier	Number of Deliveries	Quantity Factor	Marks For Quantity	Delivery Factor	Marks For Delivery	Quality Factor	Marks For Quality	Total Marks	Decision number	
Signature of Project Manager						Date					
NOTE Quantity Factor – How many times supplied the full quantity as ordered (without effecting to construction) Delivery factor- How many times delivered on or before the delivery date Quality Factor- How many times quality ok											
Marks for Quantity- $\frac{\text{Quantity factor}}{\text{Number of delivery}} \times 20$											
Marks for Delivery - $\frac{\text{Delivery Factor}}{\text{Number of delivery}} \times 30$											
Marks for Quality - $\frac{\text{Quality Factor}}{\text{Number of delivery}} \times 50$											
Total marks = Marks for Quantity + Marks for delivery + Marks for Quality											
Criteria for Reselecting											
Total marks		Decision							Decision Number		
Equal or exceed 75		Supplier will be kept continuously							1		
Between 50-74		Supplier will be kept continuously after sending notifications							2		
Below 50		Supplier will not be kept							3		
For Control stamp only			Reviewed by :MR				Approved by: PR				

Figure AXVI.1: Current Supplier Evaluation Form – Materials

Source: Author.

b) Services

Supplier Evaluation – Services	Issue 01 Date of Issue 2015 – 06 –01 Page 1 of 1																																					
<i>To be filled by each and every time of services purchased (Only for technical services)</i> <i>To be filled by all locations for services purchased directly or through Head office</i> <i>And</i> <i>Results are to be communicated to Head office for services purchased through head office</i>																																						
Year and Month --- Location -																																						
<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td colspan="3">Type of service</td> </tr> <tr> <td colspan="3">Name of Service Provider:</td> </tr> <tr> <td colspan="2">Agreement No (If any) :</td> <td>Period of time /Date :</td> </tr> <tr> <td style="text-align: center;">DESCRIPTION</td> <td style="text-align: center;">Maximum Marks</td> <td style="text-align: center;">Marks given</td> </tr> <tr> <td>1, Quality of the Service performed</td> <td style="text-align: center;">40</td> <td></td> </tr> <tr> <td>2, Meeting the completion period / date</td> <td style="text-align: center;">20</td> <td></td> </tr> <tr> <td>3, Attendance during the contract period</td> <td style="text-align: center;">10</td> <td></td> </tr> <tr> <td>4, Reaction to quality problems</td> <td style="text-align: center;">10</td> <td></td> </tr> <tr> <td>5, Cleanliness of work place</td> <td style="text-align: center;">10</td> <td></td> </tr> <tr> <td>6, Cooperation with supervisors</td> <td style="text-align: center;">10</td> <td></td> </tr> <tr> <td style="text-align: right;">TOTAL MARKS</td> <td style="text-align: center;">100</td> <td></td> </tr> <tr> <td colspan="3" style="padding: 5px;"> *Decision Number – (1 or 2 or 3) Remarks Signature _____ Date _____ </td> </tr> </table>			Type of service			Name of Service Provider:			Agreement No (If any) :		Period of time /Date :	DESCRIPTION	Maximum Marks	Marks given	1, Quality of the Service performed	40		2, Meeting the completion period / date	20		3, Attendance during the contract period	10		4, Reaction to quality problems	10		5, Cleanliness of work place	10		6, Cooperation with supervisors	10		TOTAL MARKS	100		*Decision Number – (1 or 2 or 3) Remarks Signature _____ Date _____		
Type of service																																						
Name of Service Provider:																																						
Agreement No (If any) :		Period of time /Date :																																				
DESCRIPTION	Maximum Marks	Marks given																																				
1, Quality of the Service performed	40																																					
2, Meeting the completion period / date	20																																					
3, Attendance during the contract period	10																																					
4, Reaction to quality problems	10																																					
5, Cleanliness of work place	10																																					
6, Cooperation with supervisors	10																																					
TOTAL MARKS	100																																					
*Decision Number – (1 or 2 or 3) Remarks Signature _____ Date _____																																						
<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="text-align: center;">Total marks</td> <td style="text-align: center;">Decision</td> <td style="text-align: center;">Decision Number*</td> </tr> <tr> <td>Equal or exceed 75</td> <td>Service provider will be selected for next year /time</td> <td style="text-align: center;">1</td> </tr> <tr> <td>Between 50-74</td> <td>Service provider will be selected for next year /time after notifications</td> <td style="text-align: center;">2</td> </tr> <tr> <td>Below 50</td> <td>Service provider will not be selected for next year/time</td> <td style="text-align: center;">3</td> </tr> </table>			Total marks	Decision	Decision Number*	Equal or exceed 75	Service provider will be selected for next year /time	1	Between 50-74	Service provider will be selected for next year /time after notifications	2	Below 50	Service provider will not be selected for next year/time	3																								
Total marks	Decision	Decision Number*																																				
Equal or exceed 75	Service provider will be selected for next year /time	1																																				
Between 50-74	Service provider will be selected for next year /time after notifications	2																																				
Below 50	Service provider will not be selected for next year/time	3																																				
For Control stamp only	Reviewed by :MR	Approved by: PR																																				

Figure AXVI.2: Current Supplier Evaluation Form – Services

Source: Author.

Annexure XVII: Proposed Supplier Evaluation Criterion

The rating matrix was created based on the four key criteria for choosing a supplier to deliver material and services: price, quality, delivery, reliability and communication. On a scale of 1 to 4, factors are to be evaluated. Below the rating scale is a rating matrix with extra rating guidance in the form of an example.

Supplier Evaluation Form			
Name of the Supplier:			Date:
Material/ Service:			
Aspect	Total weight	Factors	Weight
Delivery	0.25	Location	0.3
		Freight terms	0.2
		Any trade restrictions	0.1
		Lead time	0.3
Total delivery score = Factor weight x Rating			
Cost	0.25	Quoted price	0.7
		Payment terms	0.2
		Any related bank costs	0.1
Total cost score = Factor weight x Rating			
Quality	0.25	Certifications and references	0.8
		Customer service	0.2
Total quality score = Factor weight x Rating			
Reliability	0.15	Previous relationships	0.3
		Warranty/post-purchase service	0.2
		Technical expertise/ knowledge in the industry	0.2
		Ethical standard	0.1

		Political situation	0.1
		Currency situation	0.1
Communication	0.1	Service response	0.40
		Sharing information	0.30
Total reliability score = Factor weight x Rating			
Total score = Sum (Factor weight x Attribute weight x rating)			
Decision			
Comments			
Prepared by:		Reviewed by:	Approved by:
.....			

Figure AXVII.1: Proposed Supplier Evaluation Form – Material and Services

Source: Author.

Table AXVII.2: Example for Rating Matrix

Aspect	Factor	1 (Poor)	2 (Fair)	3 (Good)	4 (Excellent)
Delivery	Location	Very far (e.g., USA)	Far (e.g., Dubai)	Proximity (within the country)	Very close proximity (within 15kms)
	Freight terms (refers to shipping terms of supplier)	Poor (E.g., ex-works)	Fair (E.g., CIF)	Good (E.g., DPU)	Excellent (E.g., DDP)
	Any trade restrictions	FTA	Low	Moderate	High
	Lead time	30+ days	26-30 days	21-25 days	15-20 days
Cost	Quoted price	High	Acceptable	Low	Very low
	Payment terms	Cash payment at the point of order	Cash payment at the time of delivery	Payment within one month after the delivery	Payment within 45 days after the delivery
	Any related bank costs	High	Acceptable	Low	Very low
Quality	Certifications	No certifications	Local certifications, e.g., SLS	International certifications, e.g., ISO	International, local and other applicable certifications
	Customer service	Never attends or poor promptness of replies	Occasionally replies promptly	Responds most times promptly	Always responds promptly
Reliability	Previous relationships	Poor (Eg: defects low quality)	Fair (accepting returns)	Good (providing refunds)	Excellent (on-point order delivery)
	Warranty policies/post-purchase service	Non-favourable to buyer	Neutral	Favourable	Very favourable (E.g., supplier

					responsible for late shipments)
	Technical expertise/ knowledge of the industry	Lack of knowledge	Fairly understanding	Knowledge of the industry	Expertise in the industry
	Ethical standards	Frauds in business	Fair treatment	Ethical in business	Highly ethical
	Political situation	Poor	Fair	Good	Excellent
	Currency situation	Non-favourable to buyer	Neutral	Favourable	Very favourable
Communication	Service response	Within a month	Within a week	Within few days	On the same day
	Sharing information	Lack of information	Access to information	Access to information	Highly accessible to information

Source: Author.

Annexure XVIII: Proposed Supplier Relationship Management Guidelines

The company must always adhere to the general guidelines regarding supplier relationship management.

- i. The company's procurement department is in charge of establishing, sustaining, and managing partnerships with suppliers. Employees from different divisions of the organisation are permitted to get in touch with suppliers to locate whether products or services are available and to get technical details about what they offer. However, employees from other departments other than purchasing are not permitted to make any commitment on behalf of the company.
- ii. It is the responsibility of every employee engaged in procurement activities to represent the business professionally and maintain goodwill with suppliers. Therefore, these employees must always ensure that corporate conduct is moral and unbiased.
- iii. All suppliers must get courteous, truthful, and equitable treatment. All registered and approved suppliers must be provided with a full and equitable opportunity to compete for the right to supply the company with goods or services.
- iv. Any conflicts with suppliers must be resolved using a business-oriented strategy and per the guidelines, which support efficiency and cost effectiveness and uphold the company's interests.
- v. Except for regular and non-critical concerns out of necessity, all critical discussions with suppliers must be in writing.

Supplier Registration and Approval

- i. The procurement department will oversee the registration process for suppliers. More specifically, the supplier registration activities will be under the control of the procurement manager.
- ii. A registration form that includes the supplier's trade license, references, etc., must be completed by all suppliers of materials and services who seek to register themselves with the company or those suppliers the company wishes to register.

- iii. To analyse further options for competitive prices, the procurement executive regularly works with the procurement manager to discover new materials and services suppliers. The procurement manager and procurement executive have developed a set of criteria that must be met for suppliers to be approved.
- iv. The company's managing director and procurement manager will promote the process of partnering with suppliers to achieve the highest quality at the best costs. The idea of partnering refers to creating strategic alliances with suppliers the business trusts and are capable of supplying goods and services that satisfy quality and timeliness criteria while staying within reasonable price ranges.
- v. The list of authorized suppliers must be updated to include the findings of supplier evaluations and audits.

Negotiations with Suppliers

- i. The procurement and material manager may negotiate with suppliers for all high-value purchases and some low-value purchases (procurement through quotations).
- ii. The procurement manager shall ensure that crucial company needs, such as quality, adherence to specifications, and delivery timelines, are not compromised during the discussions to acquire cheaper rates.
- iii. All negotiation meetings must be recorded, and the minutes must be given to the appropriate supplier so that all negotiated prices, terms, and conditions are clear.
- iv. Before making a final decision, the relevant supplier must submit the business a new quotation if conversations result in modifications to the prices, terms, or conditions.

Supplier Selection

- i. The manager of procurement should make the ultimate supplier selection decision.
- ii. Before making a final choice, the procurement manager will determine whether more discussions, supplier site visits, or post-qualification procedures are required.

Annexure XIX: Proposed Accounting Software

a) Requirement Identification

A brief survey was conducted among ten employees in the finance, accounts, and procurement departments at the managerial and operational levels. This supported in identifying the priority level that needed to be focused on in fulfilling the current gap in payables management. The survey revealed errors in descending order and the most suitable solution from their perspective. The questions and the summary of the results are as follows:

Name: _____ Years of experience: _____

Position: _____ Date: _____

1. What is your department
 - Finance department
 - Accounts department
 - Procurement department
2. What is your designation?
 - Manager
 - Senior Executive
 - Executive
 - Trainee
3. What are the types of problems associated with payables management?
 - Missing invoices
 - Incorrect entries
 - No creditor ageing analysis
 - Outstanding payments
4. What are the types of missing invoices?
 - Purchase order
 - Payment Invoice
 - Delivery note
 - Goods received note

5. What are the types of incorrect entries?
- Human errors due to manually entering data
 - Incorrect entries in material returns
 - Incorrect entries related to prices
 - Misunderstanding one supplier with another
- The most suitable method to overcome the problems associated with payables management
- Introducing an accounting software
 - Training on payables management
 - Increasing the number of staff with expertise in payables management
 - Supervision

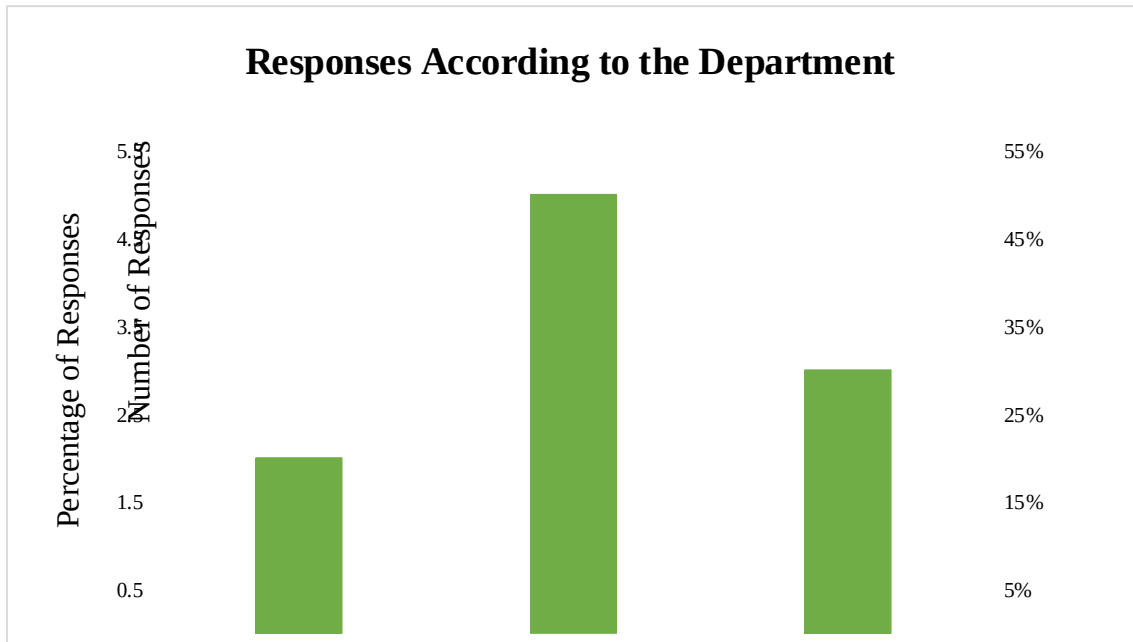


Figure AXIX.1: Responses Based on Department

Source: Author.

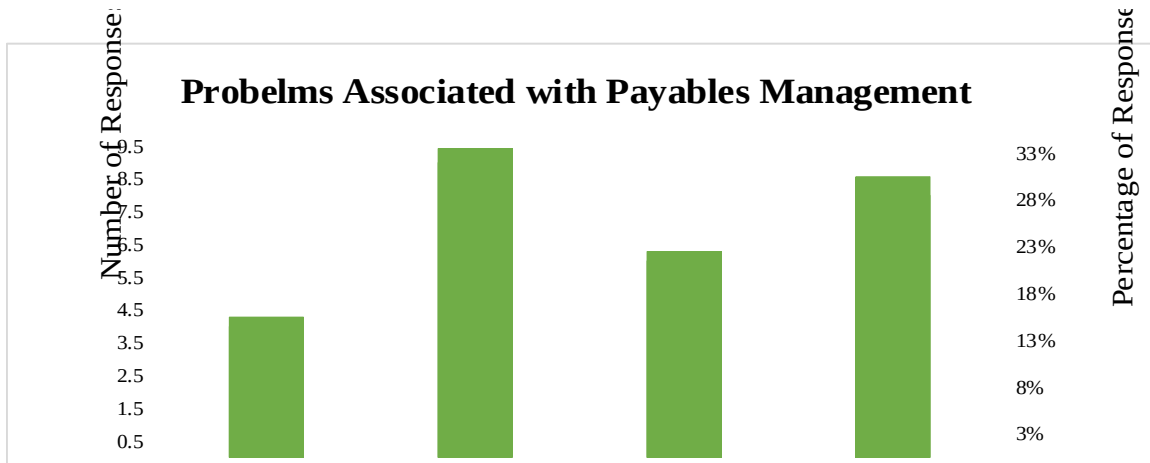


Figure AXIX.2: Problems with Payables Management

Source: Author.

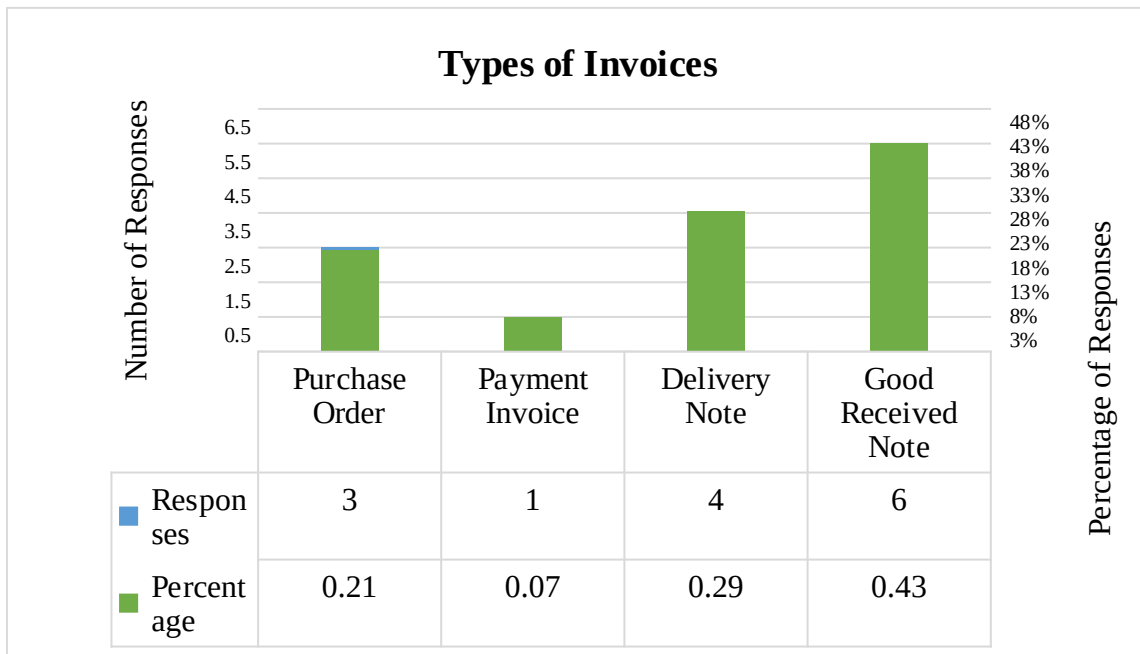
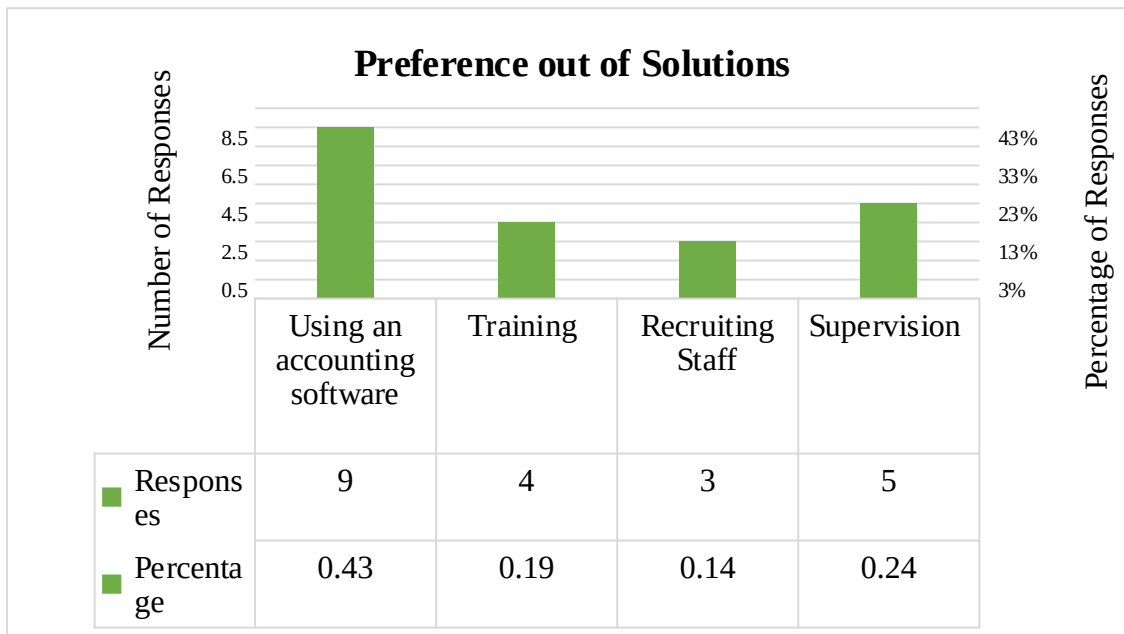


Figure AXIX.3: Types of Invoices

Source: Author.



Figure

Figure AXIX.4: Preference for Solutions

Source: Author.

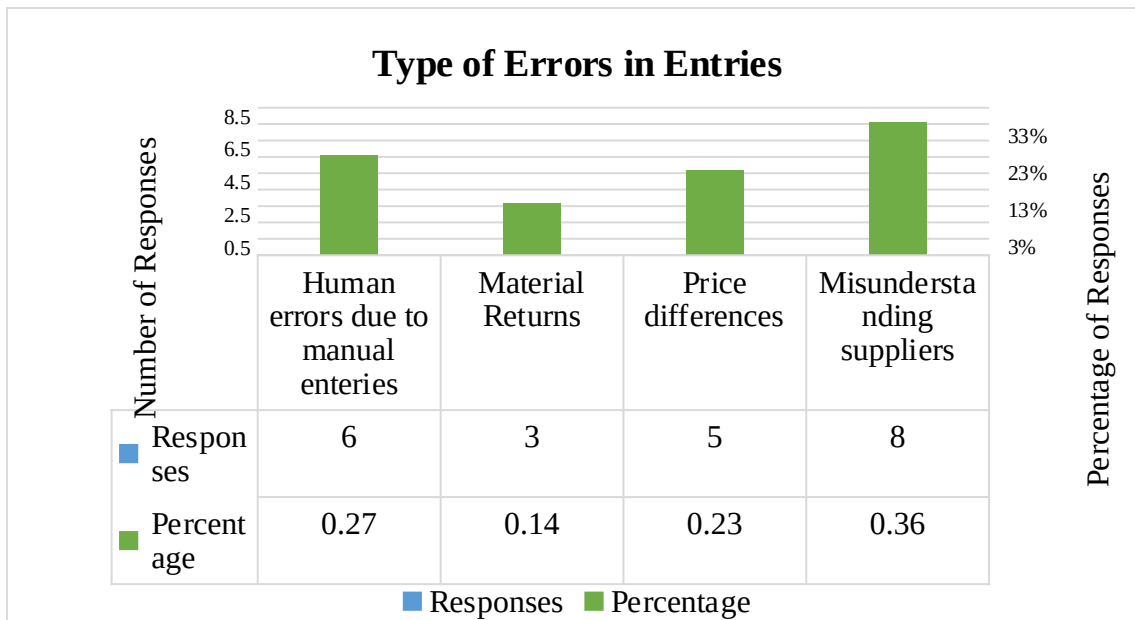


Figure AXIX.5: Type of Errors in Entries

Source: Author.

b) Objectives set to achieve by introducing quick book software

- i. To reduce payment for payables before the end of the credit period from 100% to 8%
- ii. To reduce the incorrect entries done by staff in payables management from 100% to 2%

Table XIX.1: Hardware Requirements to Introduce Quick Book Software

Computer	Dual core with minimum 2.0GHZ
RAM	02 GB for a client and 04 GB for multi-users
Hard Disk	03 GB Free Space
Backup	Flash drive or DVD writer. It is important to take backups every day.
Operating System Client	Windows10. Both 32bit and 64bit
Operating System Server	Windows10, Windows8(SP1), Windows7(SP1) both 32bit and 64bit or Windows Server 2008R2 or 2012
Network	For multi-user Local Area Networks with 100 MBPS or more speed
Printing	Any printer supported by the above operating systems
Cheque Printing	Standard printer with standard or continuous cheque leaves
Installation	Installation: DVD ROM drive to install the package
Remote Access	Terminal Server or Similar software on a server computer with VPN or broadband connection with Appropriate Bandwidth

Source: Author.

c) QuickBooks Software Training Program Overview

The author suggests conducting a one-day training program on the newly introduced quick book software provided free of charge by the dealer of the software. This intends to ensure employees are knowledgeable and comfortable in using the software to manage payables and address the identified performance gap. The dealer and online training provide training through Zoom meetings using a PPT and exercises related to managing payables. The accomplishment of the predetermined training objectives is supposed to indicate the training program's efficacy. The performance of team members will be regularly monitored by senior employees who will be selected as mentors after the training program.

Table XIX.2: Overview of the Training Program

Training Type	Off-the-job training
Delivery Method	Online (Zoom meeting)
Training Duration	6 hours
Training Time	9.00 AM to 4.00 PM
Trainer Type	External
Maximum Number of Participants	10 participants
Training Fee	Included in the software amount
Training Supporting Materials	PPT

Source: Author.

Outline of the program contents

- i. Purchase Order
- ii. Direct Emailing PO to Supplier
- iii. GRN (Goods Received Note)
- iv. Supplier Settlements
- v. Accruals Entry
- vi. Purchase Returns
- vii. Supplier Direct Mailers
- viii. Creditors Ageing Summary and Detail
- ix. Creditors Balance Summary and History
- x. Unpaid Bills Report
- xi. Supplier Contact List
- xii. Purchases by Item Summary and Detail
- xiii. Purchases by Supplier Summary and Detail
- xiv. Purchases by Division Summary and Detail
- xv. Outstanding Purchase orders by Supplier
- xvi. Expenses Analysis by Supplier Summary and Detail
- xvii. Purchases Graph by Supplier / Item
- xviii. VAT Management

d) Feedback Form

Feedback Form					
Date					
Training	Introducing QuickBooks Software in Payables Management				
Name					
Trainer					
Please help us to improve our offerings and service by indicating your impression of the items listed below. 5 Strongly Agree 4 Agree 3 Neutral 2 Disagree 1 Strongly Agree					
	1	2	3	4	5
Curriculum					
1. The subject matter was informative					
2. Materials were organized and easy to follow					
3. The training objectives were communicated and achieved					
Trainer					
1. The trainer was knowledgeable					
2. The trainer did a good job of teaching the material.					
3. Adequate time was provided for attendee questions					
4. The trainer showed respect for the participants					
Procedure and Communication					

1. I knew about the class appropriately in advance					
2. I was allowed adequate time for breaks					
Overall Impression					
1. The knowledge gained will help in my job and the accomplishment of my goals.					
2. Others will benefit from my knowledge					
3. The training was worthwhile					
Any additional comments?					

Figure XIX.6: Feedback Form

Source: Author.

Monthly KPI Form				
Department : Employee Code : Employee Name : Designation :				
KPIs	Weightage	Action Plan		Dependencies
		Actions	Monthly Target	
Accuracy of entries				
Availability of invoices				
Settling outstanding payments on due date				
Total	100%			

Supervisor

.....

.....

.....

Comments:

Employee's Comments:

.....

.....

.....

I agree with my KPIs

.....

(Signature of the employee)

(Date)

Figure XIX.7: Monthly KPI Form

Source: Author.

Annexure XX: Tangible Benefits

Table AXX.1 Reduced Idle Inventory

Total annual idle inventory (2018-2022)	5,389,498
Number of years considered	5
Average annual idle inventory	1,077,900
Current idle inventory %	54%
Expected idle inventory %	5%
Reduced idle inventory % due to project	91%
Reduced financial impact due to project	980,889

Source: Author.

Table AXX.2: Reduced Overdue Receivables

Total annual overdue receivables (2018-2022)	24,183,188
Number of years considered	5
Average annual overdue receivables	4,836,637
Current overdue receivables %	38%
Expected overdue receivables %	20%
Reduced overdue receivables % due to project	47%
Reduced financial impact due to project	2,273,219

Source: Author.

Table AXX.3: Reduced Early Payment to Suppliers

Total annual early payment to suppliers (2018-2022)	8,784,377
Number of years considered	5
Average annual early payments to suppliers	1,756,875
Current early payments to suppliers %	41%
Expected early payment % to suppliers	10%
Reduced early payment to suppliers % due to project	76%
Reduced financial impact due to project	1,335,225

Source: Author.

Annexure XXI: Estimated Costs to Implement Solutions

a) Inventory Management

- i. Design a new inventory management process – No cost.
- ii. Introduce daily project report format to identify the inventory requirement – No cost.
- iii. Conduct a workshop to introduce ABC analysis to organise inventory based on importance to the company.

Table AXXI.1: Cost to Conduct a Workshop on ABC Analysis

Cost Item	Amount (LKR)	
Venue – Head office		No cost
Facilitator – Internal		No cost
<i>Refreshments for 15 participants + facilitator</i>		
Morning Break (Tea – LKR 50*16)	800	
Lunch Break (Rice and curry 500*16)	8000	
Evening Break (Tea – LKR 50*16)	800	9,600
<i>Stationary for 15 participants</i>		
10 A4 sheets each (Rs. 5*10 sheets*15 participants)	750	
Pens (Rs.25*15 participants)	375	
Plastic file cover (Rs.190*15 participants)	2850	
Double side printed presentation slides 2slides per page (Rs.25*6*15 participants)	2250	6225
Total		15,825

Source: Author.

- iv. Proposed Store Policy - Emailed as a PDF to office staff and hard copies given to storekeepers in each construction site.

Table AXXI.2: Cost for Proposed Store Policy

Cost Item	Amount (LKR)
Double side printed 5 pages, including cover pages (Rs.15*3)	45
Total cost for three storekeepers (estimated three projects)	135

Source: Author.

b) Receivables Management

- i. Conduct training for quantity surveyors on taking measurements and invoice preparation practices.

Table AXXI.3: Cost for Proposed Training for Quantity Surveyors

Cost Item	Amount (LKR)
One side printed survey (Rs.15*12 participants)	180
One side printed training evaluation form (Rs.15*12 participants)	180
Transport allowance (Rs.1500*2 days*12 participants)	36,000
4. Training Fee (including refreshments and printed material)	250,000
Total Cost for the Training	286,360

Source: Author.

- ii. Develop a receivable policy and conduct a test on the sample policy- Emailed as a PDF to involved staff.
- iii. Design a well formulated quality plan for renovation projects – Emailed to the involved staff.
- iv. Introduce work breakdown structure method to identify each task in renovation projects – No cost.

 CHPB CENTRE FOR HOUSING PLANNING & BUILDING கிழக்கு வட்டார வளர்ச்சி மற்றும் கட்டுமானத் திட்டத்துறை CENTRE FOR HOUSING PLANNING & BUILDING கட்டடம் வீடமைப்புத் திட்டத்துறை	 ශ්‍රී ලංකා ප්‍රජාතාන්ත්‍රික සමාජවාදී ජනරජය STATE OF SRI LANKA	 ඉදිකිරීම් කර්මාන්ත සංවර්ධන අධිකාරිය CONSTRUCTION INDUSTRY DEVELOPMENT AUTHORITY நிர்மாணக் கைத்தொழில் அபிவிருத்தி அதிகாரசபை
ශ්‍රී ලංකා ප්‍රජාතාන්ත්‍රික සමාජවාදී ජනරජය STATE MINISTRY OF RURAL HOUSING AND CONSTRUCTION & BUILDING MATERIALS INDUSTRIES PROMOTION கிழக்கு வீடமைப்பு மற்றும் நிர்மாணத்துறை மற்றும் கட்டிடப் பொருட்கள் தொழில் மேம்பாட்டு இராஜாங்க அமைச்சு		
මගේ අංකය எனது இல My No.	ඔබේ අංකය உமது இல Your No.	දිනය திகதி Date
22/11/2022		

Mr. Jayendra Weerakoon
R.E. Weerakoon Constructions (Pvt) Ltd,

Dear Sir,

Quantity Surveying training program for R.E. Weerakoon Constructions

With reference to the request Letter, dated on 21.11.2022, regarding the above subject, we will accept your request & program will be scheduled to hold on December 2022, 2 day, At CHPB, No.33, Parliament road, Palawatta, Battaramulla.

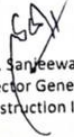
The course contents are as follows;

1. Manuals and Certificates
2. Resource personal Lecture & Transport Charges
3. Refreshment for Participants
4. Premises Rental Charges
5. Development cost & institutional involvements
6. Miscellaneous

Total Estimated budgeted cost for the program for maximum 12 Persons, is RS.250, 000.00 (Two hundred and fifty thousand rupees only). Seeking your approval to proceed further.

Cheques should be written in favor of "Chairman – Construction Industry Development Authority" AC 77454975 BOC Independence square (Torrington), on or before the date of commencing of program.

Thank you
Yours Faithfully,


 Eng. Sanyeeewa Jayasundara
 Director General
 Construction Industry Development Authority (CIDA)

කොට්ඨාස 33, පාර්ලිමේන්තු පාර, පලාවත්ත, බත්තරමුල්ල. இல.33, பரலமத்துறை வீதி, பெலவத்தப்பத்தரமுல்லை. No.33, Parliament Rd, Palawatta, Battaramulla.
 Telephone: 2787661, 2785628, 2785712 Fax: 2785629 E-mail: cida.chpb@gmail.com Web site: http://www.cida.gov.lk
 පුරාණ සංරක්ෂණ : "සැව්ලිපයා", 123, විජයරාම මාවත, කොළඹ 07. Head Office: "Saviripaya", 123, Wijerama Mawatha, Colombo 07.
 Telephone: 2686092, 2686856, 2699801 Fax: 2699738 E-mail: cida@stinet.lk Website: http://www.cida.gov.lk

Established as the Construction Industry Development Authority under the Construction Industry Development Act No.33 of 2014
 Successor to Institute for Construction Training and Development (ICTAD)



Figure XXI.1: Quotation for Quantity Surveyors Training

Source: Primary data.

c) Payables Management

- i. Introduce an improved supplier evaluation process – No cost.
- ii. Develop well formulated supplier evaluation criteria sheets – Emailed to involved staff as a PDF.
- iii. Proposed Supplier Relationship Management Guidelines – Soft copy provided – No cost.
- iv. Introducing QuickBooks Software to the Accounts Department.

Table AXXI.4: Cost to Introduce QuickBooks Software to the Accounts Department

Cost Item	Amount (LKR)
QuickBooks Pro Version	74,000
Monthly KPI Form (Rs.15*12 months*10 participants)	1,800
Total Cost	75,800

Source: Author.



AccSoft Solutions (Pvt) Ltd

26, Pamankada Lane, Off W.A.De Silva Mawatha
Colombo-06, Sri Lanka.

Telephone: 53 53 530, info@accsoft.lk, www.accsoft.lk

09th December 2022

Ms. Jayendra
RE Weerakoon Constructions
Kalutara

Dear Madam

Proposal for Supply & Implementation of – QuickBooks

We thank you for giving us the opportunity for submitting a proposal for supply & implementation of QuickBooks. Please find enclosed our quotation along with AccSoft company profile, Major customer list and a QuickBooks Brochure.

QuickBooks is accredited by the “Institute of Chartered Accountants in England and Wales” and is marketed in many countries including USA, Canada & Australia etc. World-wide QuickBooks is used by over 5 million users and in Sri Lanka it's the number one selling Accounting Software for small to medium companies with over 3000 customers.

AccSoft Solutions (Pvt) Ltd is an **Authorized Master Re-seller for QuickBooks and Accredited Training Partner** in Sri Lanka. For confirmation please visit following link.

<http://www.quickbooks.co.za/reseller/find-a-reseller/reseller-details/?id=137>

If you need a further clarification or demonstration please contact the undersigned on 0777 367 786. Favourable and a prompt response much appreciated.

Thanking You,

Director

1) Financial Consideration	Price	
QuickBooks Genuine Licensed Software		
Option 01 UK Edition (VAT compatible with VAT Reports)		
QuickBooks PRO Version – Single User (Onetime payment)	Rs.74,000	
QuickBooks PRO Version – Each Additional User (Onetime payment)	Rs.48,000	
Option 02 UK Edition (VAT compatible with VAT Reports)		
QuickBooks PREMIER Version – Single User (Onetime payment)	Rs.96,000	
QuickBooks PREMIER Version – Each Additional User (Onetime payment)	Rs.68,000	
Note: For out of Colombo training and support will be provided through zoom and team viewer. If required to visit Outstation additional charges applicable		
Note: Option 1 support maximum up to five concurrent users per company database. Option 2 support up to thirty concurrent users on Purchasing additional licenses.		
Major differences in PRO & PREMIER Versions	PRO	PREMIER
Sales order & back order tracking (Proforma invoice)	No	Yes
Previous bank reconciliations	Last month	All months
Item assembly	No	Yes
Attached images to stock master file	No	Yes
One Click Journal Entry Reversal	No	Yes
Multiple Price Level	Basic	Advanced
Important! AccSoft Solutions (Pvt) Ltd is an Authorized Master Re-seller for QuickBooks in Sri Lanka. For details please refer covering email. **Prices are inclusive of free training as mentioned page 5 & one year free service maintenance. #Version information next page		

Payment Terms

50% on confirmation and balance after implementation. We will start implementation within two days from receiving advance payment. We shall deliver the Genuine Licence by email on realization of full payment only. Cheques should only be drawn in favour of “AccSoft Solutions (Pvt) Ltd”.

Figure XXI.2: Quotation for Accounting Software

Source: Primary data.